

AN ARITHMETIC LEFSCHETZ PROGRAMME OVER $\mathbb{Z}$: A METRIZED BIVARIANT CORRESPONDENCE SQUARE, ITS DETERMINANT PAIRING AND ITS CORNER TRACE CONDITIONAL ROW-(D) EVALUATION DRAFT

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ABSTRACT. We construct the first three ingredients of Weil’s argument over $\mathbb{Z}$ in an explicit periodic–cohomological–nuclear setting. The noncollapsed spherical square carries intrinsic periodic section cohomology, a coefficient-one determinant and derived prime contact. Witt Frobenius gives a faithful multiplicative correspondence family whose dynamic contact is $[\mathbb{Z} \xrightarrow{\Phi_n(1)} \mathbb{Z}]$ and has degree $\Lambda(n)$. Its monoidal realization in Meyer’s Poisson quotient yields the complete nuclear Lefschetz character, including the finite, polar and archimedean terms.

For the fourth ingredient we identify the primitive operator and give it a balanced channel factorization into a positive reference minus a positive load, from which the reference inverse, the threshold step and the one-step criterion are all read off. We prove full-space positivity for $0 < T \leq \log 2$; that the threshold step is exact, the old core receiving no correction from the newborn contact and the new directions being the annulus plus exactly two Tate modes; that the threshold condition is equivalent to a *unit* capacity bound in the output-defect metric; and that the attendant range condition is a consequence of the quantitative estimate rather than a separate hypothesis. We also prove a negative result which shapes the rest: comparing the two positive halves by size alone fails by a factor of exactly four in the limit, with a deficit of leading order $3\sqrt{N}$, so no sharpening of the coercivity constant can close it. At every arithmetic threshold we construct the finite Gamma–Tate amplitude, conservative Tate terminal, positive unreset Gamma storage and exact correlated rational leakage; the reciprocal half-integer band theorem pays every connected leakage cluster and reduces disconnected-band recombination to an exact constant/mean-zero Schur block.

Row (d) is thereby reduced to two named inputs: a uniform Schur angle bound for that block, and a transfer comparison identifying the Gamma–Tate source defect with the newborn-cell defect at unit constant. The second is finite-dimensional at each threshold; we evaluate it, and it fails — the comparison constant is below one and decays with the threshold, with the finite-dimensional evaluation bounding the true constant from above. The source model therefore does not reach the exact condition, although the exact condition is itself satisfied on every threshold tested. Proving it directly, in the output-defect metric where it carries constant one, is the remaining problem. Rows (a)–(c) and the certified initial range of row (d) are unconditional; the global Hodge inequality and the Riemann Hypothesis remain open.

1. WHERE THIS COMES FROM

1.1. The two companion papers. The companion paper *Branch Selection and Global Polarization in the Connes–CCM Route to RH: Finite Prime–Gamma Inertia, Tate Boundary Geometry, and the Divisor-Sensitivity Obstruction* develops the finite semilocal chain of Connes, Consani and Moscovici [10] together with Suzuki’s screw-function realization [24, 25], and reduces RH to a strict inequality on a cofinal family of finite rows. Its own contribution is a branch-selection

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theorem and an inertia calculus — Haynsworth additivity, Schur complements, scalar innovations, an adaptive Krylov hierarchy — carried out on the finite prime–Gamma coordinate, and a Tate boundary-geometry analysis of the archimedean fibre. It closes with an unconditional theorem, requiring nothing beyond Hardy’s 1914 result: the ordinary absolutely continuous completion destroys exactly the resonant data the argument needs. Its concluding section names two open programmes: a geometric intersection product with a Hodge index theorem, and a divisor-sensitive resonant completion.

The second companion, *Obstructions to Weil Positivity: A Catalogue of No-Go Theorems, Falsifiers and Countermodels in the Prime–Gamma Doob Coordinate*, does the complementary work: it records, with proofs, the mechanisms which cannot supply the missing inequality.

Both converge on the same reading. The obstruction is not a gap in the analysis. It is that a source object with the structure of a continuum is repeatedly asked to factor through a target with a finiteness constraint — atomic support against a dense group, point spectrum against absolutely continuous spectrum, a divisible group against a finitely generated one. The present paper takes the first of the two named programmes.

1.2. What this paper attempts. Weil’s proof of the Riemann Hypothesis for function fields [26, 27] rests on four ingredients over $\mathbb{F}_q$:

- (a) the surface $C \times_{\mathbb{F}_q} C$;
- (b) the Frobenius correspondences $\Gamma_n \subset C \times C$;
- (c) the identity $\Gamma_n \cdot \Delta = N_n$ between intersection numbers and the zeta counting function;
- (d) the Hodge index theorem, in the Castelnuovo–Severi form.

Together these give $|a_n| \leq 2g\sqrt{q}$. The programme of this paper is to ask what each of (a)–(d) would be over $\mathbb{Z}$, and to build them. The ambient framework is that of Connes and Consani [3, 5, 6, 7], whose Riemann–Roch strategy adapts Weil’s argument to characteristic zero along the lines of Mattuck–Tate and Grothendieck [20, 15], and whose Scaling Site supplies the correspondences Ψ_λ we use throughout.

Two structural facts fix the shape of the programme before any construction begins.

- **(b) is not imported.** Deninger’s foliated dynamical systems for arithmetic schemes [14] produce, for $\text{Spec } \mathbb{Z}$, a flow whose periodic orbits are in bijection with the primes, of length $\log p$. Morishita [22] constructs a continuous, Galois-equivariant, flow-anti-equivariant map from these systems to the Connes–Consani adelic spaces, sending the periodic packet at p onto the corresponding circle. But that map is built from rational Witt vectors, and at a point over p the residue field is $\overline{\mathbb{F}}_p$, whose multiplicative group $\mu_{(p)}$ has order prime to p . Every such map therefore annihilates the p -adic component — which is precisely the transverse direction whose kernel trace produces the local factor $|1 - u|_p^{-1}$. The bridge transports which orbits exist and nothing about how they meet. Section 7 instead builds row (b) inside row (a), using the Witt Frobenius and the cotangent contact already native to the arithmetic curve.
- **(d) cannot be imported from the explicit formula.** One may compute the signature of the corner pairing directly and find that it is equivalent to RH; that computation is recorded in §3.5. An equivalence derived from the explicit formula does no work, because Weil’s Hodge index is an *input* from geometry which then *implies* the zeta bound. It cannot be obtained by reading the desired sign back from the explicit formula. A proof must instead come either from an independent geometric index theorem or from a source-defined positive factorization whose construction does not use that sign.

Accordingly the burden falls first on (a). Section 2 constructs and audits the intrinsic periodic–Deligne–nuclear spherical package, including its total coefficient cohomology, normalized determinant, independent code certificate and reduced derived contact. Section 7 constructs row (b) on that carrier. The intervening sections study the analytic corner trace, and Section 8 constructs the nuclear cohomological completion that proves row (c). The fourth-row construction and its final open inequality are stated in Section 9.10.

1.3. The four rows over $\mathbb{F}_q$ and over $\mathbb{Z}$. The following table is the map the rest of the paper fills in. The left half is Weil’s, complete since 1948; the right half is what the present programme has over $\mathbb{Z}$, with its status stated as sharply as the results below allow. It is included because the distance between the two halves, and its precise shape, is the actual subject of this work.

Over $\mathbb{F}_q$ (Weil)	Over $\mathbb{Z}$ (this programme)	Status over $\mathbb{Z}$
(a) The surface $C \times_{\mathbb{F}_q} C$: smooth, projective, with finitely generated Néron–Severi group.	The noncollapsed spherical square $\mathcal{S}$; invertible Arakelov line objects; intrinsic periodic coefficient cohomology on the canonical integer-radius sector with Künneth, regular section-moduli cotangent and coefficient-one determinant; an independent spherical code certificate; reduced derived prime contact; and a rank-three module over the nuclear Dirichlet algebra (§2).	<i>Constructed in the intrinsic periodic–Deligne–nuclear category.</i> The total cohomology is that of the geometrically induced periodic coefficient object on the square; it is not, and by the proved growth obstruction cannot be, the raw functor of all bounded spherical sections.
(b) The Frobenius correspondences $\Gamma_n \subset C \times C$: honest algebraic cycles, with an intersection theory already available.	The faithful Witt cohomological correspondences Γ_n on $\mathcal{S}$: constant Witt coefficient dynamics, nuclear realization, metrized divisor line, and the derived lambda-characteristic contact in one constrained graph-image category (§7).	<i>Complete as cohomological correspondences.</i> This is relative to row (a) in the stated category. The contact is derived from F_n , agrees in Perf_{IDN} with row (a), and has determinant mass $\Lambda(n)$; no undecorated Chow lift is asserted.
(c) $\Gamma_n \cdot \Delta = N_n$: an intersection number equals the counting function of the zeta function.	The nuclear diagonal character of the completed Frobenius family equals the two finite contact sums plus the Fourier–archimedean term (§8).	<i>Complete as a cohomological/nuclear Lefschetz identity.</i> An ordinary Cartier/Chow promotion is not asserted.
(d) Castelnuovo–Severi: the intersection form on NS has exactly one positive eigenvalue. An <i>input</i> from geometry, which then implies the bound.	The corner pairing has $n_+ = 1 + \#P$, whence $n_+ = 1 \iff \text{RH}$ (§3.5). Its nuclear form has infinite rank and cannot factor through the two-dimensional ruling quotient (§9). The associated full primitive operator is proved positive for every $0 < T \leq \log 2$. The balanced channel factorization, the exactness of the threshold step, the unit output-capacity form of the threshold condition and the regularized one-step criterion are proved. At every arithmetic threshold the Gamma–Euler–Tate source, reciprocal bands, connected-cluster payment and constant/mean-zero Schur reduction are constructed.	<i>Construction complete; global positivity open, and the source route measured to fall short.</i> The theorem through $T = \log 2$ is unconditional, as is the scalar no-go (Cor. 9.20). Two inputs would suffice: Conjecture 9.37, the uniform Schur-angle estimate; and Problem 9.44, the transfer comparison carrying it to the exact condition. The second is evaluated in Remark 9.45 and <i>fails</i> , with the finite-dimensional evaluation bounding the true constant from above; so the source model does not reach the exact target. The exact target is itself satisfied on every threshold tested, and proving it directly is the remaining problem. Unconditional global row (d) is not claimed.

Two asymmetries in this table are worth naming, because together they locate the whole difficulty.

First, row (a) supplies a literal noncollapsed carrier, genuine divisor lines, an intrinsic periodic coefficient object, its total derived global sections, regular section-moduli cotangent, coefficient-one normalized determinant and reduced derived contact in one category. The universal framed code supplies an independent spherical determinant certificate. Its finiteness is local freeness over a nuclear arithmetic algebra, not finite abelian rank; the latter cannot realize the full bilinear kernel $\Lambda(mn)$ by Theorem 2.22. In Weil's column (c) is a theorem in the ordinary intersection theory of a surface and (d) is an input. Over $\mathbb{Z}$, row (c) is proved below as a categorical trace theorem in the nuclear cohomological completion of the row-(b) correspondences. It is not promoted back to an ordinary Cartier/Chow intersection on $\mathcal{B}_S$. Row (d) remains an open geometric input.

Second, the preliminary Dirichlet assembly of Theorem 7.9 is one-sided: the sum runs over positive shells $k \geq 1$. The completed row-(c) object of Theorem 8.3 is instead the summable nuclear supercharacter of the two Poisson quotients and is intrinsically two-sided. These must not be confused with the failed operation of merely symmetrizing a logarithmic derivative. That failure is sharp enough to be worth stating as a result, since it is the reason the construction must pass through a graded quotient at all.

Proposition 1.1 (Bare symmetrization is blind to the zeros). *Writing $G = \log \Gamma_{\mathbb{R}}$, the functional equation $\xi(s) = \xi(1-s)$ gives, after cancelling the polar contributions,*

$$(1) \quad \frac{\zeta'}{\zeta}(s) + \frac{\zeta'}{\zeta}(1-s) = -G'(s) - G'(1-s) = \log \pi - \frac{1}{2}\psi\left(\frac{s}{2}\right) - \frac{1}{2}\psi\left(\frac{1-s}{2}\right),$$

an elementary function of Γ alone. In particular the symmetrized logarithmic derivative has no pole at any zero of ξ , and every functional constructed from it is independent of the zeros.

Proof. At a zero ρ of ξ the two summands on the left have poles with opposite residues, which therefore cancel; the displayed identity is logarithmic differentiation of the functional equation. The right-hand side involves only Γ and π . $\square$

The Poisson quotient super-representation does something different: it retains the two graded quotient representations *before* taking their supercharacter, and the zeros survive in the grading rather than in a pole. Proposition 1.1 therefore accounts for the negative principal-difference calculation below while permitting Theorem 8.3; it also explains why the spectral signature cannot be used as the geometric input required in row (d), and it is the precise sense in which (1) is a no-go rather than a simplification.

1.4. A standing constraint. Throughout, no definition may use a zero of ξ , a Li coefficient, the sign of the Weil quadratic form, or a positive part extracted from that form. Zeros may appear in a theorem which is a consequence of a zero-free definition; they may not appear in the definition itself. This constraint is what makes the difference between a construction and a restatement, and several of the results below are negative precisely because a candidate construction violated it.

2. THE DELIGNE–NUCLEAR SPHERICAL SQUARE

This section constructs the object used as row (a). There are two points at which the characteristic-zero problem forces a departure from the literal wording of the classical row. First,

the set of all bounded sections of a spherical tensor product is much too large to have a surface Riemann–Roch dimension. Second, the full bilinear contact kernel $\Lambda(mn)$ has arbitrarily large prime rank and therefore cannot factor through a finitely generated abelian Néron–Severi group carrying that kernel. Both assertions will be proved below. The replacements are, respectively, intrinsic periodic coefficient cohomology on the same square and finite local freeness over a nuclear Dirichlet algebra. A negabinary code supplies an independent spherical determinant certificate for the first replacement. These are not hypotheses added to hide the two failures: they are constructions on the same spherical carrier, and the two failures explain why these are the strongest mutually compatible requirements.

2.1. The categorical contract. Write $\mathbb{S}$ for the sphere Γ -ring and HR for the Eilenberg–Mac Lane Γ -ring of a commutative ring R . We use the closed symmetric monoidal category of Segal Γ -sets and its derived category of simplicial Γ -sets. The Day smash product is denoted by $\wedge_{\mathbb{S}}$. These conventions are those of [4, §§2,7]; in particular, the assembly morphism

$$(2) \quad \alpha_{M,N} : M \wedge_{\mathbb{S}} N \longrightarrow \widetilde{M} \circ N$$

is functorial in both variables.

Definition 2.1. An *intrinsic periodic–Deligne–nuclear arithmetic square* consists of:

- (1) a noncollapsed spherical square $\mathcal{Y}_{\mathbb{S}}$;
- (2) invertible divisor objects with internal principal descent and a functorial effective cone;
- (3) on the canonical positive integer-radius sector, a geometrically induced coefficient object on the square, intrinsic derived global sections, and perfect cotangents of the regular section moduli, with one-ruling linear growth and mixed Künneth quadratic growth, followed by its unique continuous numerical extension;
- (4) the resulting continuous determinant line, its spherical code certificate, the reduced derived prime-contact line and their quotient-defined Green factor in one symmetric monoidal category;
- (5) an explicit finite locally free correspondence module over a nuclear arithmetic coefficient algebra, with exact composition and von Mangoldt contact.

Completeness below means completeness of these five specified structures and their comparison maps. It does not identify coefficient cohomology with the functor of all raw bounded spherical sections; that identification is impossible by Theorem 2.6. Nor does it mean that the two incompatible properties ruled out in Theorems 2.6 and 2.22 have been assumed.

The definition is independent of the zeros of ξ and of every positivity statement. It is therefore eligible as input to the later rows rather than a reformulation of their desired conclusion.

2.2. The literal spherical carrier and its divisor lines. Let $X = \overline{\text{Spec } \mathbb{Z}}$ with the sheaf $\mathcal{O}_X$ of $\mathbb{S}$ -algebras constructed by Connes and Consani [4, Proposition 6.3]. Thus X has the usual finite points and the archimedean closed point, and $\mathcal{O}_X$ is obtained by imposing the ordinary absolute-value bound at infinity. Define

$$(3) \quad \mathcal{Y}_{\mathbb{S}} := X \times_{\mathbb{S}} X, \quad \mathcal{O}_{\mathcal{Y}} := \text{pr}_1^{-1} \mathcal{O}_X \wedge_{\mathbb{S}} \text{pr}_2^{-1} \mathcal{O}_X.$$

This formula is also a construction of the product: on every pair of open charts it takes the Day smash of the two section Γ -rings, uses the smashes of the restriction maps, and then sheafifies levelwise. Thus the relative smash in (3) is, by definition, the sheafification of that presheaf; the sheaf axiom is not being assumed for an un-sheafified Day product.

Proposition 2.2 (Noncollapse). *The two projections in (3) do not identify $\mathcal{Y}_{\mathbb{S}}$ with X .*

Proof. At the generic stalk, filtered colimits commute with the Day colimits and the structure Γ -ring is $H\mathbb{Z} \wedge_{\mathbb{S}} H\mathbb{Z}$. Connes and Consani prove that this smash is not isomorphic to $H\mathbb{Z}$ [4, Proposition 7.4]. More explicitly, after assembly the two-variable Laurent polynomial class $(t_1 - 1)(t_2 - 1)$ restricts to zero under both coordinate marginals but is nonzero before either restriction. For $H\mathbb{Z}$ the pair of marginals is bijective. An isomorphism of the square with either factor would induce an isomorphism on the generic stalk and contradict that distinction. $\square$

An Arakelov divisor on X is written $D = D_f + a\{\infty\}$. Connes–Consani associate to it a sheaf $\mathcal{O}_X(D)$ of $\mathcal{O}_X$ -modules, and multiplication by q^{-1} gives the canonical isomorphism

$$(4) \quad \mathcal{O}_X(D) \xrightarrow{\sim} \mathcal{O}_X(D + \operatorname{div}(q)), \quad q \in \mathbb{Q}^\times.$$

Their multiplication theorem says that $\mathcal{O}_X(D) \wedge_{\mathcal{O}_X} \mathcal{O}_X(E) \simeq \mathcal{O}_X(D + E)$ whenever the relevant archimedean radii are rational; for arbitrary radii its strict nonunital version has the same exact tensor law [4, Proposition 6.4 and Remark 6.5].

We shall use the integral, prime-generated, rational-radius sector $\operatorname{Div}_{\operatorname{ext}}(X)$. It contains all finite prime rulings needed for the contact formula. For D, E in this sector put

$$(5) \quad \mathcal{L}(D, E) := \operatorname{pr}_1^* \mathcal{O}_X(D) \wedge_{\mathcal{O}_{\mathcal{Y}}} \operatorname{pr}_2^* \mathcal{O}_X(E).$$

Proposition 2.3 (Divisor tensor law and principal descent). *The object $\mathcal{L}(D, E)$ is invertible and there are coherent isomorphisms*

$$(6) \quad \mathcal{L}(D, E) \wedge \mathcal{L}(D', E') \simeq \mathcal{L}(D + D', E + E'), \quad \mathcal{L}(D, E)^{-1} \simeq \mathcal{L}(-D, -E).$$

The assignment descends through principal divisors.

Proof. Associativity and symmetry of relative smash reduce the first isomorphism to the one-dimensional multiplication theorem in each factor. Taking $(D', E') = (-D, -E)$ gives the unit and proves invertibility. For principal changes (q, r) , smash the two explicit maps (4); their composites and tensor products are multiplication by the corresponding products of rational numbers. Hence the unit, associativity, symmetry and descent diagrams commute before passage to isomorphism classes. $\square$

Thus the divisor classes used below are represented by actual invertible objects on the literal square. No freely declared prime lattice is being substituted for a Picard category.

For precision, let $\operatorname{Pic}_{\operatorname{ext}}^{\operatorname{fact}}$ be the *factorized external Picard groupoid*: an object retains the two one-dimensional divisor modules together with their external product (5), and a morphism is a pair of module isomorphisms. Its forgetful functor lands in the Picard groupoid of the square. Retaining the factorization is essential: no assertion that these objects exhaust the full Picard groupoid of $\mathcal{B}_{\mathbb{S}}$ is used.

Proposition 2.4 (External classes and effectivity). *The degree identifies the isomorphism classes of the one-dimensional factorized divisor modules in the rational-radius sector with $\log \mathbb{Q}_{>0} \subset \mathbb{R}$. Consequently*

$$(7) \quad \pi_0 \operatorname{Pic}_{\operatorname{ext}}^{\operatorname{fact}} \simeq (\log \mathbb{Q}_{>0})^2.$$

Its degree image has closure $\mathbb{R}^2$; we denote this completed numerical degree space by $N_{\operatorname{deg}}^1 \simeq \mathbb{R}^2$. The actual effective monoid maps to $(\log \mathbb{Q}_{>0} \cap \mathbb{R}_{\geq 0})^2$, whose closure is $\mathbb{R}_{\geq 0}^2$. Tensor product adds the coordinates, and the code transition maps preserve the actual effective monoid and its closure.

Proof. For $D = D_f + a\{\infty\}$ in the rational-radius sector, $a \in \log \mathbb{Q}_{>0}$. Multiply by the unique positive rational whose valuations cancel D_f . The result is $\deg(D)\{\infty\}$, and its degree still belongs to $\log \mathbb{Q}_{>0}$. Conversely, if $a\{\infty\}$ and $b\{\infty\}$ differ by a principal divisor with zero finite part, the rational function is ± 1 , so $a = b$. This is also the explicit content of the divisor-module classification [4, Proposition 6.4(ii)].

For $c\{\infty\}$, the level-one global sections are $\{n \in \mathbb{Z} : |n| \leq e^c\}$. A nonzero section exists exactly when $c \geq 0$. Apply this independently to the two retained factors. Since $\log \mathbb{Q}_{>0}$ is dense in $\mathbb{R}$, taking closures gives the asserted numerical space and cone. The tensor and transition assertions follow from (6) and from zero-extension of digit coordinates. $\square$

2.3. Why all bounded sections cannot be the surface cohomology. For $m \geq 0$, let

$$H_m = (H\mathbb{Z})_{[-m, m]}$$

be the bounded Γ -submodule: at level k_+ it consists of integer k -tuples whose ℓ^1 norm is at most m . These are precisely the archimedean bounded-section modules appearing after principal normalization of an integral divisor. Put

$$B_s = (1, -2, 4, \dots, (-2)^{s-1})$$

Let M_s be the largest absolute value of a subset sum of B_s , and put

$$(8) \quad L_s := \|B_s\|_1 = \sum_{j=0}^{s-1} 2^j = 2^s - 1.$$

For a tolerant $\mathbb{S}$ -module, $\dim_{\mathbb{S}}$ denotes the least cardinality of a level-one subset from which every level-one element is an admissible zero-one sum, as in the dimension definition of [7, Appendix A].

Lemma 2.5 (Negabinary separation and admissible length). *All subset sums of B_s are distinct and form an interval of 2^s consecutive integers. More precisely, for $k \geq 0$,*

$$(9) \quad M_{2k} = \frac{2(4^k - 1)}{3}, \quad M_{2k+1} = \frac{4^{k+1} - 1}{3}.$$

The largest length for which the digit vector itself is an admissible element of H_m is

$$(10) \quad r(m) := \max\{s \geq 0 : L_s \leq m\} = \lfloor \log_2(m+1) \rfloor,$$

and $r(m) = \log_2 m + O(1)$ as $m \rightarrow \infty$.

Proof. Suppose that two subsets have the same sum and subtract their incidence vectors. If j is the least index with nonzero coefficient, division by $(-2)^j$ gives an odd number equal to an even number, a contradiction. Thus all subset sums are distinct. If I_s is their set, then $I_{s+1} = I_s \cup (I_s + (-2)^s)$. Starting from $I_0 = \{0\}$, the two intervals are adjacent at every step, with their order alternating with the parity of s ; induction proves contiguity. The largest positive sum is the sum of the even powers and the absolute value of the most negative sum is the sum of the absolute values of the odd powers. Summing the two geometric series gives (9). Independently, the definition of the bounded module uses the norm of the whole digit vector, not the largest subset sum; equation (8) therefore gives the exact formula (10) and its asymptotic. Notice that $M_s \leq m$ alone would not suffice: for example $M_4 = 10$ whereas $L_4 = 15$. $\square$

Theorem 2.6 (Raw spherical Künneth obstruction). *For $m, n \geq 1$, the Connes–Consani generator dimension satisfies*

$$(11) \quad \dim_{\mathbb{S}}(H_m \wedge_{\mathbb{S}} H_n) \geq 2^{r(n)} - 1,$$

and symmetrically with m, n exchanged. If $n_t = \lfloor e^{tb} \rfloor$ with $b > 0$, the right side is $\exp(tb + O(1))$. Hence raw bounded sections cannot have quadratic $O(t^2)$ surface growth.

Proof. Use $1 \in H_m(1_+)$ and $B_s \in H_n(s_+)$, where $s = r(n)$. The latter membership is now exactly $\|B_s\|_1 = L_s \leq n$. For each subset $J \subset \{0, \dots, s-1\}$ the Day-colimit presentation gives a level-one element z_J by retaining exactly the pairs $(1, j)$ with $j \in J$. Under (2),

$$\alpha(z_J) = [b_J], \quad b_J = \sum_{j \in J} (-2)^j,$$

in the reduced free abelian group $\mathbb{Z}[\mathbb{Z}]/\mathbb{Z}[0]$. By Lemma 2.5, the $2^s - 1$ classes with $J \neq \emptyset$ are distinct basis vectors and hence linearly independent.

If a level-one set F generates the smash by admissible zero-one sums, each z_J is such a sum of elements of F . Assembly sends admissible sums to ordinary sums in its special additive target. The subgroup generated by $\alpha(F)$ therefore contains $2^s - 1$ independent vectors, while its rank is at most $\#F$. This proves (11). Lemma 2.5 gives $2^{r(n)} = \exp(tb + O(1))$, which dominates every quadratic polynomial. $\square$

The theorem fixes the logical meaning of the next construction. We do not rename the full Day smash. We linearize a distinguished, functorial family of its genuine sections and take the cotangent space of that linearization.

2.4. The initial negabinary section code. The Day presentation is Boolean at this point: a map to 1_+ either retains or discards each input pair. We require the radix to be negative so that one Boolean tower contains sections of both signs without an additional sign coordinate. Call a negative integer b an *expanding Boolean radix* when $|b| > 1$, and order these radices by absolute value. This ordered set has the unique initial element -2 . Equivalently, -2 is the unique negative integral solution of

$$(12) \quad 1 + 1 = X + X^2.$$

Thus evaluation $X \mapsto -2$ defines a morphism from the quotient of the free spherical algebra by (12) to $H\mathbb{Z}$. The other integral solution $X = 1$ repeats the unit and gives no expanding digit system. Hence the initial expanding negative integral Boolean-radix rule selects -2 without an auxiliary prime, finite field or interpolation base. The radices $-3, -4, \dots$ are not alternative outputs of this rule because none is initial. This is canonicity inside the explicitly stated ordered class of radices; no assertion that the spherical carrier itself uniquely selects that class is used.

Let $G_r = \mathbb{F}_2^r$. For $r = r(m)$ define

$$(13) \quad c_m : G_r \longrightarrow H_m(1_+), \quad (\epsilon_i) \longmapsto \sum_{i=0}^{r-1} \epsilon_i (-2)^i.$$

Every represented integer has absolute value at most $M_r \leq L_r \leq m$, so the map lands in $H_m(1_+)$; injectivity is the first assertion of Lemma 2.5. Let

$$(14) \quad Q_m := \text{Spec Sym}_{\mathbb{F}_2}(G_r^\vee) \simeq \mathbb{A}_{\mathbb{F}_2}^r.$$

Its $\mathbb{F}_2$ -points are precisely the code parameters. We call it the *linear code envelope*; it is not claimed to represent all bounded spherical sections over arbitrary coefficient rings.

For $r = r(m)$ and $s = r(n)$, and for a matrix $\epsilon = (\epsilon_{ij}) \in \mathbb{F}_2^{r \times s}$, let $v_\epsilon : r_+ \wedge s_+ \rightarrow 1_+$ retain (i, j) exactly when $\epsilon_{ij} = 1$. The Day presentation defines

$$(15) \quad z_\epsilon = [r_+, s_+, v_\epsilon, B_r \wedge B_s] \in (H_m \wedge_{\mathbb{S}} H_n)(1_+).$$

Here $B_r \in H_m(r_+)$ and $B_s \in H_n(s_+)$ by (8)–(10); this is the admissibility condition required by the Day representative.

Theorem 2.7 (Separation of the mixed code). *The map $\epsilon \mapsto z_\epsilon$ is injective. In particular the $2^{r(m)r(n)}$ points of*

$$(16) \quad Q_{m,n} := \text{Spec Sym}_{\mathbb{F}_2}((G_r \otimes_{\mathbb{F}_2} G_s)^\vee)$$

label distinct genuine sections of the spherical mixed module.

Proof. Put $D_m = (\log m)\{\infty\}$ and $D_n = (\log n)\{\infty\}$. The external-product morphism sends (15) to a global section of $\mathcal{L}(D_m, D_n)$. On restriction to the generic finite chart it is the same Day representative in $H\mathbb{Z} \wedge_{\mathbb{S}} H\mathbb{Z}$. It is therefore enough to separate the restrictions: equality of two global sections would imply equality on that chart. There the assembly formula is

$$(17) \quad \alpha(z_\epsilon) = \sum_{i=0}^{r-1} (-2)^i \left[\sum_{j=0}^{s-1} \epsilon_{ij} (-2)^j \right] \quad \text{in } \mathbb{Z}[\mathbb{Z}]/\mathbb{Z}[0].$$

For a nonzero exponent b , its coefficient is the subset sum of the $(-2)^i$ over the rows whose inner sum equals b . Negabinary uniqueness recovers that subset of row indices. The exponent b then recovers, again by uniqueness, the selected columns in every one of those rows. A nonempty row cannot have inner sum zero, since zero already represents the empty subset. After all nonzero exponents are recovered, the remaining row indices are exactly the empty rows. Hence (17) determines every entry of ϵ . Equality before assembly would imply equality after assembly, so the original sections are distinct. $\square$

The precise status of the affine envelope is important. For an $\mathbb{F}_2$ -algebra R , define the framed digit-deformation functor

$$(18) \quad \mathfrak{P}_{m,n}(R) := (G_r \otimes_{\mathbb{F}_2} G_s) \otimes_{\mathbb{F}_2} R.$$

It remembers the ordered digit-pair coordinates, while Theorem 2.7 supplies an injection from its $\mathbb{F}_2$ -points to genuine spherical sections.

Proposition 2.8 (Universal framed linear envelope). *The scheme $Q_{m,n}$ represents $\mathfrak{P}_{m,n}$. For every finite-dimensional $\mathbb{F}_2$ -space W , fibrewise linear natural transformations from $\mathfrak{P}_{m,n}$ to $R \mapsto W \otimes R$ are in canonical bijection with linear maps $G_r \otimes G_s \rightarrow W$. Under this bijection the identity is initial among framed linear realizations of the digit parameters. Its cotangent at zero is canonically $(G_r \otimes G_s)^\vee$.*

Proof. For every R ,

$$Q_{m,n}(R) = \text{Hom}_{\mathbb{F}_2\text{-alg}}(\text{Sym}((G_r \otimes G_s)^\vee), R) \simeq (G_r \otimes G_s) \otimes R.$$

The transformation assertion is the Yoneda correspondence for vector schemes. Taking the augmentation ideal at the zero vector gives the last assertion. $\square$

This universal property is deliberately framed: it is universal after the ordered digit observables have been selected. It is not a moduli theorem for all bounded sections. Indeed c_m cannot be additive: $c_m(e_0) = 1$ while $2e_0 = 0$ in G_r and $2 \cdot 1 \neq 0$ in the integral section group. Nor may one replace Q_m by the finite Boolean scheme $x_i^2 = x_i$ without changing the tangent theory: that scheme is étale and has zero cotangent. Thus the nonzero object below is exactly the universal *linear relaxation of the framed code*; it is not claimed to be the cotangent of the unlinearized section functor.

After a principal normalization $D \sim a\{\infty\}$ in the actual sector, $a \in \log \mathbb{Q}_{>0}$. For an integer scale $t \geq 1$, the bounded module for tD is $H_{[e^{ta}]}$. Proposition 2.3 transports (13) and (15) back

to actual global sections of $\mathcal{L}(tD, tE)$. Thus the code is attached to the divisor line on the carrier, not merely to an unrelated finite vector space.

2.5. The framed code cotangent and its Künneth theorem. Let 0 denote the zero code point. Since the envelopes are affine spaces,

$$(19) \quad \Omega_m := \mathfrak{m}_0/\mathfrak{m}_0^2 \simeq G_{r(m)}^\vee, \quad \Omega_{m,n} \simeq (G_{r(m)} \otimes G_{r(n)})^\vee.$$

They are smooth at 0, so there is no higher cotangent obstruction. Let V_m and $V_{m,n}$ be the real vector spaces freely generated by the distinguished coordinate bases of Ω_m and $\Omega_{m,n}$; this is a basis lift, not a nonexistent scalar extension $\mathbb{F}_2 \rightarrow \mathbb{R}$.

Theorem 2.9 (Cotangent Künneth). *There are canonical basis-preserving isomorphisms*

$$(20) \quad V_{m,n} \simeq V_m \otimes_{\mathbb{R}} V_n, \quad \dim_{\mathbb{R}} V_{m,n} = r(m)r(n).$$

If $m \leq m'$ and $n \leq n'$, zero-extension in the new high digit coordinates gives split injections $V_{m,n} \hookrightarrow V_{m',n'}$, and these injections commute with (20).

Proof. The coordinate functions at zero of $Q_{m,n}$ are indexed by Cartesian pairs (i, j) . Sending the (i, j) cotangent coordinate to the tensor of the i th and j th one-dimensional cotangent coordinates gives the first map and its inverse on bases. The dimension formula follows. Monotonicity of $r(m)$ makes zero-extension well defined; coordinate restriction is the split cotangent projection, so duality gives the asserted tangent injection. All maps are defined on the same Cartesian basis, which proves their compatibility and transitivity. $\square$

For actual effective rational-radius divisors D, E of positive degrees $a, b \in \log \mathbb{Q}_{>0}$ and integer $t \rightarrow \infty$, set

$$(21) \quad m_t = \lfloor e^{ta} \rfloor, \quad n_t = \lfloor e^{tb} \rfloor, \quad \mathcal{H}_t^{\text{cot}}(D, E) := V_{m_t, n_t}.$$

Corollary 2.10 (Coefficient-one framed-code dimension). *The limit through the positive integers*

$$(22) \quad h_{\text{cot}}(D, E) := (\log 2)^2 \lim_{t \rightarrow \infty} \frac{\dim_{\mathbb{R}} \mathcal{H}_t^{\text{cot}}(D, E)}{t^2}$$

exists and equals ab .

Proof. Equation (10) gives

$$r(m_t) = \frac{ta}{\log 2} + O(1), \quad r(n_t) = \frac{tb}{\log 2} + O(1).$$

Multiply the two expansions, use Theorem 2.9, divide by t^2 and multiply by $(\log 2)^2$. $\square$

The normalization is forced after imposing degree matching on this code. In one ruling the only constant c for which $cr(\lfloor e^{ta} \rfloor)/t \rightarrow a$ for every $a > 0$ is $c = \log 2$. Künneth multiplicativity then forces the product constant $(\log 2)^2$ in (22). This is the precise sense in which the coefficient one is canonical for the chosen digit code. It is not a metric supplied by the determinant functor before the degree-matching condition is imposed. No interpolation base has been chosen. The formula, initially proved on $(\log \mathbb{Q}_{>0})_{>0}^2$, extends uniquely by continuity to the completed numerical cone $\mathbb{R}_{>0}^2$.

Principal invariance is equally explicit. Multiplication by the unique positive rational cancelling the finite coefficients sends D to its purely archimedean normalization. The remaining rational unit is -1 ; it changes the signs of represented sections but fixes the zero code point and every digit index. Degree, $r(m_t)$, the tangent lattice and all transition maps therefore depend only on the metrized divisor class.

For a finite prime-ruling divisor

$$(23) \quad x = \sum_p x_{p,1} e_{p,1} + \sum_p x_{p,2} e_{p,2}, \quad d_i(x) = \sum_p x_{p,i} \log p,$$

the preceding construction applied to the total divisor in each ruling gives the following formula on the effective monoid. Since d_1, d_2 are additive, this polynomial has a unique quadratic extension to the Grothendieck group. For a general class we define the based line by the same metric formula; on the effective cone it agrees with the determinant limit already constructed:

$$(24) \quad F_{\text{code}}(x) = d_1(x)d_2(x), \quad B_{\text{code}}(x, y) = d_1(x)d_2(y) + d_2(x)d_1(y).$$

2.6. The nuclear arithmetic coefficient algebra. Let

$$(25) \quad \mathcal{C}_{\mathbb{R}} := \left\{ a = (a_n)_{n \geq 1} : q_k(a) := \sum_{n \geq 1} |a_n| n^k < \infty \text{ for every } k \geq 0 \right\}$$

with Dirichlet convolution $(a * b)_r = \sum_{mn=r} a_m b_n$.

Proposition 2.11 (Nuclear Dirichlet algebra). *$\mathcal{C}_{\mathbb{R}}$ is a complete commutative unital nuclear Fréchet algebra. Its point masses satisfy*

$$(26) \quad \delta_m * \delta_n = \delta_{mn}.$$

The von Mangoldt functional

$$(27) \quad \ell(a) = \sum_{n \geq 1} a_n \Lambda(n)$$

is continuous.

Proof. Absolute convergence and $(mn)^k = m^k n^k$ give

$$q_k(a * b) \leq q_k(a) q_k(b),$$

which proves joint continuity and the algebra law. Completeness follows by viewing (25) as the projective limit of the Banach spaces $\ell^1(\mathbb{N}, n^k)$. After conjugating the bonding map $\ell^1(\mathbb{N}, n^{k+2}) \rightarrow \ell^1(\mathbb{N}, n^k)$ to ordinary ℓ^1 , it is the diagonal operator with entries n^{-2} . Its rank-one expansion has nuclear norm at most $\sum_n n^{-2} < \infty$. The projective-limit criterion for countably normed spaces therefore proves nuclearity [23, Chapter 4]. Formula (26) is immediate. Finally $\Lambda(n) \leq \log n \leq n$ for $n \geq 2$, so $|\ell(a)| \leq q_1(a)$. $\square$

The augmentation

$$(28) \quad \varepsilon : \mathcal{C}_{\mathbb{R}} \longrightarrow \mathbb{R}, \quad \varepsilon(a) = a_1,$$

is a continuous algebra map, since $(a * b)_1 = a_1 b_1$.

2.7. One arithmetic perfect category. Let $H_{\text{sol}}(\mathcal{C}_{\mathbb{R}})$ denote the Eilenberg–Mac Lane E_{∞} -algebra of the solidification of the condensed Fréchet algebra $\mathcal{C}_{\mathbb{R}}$. Solidification is symmetric monoidal by [12, Theorem 6.2]; hence the continuous commutative convolution algebra of Proposition 2.11 determines a commutative algebra object after solidification, and its connective Eilenberg–Mac Lane object is an E_{∞} -algebra. This is the meaning of the notation; no discrete replacement of the Fréchet topology is made. On $\mathcal{B}_{\mathbb{S}}$ put

$$(29) \quad \mathcal{O}_{\mathbb{Z}} = \mathcal{O}_{\mathcal{B}} \wedge_{\mathbb{S}} H\mathbb{Z}, \quad \mathcal{O}_{\mathbb{R}} = \mathcal{O}_{\mathcal{B}} \wedge_{\mathbb{S}} H\mathbb{R}, \quad \mathcal{O}_{\mathcal{C}} = \mathcal{O}_{\mathcal{B}} \wedge_{\mathbb{S}} H_{\text{sol}}(\mathcal{C}_{\mathbb{R}}).$$

The maps $\mathbb{Z} \rightarrow \mathbb{R}$ and (28) induce maps to $\mathcal{O}_{\mathbb{R}}$. Write $\mathcal{Y}_{\mathbb{Z}}$, $\mathcal{Y}_{\mathbb{R}}$ and $\mathcal{Y}_{\mathbb{C}}$ for the same carrier with these three structure sheaves. Define

$$(30) \quad \text{Perf}_{DN}(\mathcal{Y}_{\mathbb{S}}) = \text{Perf}(\mathcal{O}_{\mathbb{Z}}) \times_{\text{Perf}(\mathcal{O}_{\mathbb{R}})}^h \text{Perf}(\mathcal{O}_{\mathbb{C}}).$$

Concretely, an object is a triple $(M_{\mathbb{Z}}, M_{\mathbb{C}}, \alpha)$ of perfect modules and an equivalence

$$\alpha : M_{\mathbb{Z}} \otimes_{\mathcal{O}_{\mathbb{Z}}} \mathcal{O}_{\mathbb{R}} \xrightarrow{\sim} M_{\mathbb{C}} \otimes_{\mathcal{O}_{\mathbb{C}}} \mathcal{O}_{\mathbb{R}}.$$

Tensor products, duals and cofibres are taken componentwise and the real comparison maps are tensored or mapped accordingly. This description proves directly that (30) is stable and symmetric monoidal and that its perfect objects are dualizable. It is the arithmetic integral–real gluing familiar from Deligne cohomology, with the nuclear coefficient component added.

Only the full stable symmetric monoidal subcategory generated by invertible divisor lines, finite free modules and their finite cofibres is used below. For these objects perfectness is local and immediate from a finite free resolution; tensor products and descent maps are computed componentwise. Thus no exactness assertion for an infinite completed tensor product enters any proof. Relative affine objects such as (31) mean, by definition, the opposites of the displayed commutative free algebras; their functor of points is the free-algebra adjunction (32).

The line (5) extends scalars to the three components; the two realifications agree tautologically. Proposition 2.3 therefore gives invertible objects, with the same principal descent, in (30).

At scale t , let $V_{t,\mathbb{Z}}(D, E)$ be the free integral lattice on the ordered digit-pair coordinate basis of (21), and let $V_t(D, E)$ be its real span. There is an integral affine envelope

$$(31) \quad \mathcal{Q}_{t,\mathbb{Z}}(D, E) = \underset{\mathcal{Y}_{\mathbb{Z}}}{\text{Spec Sym}}_{\mathcal{O}_{\mathbb{Z}}}(\mathcal{O}_{\mathbb{Z}} \otimes_{\mathbb{Z}} V_{t,\mathbb{Z}}(D, E)).$$

Its special fibre over $\mathbb{F}_2$ is the product of the carrier fibre with the affine code envelope (16); in particular its parameter fibre is exactly Q_{m_t, n_t} . Define $\mathcal{Q}_{t,\mathbb{C}}$ by the same formula over $\mathcal{O}_{\mathbb{C}}$ with V_t .

The free-algebra adjunction characterizes this construction: for an affine $f : T \rightarrow \mathcal{Y}_{\mathbb{Z}}$,

$$(32) \quad \text{Hom}_{\mathcal{Y}_{\mathbb{Z}}}(T, \mathcal{Q}_{t,\mathbb{Z}}) \simeq \text{Hom}_{\mathcal{O}_T}(f^*(\mathcal{O}_{\mathbb{Z}} \otimes V_{t,\mathbb{Z}})^{\vee}, \mathcal{O}_T).$$

Thus it is the relative vector bundle universally linearizing the framed digit coordinates of Proposition 2.8; it is not identified with a moduli space of all sections.

If e_t is the zero section, the polynomial-algebra calculation of the relative cotangent complex gives

$$e_t^* L_{\mathcal{Q}_{t,\mathbb{Z}}/\mathcal{Y}_{\mathbb{Z}}} \simeq \mathcal{O}_{\mathbb{Z}} \otimes_{\mathbb{Z}} V_{t,\mathbb{Z}}, \quad e_t^* L_{\mathcal{Q}_{t,\mathbb{C}}/\mathcal{Y}_{\mathbb{C}}} \simeq \mathcal{O}_{\mathbb{C}} \otimes_{\mathbb{R}} V_t$$

in degree zero: a symmetric algebra is free, so its cotangent complex is its module of differentials and has no higher term. Define

$$(33) \quad \mathbf{H}_t^{\text{cot}}(D, E) = (\mathcal{O}_{\mathbb{Z}} \otimes_{\mathbb{Z}} V_{t,\mathbb{Z}}(D, E), \mathcal{O}_{\mathbb{C}} \otimes_{\mathbb{R}} V_t(D, E), \alpha_t),$$

where α_t identifies the two real digit bases. Thus (33) is the relative cotangent at the zero section of the universal framed linear envelopes. It is determined by the framed code through (32), but is not the intrinsic cotangent of the full spherical section functor. Each component is finite free, hence perfect, and Theorem 2.9 gives an isomorphism in the same category

$$(34) \quad \mathbf{H}_t^{\text{cot}}(D, E) \simeq \mathbf{H}_{t,1}^{\text{cot}}(D) \otimes \mathbf{H}_{t,2}^{\text{cot}}(E).$$

The zero-extension maps make these objects a functorial pro-system on the effective divisor category, and principal isomorphisms transport that system.

2.8. The auxiliary normalized code determinant. Apply the determinant functor for perfect complexes [19] to (33). On its real digit determinant, give the ordered wedge generator the norm

$$(35) \quad \|1_t\| = \exp(-(\log 2)^2 \dim_{\mathbb{R}} V_t(D, E)).$$

This is the degree-matched metric of the framed code. The determinant functor supplies the line and its basis; the displayed norm is the explicit normalization selected in Corollary 2.10, not the orthonormal determinant metric. For a based normed line $(L, 1)$ and $c > 0$, write $L^{(c)}$ for the same based line with norm $\|1\|^c$; this is an explicit metric operation, not a fractional algebraic tensor power.

Proposition 2.12 (Continuous normalized code determinant). *The normalized lines $\det \mathbf{H}_t^{\text{cot}}(D, E)^{\langle t^{-2} \rangle}$ converge in the Picard groupoid of based normed real lines to*

$$(36) \quad \lambda_{\text{code}}(D, E) = (\mathbb{R}1, \|1\| = e^{-ab}).$$

The limit is principal-invariant, functorial under effective inclusions and compatible with Künneth.

Proof. By the proof of Corollary 2.10,

$$\frac{(\log 2)^2}{t^2} \dim_{\mathbb{R}} V_t(D, E) = ab + O(t^{-1}).$$

Taking minus the logarithm of (35) proves convergence to (36). The transition, principal and Künneth maps preserve the ordered digit bases; the corresponding determinant diagrams therefore commute at every finite stage and after passage to the limit. $\square$

2.9. Intrinsic periodic section cohomology. The code cotangent above is a spherical certificate, but it cannot be the cotangent of the functor of all bounded spherical sections: Theorem 2.6 gives incompatible growth. We now construct the cohomology object used in the final package from an actual moduli space of sections.

Connes and Consani construct the arithmetic pullback

$$\pi : X_{\text{vis}} \longrightarrow X$$

of the Riemann sector along the extended Abel–Jacobi map. Its fibre over a finite prime p is canonically the pointed periodic curve

$$(37) \quad C_p = \mathbb{R}_+^\times / p^{\mathbb{Z}} \simeq \mathbb{R} / (\log p)\mathbb{Z},$$

whose neutral point is the standard subgroup $\mathbb{Z}[1/p] \subset \mathbb{R}$ [8, §6.3]. Form the finite-prime pair pullback

$$(38) \quad H_{\text{per}} := \coprod_{p,q} C_p \times C_q, \quad \Pi_{\text{per}} : H_{\text{per}} \longrightarrow \mathcal{Y}_{\mathbb{S}}.$$

Here the target of the underlying map is $X \times X$; its spherical structure remains the one in (3). Give the source the disjoint-union topology of the periodic sites. Inverse images preserve covers, so (38) defines a geometric morphism of the corresponding sites.

Write o_p for the neutral point and $D_p(a) := a[o_p]$ for $a \in \mathbb{Z}$. This is an actual divisor, not merely a degree label. The full divisor-class classification is $(\deg, \chi) : \text{Div}(C_p) / \text{Prin}(C_p) \simeq \mathbb{R} \times \mathbb{Z} / (p-1)\mathbb{Z}$ [5, Theorem 5.6]; below we need only the integral neutral-point sector forced by the prime rulings.

We use an ∞ -categorical enhancement, not an unspecified model structure. For a fixed (p, q) let $\mathbf{V}_{p,q}^0$ be the essentially small symmetric monoidal category generated by the actual pointed filtered complete $\mathbb{R}_{\text{max}}$ -modules $H^0(C_p, D)$, $H^0(C_q, E)$, their reduced external products and their

principal isomorphisms. Put $\mathbf{V}_{p,q} := \text{Ind}(NV_{p,q}^0)$, where N is the categorical nerve. Day extension of tensor makes this a presentable symmetric monoidal category whose tensor preserves colimits separately and is therefore closed; the original section modules embed fully faithfully. Equivalences in $\mathbf{V}_{p,q}$ are equivalences in this presentable ∞ -category; no additional class of weak equivalences is chosen.

Define a $\mathbf{V}_{p,q}$ -enriched category $\text{Sec}_{p,q}$. Its objects are pairs (D, E) in the principal-equivalence orbits of $(D_p(a), D_q(b))$, for $(a, b) \in \mathbb{Z}^2$. When both degree differences are nonnegative, put

$$(39) \quad \underline{\text{Hom}}((D, E), (D', E')) := H^0(C_p, D' - D) \hat{\boxtimes}_{\text{red}} H^0(C_q, E' - E).$$

If either degree difference is negative, use the zero pointed module. Composition is actual pointwise tropical multiplication of sections. The divisor tensor law proves associativity, the constant section is the unit, and addition of divisors makes $\text{Sec}_{p,q}$ symmetric monoidal.

Let

$$(40) \quad \mathbf{A}_{p,q} := \text{PSh}_{\mathbf{V}_{p,q}}(\text{Sec}_{p,q}^{\text{op}}, \mathbf{V}_{p,q})$$

denote the $\mathbf{V}_{p,q}$ -enriched presheaf ∞ -category, with enriched Day convolution, and write $y_{p,q}$ for enriched Yoneda. Equivalences and colimits in $\mathbf{A}_{p,q}$ are computed objectwise. Presentability, the objectwise formulas and generation by representables are the enriched presheaf results of [1, Theorem 1.2 and Corollaries 1.4–1.5]. Principal multiplication gives canonical enriched equivalences between representatives. Put $\mathcal{L}_{p,q}(D, E) := y_{p,q}(D, E)$. Then

$$(41) \quad \mathcal{L}_{p,q}(x) \otimes_{\text{Day}} \mathcal{L}_{p,q}(y) \simeq \mathcal{L}_{p,q}(x + y),$$

$$(42) \quad \mathbf{R}\underline{\text{Hom}}_{\mathbf{A}_{p,q}}(y_{p,q}(0), \mathcal{L}_{p,q}(x)) \simeq \underline{\text{Hom}}_{\text{Sec}_{p,q}}(0, x).$$

Here $\mathbf{R}\underline{\text{Hom}}$ means the intrinsic enriched mapping object of the displayed ∞ -category; it is not a derived functor computed in an undeclared model. The second identity is the ∞ -categorical enriched Yoneda equivalence. Representables are compact projective because mapping from $y_{p,q}(x)$ is objectwise evaluation, which preserves colimits. Thus there is no replacement or higher correction hidden in (41).

Proposition 2.13 (Universal coefficient internalization). *The category $\mathbf{A}_{p,q}$ is the free cocomplete $\mathbf{V}_{p,q}$ -linear coefficient category generated by the actual periodic divisor-section category. More precisely, every enriched strong symmetric monoidal functor from $\text{Sec}_{p,q}$ to a cocomplete $\mathbf{V}_{p,q}$ -linear category extends, uniquely up to a contractible choice, to a cocontinuous lax symmetric monoidal functor from $\mathbf{A}_{p,q}$.*

Proof. Enriched left Kan extension along Yoneda gives the extension. Every presheaf is a weighted colimit of representables, so cocontinuity forces it uniquely. Day convolution is the left Kan extension of tensor on $\text{Sec}_{p,q}$, proving the monoidal assertion and its coherence. $\square$

To place this coefficient category over the periodic carrier without a type error, define

$$(43) \quad \mathcal{E}_{p,q} := \text{Sh}(C_p \times C_q; \text{Spc}) \hat{\boxtimes} \mathbf{A}_{p,q}, \quad \mathcal{O}_{p,q} := \mathbf{1} \boxtimes y_{p,q}(0), \quad \tilde{\mathcal{L}}_{p,q}(D, E) := \mathbf{1} \boxtimes \mathcal{L}_{p,q}(D, E).$$

The cocomplete tensor product exists by [18, Proposition 4.8.1.17]. The spatial factor is cartesian monoidal, so its unit $\mathbf{1}$ is terminal and $\mathbf{R}\underline{\text{Hom}}(\mathbf{1}, \mathbf{1})$ is contractible; all tropical tensor data remain in $\mathbf{A}_{p,q}$. Define total coefficient cohomology by

$$(44) \quad \mathbf{R}\Gamma_{\text{per}}(C_p \times C_q, M) := \mathbf{R}\underline{\text{Hom}}_{\mathcal{E}_{p,q}}(\mathcal{O}_{p,q}, M).$$

For representable coefficient lines, the terminal sheaf factor and (41) identify this object with the actual global periodic section module. This definition makes no unproved claim that a separate tropical sheaf cohomology has vanishing higher terms.

Write $\Pi_{p,q} : C_p \times C_q \rightarrow \mathcal{Y}_{\mathbb{S}}$ for the (p, q) component of (38) and set

$$\mathcal{E}_{\mathcal{Y};p,q} := \mathrm{Sh}(\mathcal{Y}_{\mathbb{S}}; \mathrm{Spc}) \hat{\boxtimes} \mathbf{A}_{p,q}.$$

The coefficient adjunction is the explicit tensor product

$$(45) \quad \tilde{\Pi}_{p,q}^* := \Pi_{p,q}^* \hat{\boxtimes} \mathrm{id}_{\mathbf{A}_{p,q}} : \mathcal{E}_{\mathcal{Y};p,q} \rightleftarrows \mathcal{E}_{p,q} : \mathbf{R}\tilde{\Pi}_{p,q,*} := \mathbf{R}\Pi_{p,q,*} \hat{\boxtimes} \mathrm{id}_{\mathbf{A}_{p,q}}.$$

The left arrow is strong symmetric monoidal: inverse image for a geometric morphism preserves finite products and all colimits, hence is strong for the cartesian spatial tensor, while the identity is strong for Day convolution. Its right adjoint exists by presentability and is therefore canonically lax symmetric monoidal.

Let $\mathcal{E}_{\mathrm{per}}^{\mathrm{fs}}$ and $\mathcal{E}_{\mathcal{Y};\mathrm{per}}^{\mathrm{fs}}$ denote the finite-support direct sums of the source and target categories in (45). Taking the componentwise adjunctions gives the promised diagram

$$(46) \quad \tilde{\Pi}_{\mathrm{per}}^* : \mathcal{E}_{\mathcal{Y};\mathrm{per}}^{\mathrm{fs}} \rightleftarrows \mathcal{E}_{\mathrm{per}}^{\mathrm{fs}} : \mathbf{R}\tilde{\Pi}_{\mathrm{per},*}, \quad \tilde{\Pi}_{\mathrm{per}}^* \dashv \mathbf{R}\tilde{\Pi}_{\mathrm{per},*}.$$

This adds coefficient geometry over the fixed spherical carrier and does not replace $\mathcal{O}_{\mathcal{Y}}$. Total sections on the target are derived enriched Hom from the componentwise unit $\mathbf{1}_{\mathcal{Y}} \boxtimes y_{p,q}(0)$. Adjunction identifies them with (44); hence every source, target and monoidal structure used below is explicit.

Let $\mathrm{Pic}_{\mathbb{N}}^+ \subset \mathrm{Pic}_{\mathrm{ext}}^{\mathrm{fact}}$ be the monoid of positive factorized classes whose two radii $Q_i = \exp(d_i)$ are positive integers. A class in this monoid has the canonical, representative-independent coordinates

$$(47) \quad a_{p,i} = v_p(Q_i), \quad d_i = \sum_p a_{p,i} \log p, \quad a_{p,i} \geq 0.$$

Indeed degree is principal-invariant and unique factorization recovers all $a_{p,i}$ from it. Thus a principal change of an Arakelov divisor cannot change the periodic components used below. This is the precise input sector of the periodic cohomology; the resulting quadratic determinant is then extended to the completed numerical degree cone by continuity.

For a finite-support effective ruling class in this monoid, written in its canonical coordinates as

$$x = \sum_p a_{p,1} e_{p,1} + \sum_q b_{q,2} e_{q,2},$$

form the finite coproduct of the internal representable lines and define

$$(48) \quad \mathcal{F}_{\mathrm{per}}(x) := \mathbf{R}\tilde{\Pi}_{\mathrm{per},*} \bigoplus_{p,q}^I \tilde{\mathcal{L}}_{p,q}(D_p(a_{p,1}), D_q(b_{q,2})).$$

The prime means that only the support of x is used. Its spatial factor is the unit sheaf, but its coefficient factor is not a precomputed numerical vector space: the representable line is first constructed from the actual periodic section category and only then pushed to the square.

Theorem 2.14 (Internal periodic sections and Künneth). *There are natural equivalences*

$$(49) \quad \mathbf{R}\Gamma(\mathcal{Y}_{\mathbb{S}}, \mathcal{F}_{\mathrm{per}}(x)) \simeq \bigoplus_{p,q}^I H^0(C_p, D_p(a_{p,1})) \hat{\boxtimes}_{\mathrm{red}} H^0(C_q, D_q(b_{q,2})).$$

They are principal-invariant, lax symmetric monoidal in x , and the componentwise equivalence is the internal Künneth map.

No stabilization is used in this formula: both sides are objects of the presentable coefficient ∞ -category. Stabilization begins only after Theorem 2.15: its finite-dimensional real cotangent frame V is sent to the perfect complex $V[0] \in \text{Perf}(\mathbb{R})$, and then to its integral and nuclear realizations. Thus the degree-zero perfect complex used later is defined explicitly and is not being inferred from an unproved acyclicity assertion about the raw spherical structure sheaf.

Proof. Adjunction for (46), with global sections defined by $\mathbf{R}\text{Hom}$ from the coefficient unit, identifies the left side with (44) on $\mathcal{E}_{\text{per}}^{\text{fs}}$. Its components are open and closed, so finite-support global sections split componentwise. Apply the second equivalence of (41). By (47), principal translation on the Arakelov class leaves the prime coordinates unchanged; principal translation inside a periodic component is an isomorphism of its divisor line. Day convolution gives multiplication before direct image, and right direct image is lax symmetric monoidal. These constructions prove all compatibility diagrams. $\square$

We next exhibit the intrinsic cotangent of the section object. For $N > p$, Connes–Consani’s finite-depth module $\mathcal{E}_{N,p}$ has exactly $d = N - p + 1$ ordered extremal rays $\phi_0, \dots, \phi_{d-1}$, these rays generate the whole module, and their canonical coefficient retraction is given by residuation

$$(50) \quad \gamma_i(f) = \inf_z (f(z) - \phi_i(z))$$

[5, Lemma 5.19, Appendix A, Proposition A.2 and Lemma A.3]. For the second prime use extremals $\psi_0, \dots, \psi_{e-1}$. The functionally reduced external section space is

$$(51) \quad \mathfrak{S}_{d,e} = \left\{ \max_{i,j} \{ \phi_i(u) + \psi_j(v) + c_{ij} \} : c_{ij} \in \mathbb{R}_{\max} \right\} \subset C(C_p \times C_q, \mathbb{R}_{\max})$$

with the uniform topology.

Call a section in (51) *regular* if every pair (i, j) is the unique maximizer at some point. This is intrinsic: it says that deleting the corresponding extremal ray changes the section on an open set. Denote the regular locus by $\mathfrak{S}_{d,e}^{\text{reg}}$.

Theorem 2.15 (Regular section-moduli cotangent). *The locus $\mathfrak{S}_{d,e}^{\text{reg}}$ is nonempty and open. It is a real manifold of dimension de , and its cotangent bundle has the canonical ordered frame*

$$(52) \quad \Omega_{d,e}^{\text{per}} \simeq \mathcal{O}_{\mathfrak{S}^{\text{reg}}} \otimes_{\mathbb{R}} \mathbb{R}[\text{Ext}(\mathcal{E}_{N,p}) \times \text{Ext}(\mathcal{E}_{M,q})]^\vee.$$

External product induces the cotangent Künneth isomorphism. Principal translations act on (52) by the identity.

Proof. Lemma 5.19(iii) of [5] supplies offsets and points where each one-dimensional extremal uniquely dominates. Their Cartesian products give a coefficient vector in (51) and a unique-dominance witness for every pair, proving nonemptiness.

For a regular coefficient vector, choose one witness for every pair. The minimum of the finitely many strict gaps is positive. Perturbing every coefficient by less than one quarter of that gap preserves all witnesses, and evaluation there recovers each c_{ij} continuously. Conversely the maximum map is 1-Lipschitz in the coefficient sup norm.

The coefficients are intrinsic. Applied to the pair extremal $B_{ij} = \phi_i \boxplus \psi_j$, the residuation formula is

$$c_{ij} = \inf_{(u,v)} (f(u,v) - B_{ij}(u,v)).$$

The inequality $f \geq B_{ij} + c_{ij}$ gives one direction, and equality at an activity point gives the other. The infimum is 1-Lipschitz in f ; hence sufficiently small uniform perturbations preserve the positive gaps and regularity. The coefficient maps are therefore compatible manifold charts

and their differentials give (52). The extremals of an external tensor are exactly the Cartesian pairs: unique-dominance witnesses prove that every pair is extremal, while the pair functions generate the tensor image. This proves Künneth. A principal translation adds a fixed function and leaves every coefficient differential unchanged. $\square$

For $a \in \mathbb{Z}_{>0}$ let $d_{p,r}(a)$ be the number of extremal rays of $H^0(C_p, D_p(a))^{p^r}$. Frobenius and the exact special-module theorem give, for $ap^r > p$,

$$(53) \quad d_{p,r}(a) = ap^r - p + 1, \quad p^{-r}d_{p,r}(a) \longrightarrow a$$

[5, Lemmas 5.18–5.19]. This is an exact model of the actual filtered section module because $D_p(a)$ is the integral divisor at the neutral point. Explicitly, Lemma 5.18 identifies the r -fold Frobenius pullback with the divisor of degree ap^r , and Lemma 5.19 identifies its special module with $\mathcal{E}_{ap^r,p}$, whose ordered extremal count is $ap^r - p + 1$. Thus no asymptotic estimate is substituted for the displayed finite-rank equality.

The logarithm identifies the quotient map $\mathbb{R}_+^\times \rightarrow C_p$ with $\mathbb{R} \rightarrow \mathbb{R}/(\log p)\mathbb{Z}$. Consequently there is a unique translation-invariant measure μ_p whose pullback normalization on the universal cover is Lebesgue measure, and

$$(54) \quad \mu_p(C_p) = \log p.$$

This normalization is forced by the fixed logarithmic coordinate: replacing μ_p by a scalar multiple would no longer pull back to Lebesgue measure. In particular, the factors below are geometric fibre lengths, not adjustable weights. Define the arithmetic continuous dimension by weighting the componentwise continuous dimensions with the product of the two canonical fibre lengths. Then, for effective x ,

$$(55) \quad \text{cdim}_{\text{ar}} \mathbf{R}\Gamma(\mathcal{Y}_{\mathbb{S}}, \mathcal{F}_{\text{per}}(x)) = \sum_{p,q} (\log p)(\log q) a_{p,1} b_{q,2} = d_1(x) d_2(x).$$

This is the dimension of internal global sections of (48), not the dimension of the raw Day smash.

The claimed numerical extension is unique in the asymptotic degree cone, not by a false density assertion about $\log \mathbb{N}$ itself. For every $a > 0$ put $Q_t = \lfloor e^{ta} \rfloor$. Then $t^{-1} \log Q_t \rightarrow a$. Applying this independently in the two rulings shows that the normalized values on integer-radius classes satisfy

$$t^{-2}(\log Q_t)(\log R_t) \longrightarrow ab.$$

Hence every point of $\mathbb{R}_{\geq 0}^2$ is an asymptotic ray limit, and continuity together with bidegree-one homogeneity forces the unique extension $F(a, b) = ab$.

2.10. The intrinsic periodic–Deligne–nuclear category. Let Perf_{per} be the stable idempotent completion of the split exact category generated by the finite-depth cotangent frames (52), principal translations and Künneth maps. Its tensor product is induced by external product; Cartesian products of extremal bases make it exact and symmetric monoidal. Extend scalars by

$$(56) \quad \text{Perf}_{\text{per}, \mathcal{C}} := \text{Perf}_{\text{per}} \otimes_{\text{Perf}(\mathbb{R})} \text{Perf}(\mathcal{O}_{\mathcal{C}}).$$

The ordered-frame realization is the exact strong symmetric monoidal functor

$$(57) \quad T_{\mathcal{C}} : \text{Perf}_{\text{per}, \mathcal{C}} \longrightarrow \text{Perf}(\mathcal{O}_{\mathcal{C}}), \quad P \longmapsto \mathcal{O}_{\mathcal{C}} \otimes_{\mathbb{R}} V_{\text{Ext}}(P).$$

Coefficient-only objects are obtained by tensoring the periodic unit with an arbitrary perfect $\mathcal{O}_{\mathcal{C}}$ -module.

Define

$$(58) \quad \text{Perf}_{IDN}(\mathcal{Y}_{\mathbb{S}}) := \text{Perf}_{DN}(\mathcal{Y}_{\mathbb{S}}) \times_{\text{Perf}(\mathcal{O}_{\mathcal{C}})}^h \text{Perf}_{\text{per}, \mathcal{C}},$$

using nuclear-component evaluation on the first factor and (57) on the second.

Proposition 2.16 (Categorical gluing). *The category (58) is stable and symmetric monoidal. An object is a Deligne–nuclear perfect object, a periodic perfect object, and an equivalence identifying the nuclear component with its periodic frame realization. The coefficient-only construction gives compatible lifts of the perfect divisor, determinant and contact objects used in rows (b) and (c); the reduced contact has zero periodic component. The completed nuclear correspondence module is not asserted to be an object of this perfect pullback: it acts through the common $\mathcal{O}_C$ coefficient component.*

Proof. Small stable idempotent-complete categories admit their tensor product and stable completion. Exact strong symmetric monoidal functors preserve finite cofibres and tensor products. Homotopy limits of stable categories are stable; the componentwise tensor and duals preserve the comparison equivalence. This proves the first assertion. The concrete description is the definition of a homotopy pullback. For a Deligne–nuclear perfect object with nuclear component M_C , use the periodic unit tensored with M_C ; its image under T_C is M_C . For a contact with zero nuclear component use the zero periodic object. These constructions apply to the named perfect objects. An arbitrary complete nuclear module need not be dualizable; nevertheless its scalar action on the common nuclear component is defined without promoting it to a perfect object. $\square$

At depth (r, s) , let $V_{p,q;r,s}^{\text{per}}(x)$ be the real vector space on the ordered extremal-pair frame and let $V_{p,q;r,s,\mathbb{Z}}^{\text{per}}(x)$ be its free integral lattice. Together with its nuclear scalar extension and the intrinsic cotangent object, these define

$$(59) \quad \mathbf{H}_{p,q;r,s}^{\text{int}}(x) \in \text{Perf}_{IDN}(\mathcal{B}_{\mathbb{S}}).$$

Their finite-support direct sums give $\mathbf{H}_{r,s}^{\text{int}}(x)$. Theorem 2.15 proves Künneth and principal descent for these objects.

Proposition 2.17 (Arithmetic scalar compatibility). *The intrinsic perfect systems carry endomorphisms ρ_n^{int} with $\rho_m^{\text{int}} \rho_n^{\text{int}} = \rho_{mn}^{\text{int}}$. For $n > 1$ the integral and real components are zero, while the nuclear and periodic scalar components are left convolution by δ_n ; for $n = 1$ every component is the identity. These endomorphisms commute with principal descent, finite-support sums and Künneth.*

Proof. The augmentation (28) sends δ_n to zero for $n > 1$ and δ_1 to one. Hence the integral map just specified and left convolution on the nuclear component have identical realifications, so they define a morphism in Perf_{DN} . On the periodic factor use the coefficient-only scalar map $\text{id}_P \otimes L_{\delta_n}$; its image under T_C is the same nuclear convolution map, so the homotopy-pullback comparison square commutes. Finally $L_{\delta_m} L_{\delta_n} = L_{\delta_m * \delta_n} = L_{\delta_{mn}}$. All structural maps named in the statement are $\mathcal{O}_C$ -linear and based, proving the compatibilities. $\square$

For the comparison with the code, let $\text{Sys}_{p,q}$ be the stable category of bi-indexed sequences of based finite perfect real objects, with pointwise exact triangles. Let $\text{Sys}_{p,q}^{<2}$ be the thick subcategory of systems whose total rank is $o(p^r q^s)$. Shifts, retracts and finite cofibres preserve this estimate, so it is thick. We write

$$(60) \quad \text{Sys}_{p,q}^{(2)} := \text{Sys}_{p,q} / \text{Sys}_{p,q}^{<2}$$

for the resulting Verdier quotient. Thus “modulo subquadratic perfect systems” below refers to the explicitly defined category (60).

Give the ordered extremal determinant generator on the (p, q) component the norm

$$(61) \quad \|\omega_{p,q;r,s}\| := \exp(-(\log p)(\log q)p^{-r}q^{-s} \dim_{\mathbb{R}} V_{p,q;r,s}^{\text{per}}(x)).$$

Equations (53) and (55) prove convergence of the finite-support determinant product to

$$(62) \quad \lambda_{\text{int}}(x) = (\mathbb{R}\mathbf{1}, \|\mathbf{1}\| = e^{-d_1(x)d_2(x)}).$$

For the system-level comparison define the decomposed spherical certificate

$$(63) \quad \lambda_{\text{code}}^{\text{dec}}(x) := \bigotimes_{p,q}^I \lambda_{\text{code}}(a_{p,1} \log p, b_{q,2} \log q).$$

The bilinear formula (24) and distributivity give the canonical based isometry

$$(64) \quad \lambda_{\text{code}}^{\text{dec}}(x) \simeq \lambda_{\text{code}}(x);$$

both distinguished generators have norm $\exp(-\sum_{p,q} a_{p,1} b_{q,2} \log p \log q)$.

Theorem 2.18 (Intrinsic determinant and spherical certificate). *The line (62) is the continuous determinant of the cotangent of the regular moduli of the intrinsic sections (49). There is a canonical based isometry*

$$(65) \quad \lambda_{\text{int}}(x) \xrightarrow{\sim} \lambda_{\text{code}}(x).$$

For each fixed divisor and prime pair, the periodic cotangent system and the corresponding component of the decomposed code certificate are isomorphic in $\text{Sys}_{p,q}^{(2)}$.

Proof. The first assertion is Theorem 2.15 followed by (61). On a (p, q) component write $a = a_{p,1}$ and $b = b_{q,2}$. For the comparison choose the cofinal integer code scales

$$t_{p,r} = \left\lfloor \frac{p^r \log 2}{\log p} \right\rfloor, \quad t_{q,s} = \left\lfloor \frac{q^s \log 2}{\log q} \right\rfloor.$$

By Lemma 2.5, the one-dimensional code ranks satisfy

$$\frac{r(\lfloor e^{t_{p,r} a \log p} \rfloor)}{p^r} \longrightarrow a,$$

and likewise for (q, s, b) . Their difference from the periodic ranks in (53) is therefore $o(p^r)$ and $o(q^s)$. The difference of the mixed ranks is $o(p^r q^s)$.

Order both pair bases lexicographically and include each into the based free object of the larger rank. The two cokernels have subquadratic total rank, so the resulting zigzag is an isomorphism in the stated stable quotient. Moreover

$$\frac{(\log 2)^2}{t_{p,r} t_{q,s}} / ((\log p)(\log q) p^{-r} q^{-s}) \longrightarrow 1.$$

The complementary determinant exponent tends to zero, and both determinant norms tend to $e^{-ab \log p \log q}$. Sending the distinguished limiting generator to the distinguished limiting generator proves the componentwise isometry; tensor over the finite support of x and then compose with (64) to obtain (65). $\square$

Remark 2.19 (The strength of the comparison). Theorem 2.18 should be read as an *asymptotic based-rank comparison in the explicitly defined Verdier quotient* (60), together with a genuine convergence of determinant norms. It is not a geometric map between the negabinary sections and the periodic sections. The mechanism of the system-level half is that two based systems whose ranks agree to $o(p^r q^s)$ become isomorphic in a quotient which was defined by killing exactly that discrepancy; the substance is therefore the rank asymptotics supplied by Lemma 2.5 and (53), not the passage to the quotient.

The determinant half is stronger and is what is used downstream: both norms converge to $e^{-ab \log p \log q}$, and the isometry (65) matches distinguished limiting generators. Note also that the weights in (61) and the code normalization are fixed in advance so as to produce this common limit; the comparison certifies that two independently assembled systems reach the same leading term under those normalizations, and does not derive the normalization from the geometry.

Define

$$(66) \quad F_{\text{int}}(x) = d_1(x)d_2(x), \quad B_{\text{int}}(x, y) = d_1(x)d_2(y) + d_2(x)d_1(y).$$

The isometry (65) identifies this form with the one computed by the auxiliary code, but its source is now intrinsic section cohomology.

2.11. Derived contact and the intrinsic Green factor. We now derive the finite contact from the two prime rulings. The canonical section of $\mathcal{O}_X([p])$ pulls back to maps

$$(67) \quad s_{p,1} : \mathcal{L}(-[p], 0) \longrightarrow \mathcal{O}_{\mathcal{Y}}, \quad s_{p,2} : \mathcal{L}(0, -[p]) \longrightarrow \mathcal{O}_{\mathcal{Y}}.$$

After extension to $\mathcal{O}_{\mathbb{Z}}$, define the two derived zero-locus objects by their cofibres $\mathcal{O}_{V_{p,i}} := \text{cofib}(s_{p,i})$ and their derived incidence by

$$(68) \quad \mathcal{I}_p^{\mathbf{L}} := \mathcal{O}_{V_{p,1}} \otimes_{\mathcal{O}_{\mathbb{Z}}}^{\mathbf{L}} \mathcal{O}_{V_{p,2}}.$$

Let $\Delta : X \rightarrow \mathcal{Y}_{\mathbb{S}}$ be the diagonal. Since $\Delta^* \mathcal{L}(-[p], 0) \simeq \mathcal{O}_X(-[p])$ and likewise in the second factor, both pulled-back sections are multiplication by p in an integral local trivialization.

Proposition 2.20 (Derived prime contact). *The diagonal pullback of (68) has the canonical Koszul model*

$$(69) \quad \mathbf{L}\Delta^* \mathcal{I}_p^{\mathbf{L}} \simeq (\mathcal{O}_X/p) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (\mathcal{O}_X/p).$$

Its degree-zero reduced contact is $\mathcal{O}_X/p$ and its integral coefficient is $\mathbb{F}_p$. The latter has the canonical perfect $\mathbb{Z}$ -resolution $[\mathbb{Z} \xrightarrow{p} \mathbb{Z}]$. The degree -1 term of the full contact is another copy of $\mathbb{F}_p$; it is retained in (69) and is removed only by the explicitly named reduced truncation.

Proof. Derived pullback is symmetric monoidal and takes the two maps (67) to the same Cartier map p . Taking their cofibres proves (69). Resolve either quotient by $[\mathcal{O}_X \xrightarrow{p} \mathcal{O}_X]$. Tensoring with $\mathcal{O}_X/p$ makes the differential zero, so the derived tensor has $H^0 \simeq \mathcal{O}_X/p$ and $H^{-1} \simeq \mathcal{O}_X/p$. Finally $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{F}_p \rightarrow 0$ is the canonical free resolution of the reduced coefficient. $\square$

Extending this reduced coefficient resolution to $\mathcal{O}_{\mathbb{Z}}$ gives the perfect two-term complex

$$(70) \quad C_{p,\mathbb{Z}} = [\mathcal{O}_{\mathbb{Z}} \xrightarrow{p} \mathcal{O}_{\mathbb{Z}}].$$

After real base change multiplication by p is invertible, so this complex is canonically contractible. Hence

$$(71) \quad C_p = (C_{p,\mathbb{Z}}, 0, C_{p,\mathbb{Z}} \otimes \mathcal{O}_{\mathbb{R}} \simeq 0)$$

is a perfect object of (30). With the standard torsion metric, the determinant generator of $[\mathbb{Z} \xrightarrow{p} \mathbb{Z}]$ has norm p^{-1} : indeed the image lattice has index p in the target lattice. Therefore

$$(72) \quad -\log \|\det_{\text{tor}} C_p\| = \log p.$$

Write $\kappa_p := \det_{\text{tor}} C_p$ for this based normed determinant line. This distinction matters: C_p is a perfect torsion complex, whereas κ_p is an invertible object of the determinant Picard groupoid. For a label n , use C_p when $n = p^k$ and the zero object when n has at least two prime divisors.

This realizes $\Lambda(n)$ objectwise; it does not make the false assertion that $\mathbb{Z}/p$ is idempotent under derived tensor.

For finite ruling divisors x, y as in (23), tensor the appropriate powers of the determinant lines κ_p . No derived tensor power of $\mathbb{F}_p$ is taken. The resulting contact line $\lambda_C(x, y)$ has metric exponent

$$(73) \quad C_\Lambda(x, y) = \sum_p (x_{p,1}y_{p,2} + x_{p,2}y_{p,1}) \log p.$$

If $\delta\lambda_{\text{int}}(x, y)$ denotes $\lambda_{\text{int}}(x + y) \otimes \lambda_{\text{int}}(x)^{-1} \otimes \lambda_{\text{int}}(y)^{-1}$, define

$$(74) \quad \lambda_G(x, y) := \delta\lambda_{\text{int}}(x, y) \otimes \lambda_C(x, y)^{-1}.$$

Corollary 2.21 (Canonical contact–Green factorization). *There is a canonical isometry*

$$(75) \quad \delta\lambda_{\text{int}} \simeq \lambda_C \otimes \lambda_G,$$

whose logarithmic metric identity is

$$(76) \quad B_{\text{int}} = C_\Lambda + G.$$

All symmetry, associativity and biextension interchange diagrams commute.

Proof. Equation (75) is obtained by tensoring (74) with λ_C . Taking negative logarithms of the distinguished-generator norms and using (66) and (73) gives (76). Every structural map is built from the associator, symmetry and inverse in the determinant Picard groupoid. Mac Lane coherence, or direct cancellation of the based lines, proves all stated diagrams. $\square$

The logical direction is part of the statement: λ_G is the unique based normed line defined by the quotient (74). Thus the corollary is a canonical factorization of the intrinsic section determinant, not an independent construction of a Green current by a second geometric process. To keep this visible downstream we call λ_G the *formal complementary determinant line*: the isometry (75) holds by construction, and the content of the corollary is the coherence of the structural diagrams together with the metric identity (76), not the existence of λ_G . Nothing below may use λ_G as if it had been produced by an archimedean height or a Green’s-function computation.

On the real ruling quotient

$$(77) \quad N_{\text{rul}}^1 := \left(\bigoplus_p \mathbb{R}e_{p,1} \oplus \bigoplus_p \mathbb{R}e_{p,2} \right) / \ker(d_1, d_2) \simeq \mathbb{R}^2,$$

the matrix of B_{int} is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and has signature $(1, 1)$. This is the numerical ruling form. It is not claimed to contain the completed mixed correspondence space used in row (d).

2.12. Exact correspondences and the correct finiteness theorem. Extend the nuclear coefficient sheaf in (29) and put

$$(78) \quad \mathcal{N}_{DN} := \mathcal{O}_C e_1 \oplus \mathcal{O}_C e_2 \oplus \mathcal{O}_C e_\Gamma.$$

It is locally free of rank three over $\mathcal{O}_C$. The first two generators are the rulings. The third is an explicitly adjoined free coefficient direction for the correspondence algebra; it is not asserted to be the class of a mixed Cartier divisor on the spherical square. On this third summand define

$$(79) \quad \rho_n(ae_\Gamma) = (\delta_n * a)e_\Gamma.$$

The estimate $q_k(\delta_n * a) \leq n^k q_k(a)$ proves continuity, and (26) gives $\rho_m \rho_n = \rho_{mn}$. The diagonal contact is the continuous functional ℓ , so

$$(80) \quad \ell(\delta_n) = \Lambda(n), \quad \ell(\delta_m * \delta_n) = \Lambda(mn),$$

including even prime powers; no reduction of the arithmetic label modulo two is involved.

The cotangent objects themselves admit the compatible enrichment

$$\mathcal{O}_C \otimes_{\mathbb{R}} V_{m,n}.$$

They are finite free of rank $r(m)r(n)$, and multiplication by δ_k commutes with restrictions, zero-extensions and the Cartesian-basis Künneth map. The augmentation (28) recovers the real digit determinant, while ℓ retains the independent contact mass.

Theorem 2.22 (Finite-rank obstruction). *There is no finite-dimensional real vector space E , vectors $\gamma_n \in E$ for every $n \geq 1$, and bilinear form K satisfying*

$$(81) \quad K(\gamma_m, \gamma_n) = \Lambda(mn)$$

for all m, n . Consequently no finitely generated abelian numerical group equipped with classes γ_n and a bilinear form can realize the full two-variable kernel $\Lambda(mn)$ in this way.

Proof. Choose distinct primes $p_1, \dots, p_r$. The matrix of K on the vectors γ_{p_i} is

$$(\Lambda(p_i p_j))_{i,j} = \text{diag}(\log p_1, \dots, \log p_r),$$

because a product of two distinct primes is not a prime power. It has rank r , hence $\dim E \geq r$. Since r is arbitrary, E cannot be finite dimensional. Tensoring a finitely generated abelian group with $\mathbb{R}$ would produce such an E , proving the last assertion. $\square$

Theorem 2.22 rules out finite rank under the displayed full-kernel hypothesis; it does not classify every conceivable finite numerical theory. Formula (78) supplies one explicit replacement: three module generators suffice after the infinitely many labels are moved into the scalar nuclear algebra. No universality or uniqueness of this extension of scalars is asserted.

2.13. Completion theorem and audit.

Theorem 2.23 (Constructed intrinsic periodic–Deligne–nuclear square). *The tuple*

$$(82) \quad \mathcal{A}_{DN} = (\mathcal{Y}_{\mathbb{S}}, \mathcal{O}_{\mathcal{Y}}, \text{Perf}_{IDN}, \mathcal{L}, \mathcal{F}_{\text{per}}, \mathbf{H}^{\text{int}}, \lambda_{\text{int}}, \lambda_{\text{code}}, \lambda_C, \lambda_G, \mathcal{N}_{DN})$$

is an intrinsic periodic–Deligne–nuclear arithmetic square in the sense of Definition 2.1. No carrier, descent, section-functor, regular-cotangent, Künneth, continuous-determinant, reduced-contact, Green-factorization, composition or stated module-finiteness assertion in that definition is conditional.

Proof. The literal carrier and noncollapse are (3) and Proposition 2.2. Proposition 2.3 gives invertible divisor objects and principal descent, and Proposition 2.4 computes their effective cone. Theorem 2.14 constructs the internal derived section functor and its Künneth equivalence. Theorem 2.15 constructs the cotangent of its regular section moduli, while (55) proves the coefficient-one quadratic dimension. Proposition 2.16 places these objects, their integral and nuclear realizations and the contact in one stable symmetric monoidal perfect category, and Proposition 2.17 constructs their compatible arithmetic scalar action. Theorem 2.18 constructs the continuous determinant and its independent spherical code certificate. Proposition 2.20, equations (71)–(72), and Corollary 2.21 give the contact determinant and intrinsic Green factorization. Proposition 2.11 and (78)–(79) give exact composition, contact and local freeness of rank three. These verify the five clauses of Definition 2.1. $\square$

For clarity, Theorem 2.23 neither asserts nor needs that all raw bounded sections have quadratic dimension: Theorem 2.6 proves the contrary. It also neither asserts nor needs finite abelian numerical rank: Theorem 2.22 proves that finite rank cannot realize the full bilinear kernel $\Lambda(mn)$ required by the particular contact pairing used in rows (b) and (c). The negabinary envelope remains auxiliary. The intrinsic cohomology is the cotangent of the regular moduli of actual periodic coefficient sections on the square. No assertion equating it with all raw bounded functions is made or needed.

3. THE CORNER PAIRING

3.1. Correspondence potentials. Let Ψ_λ denote the Frobenius correspondence of slope $\lambda > 0$ on the square of the Scaling Site [5, 6], whose tropical hypersurface on the real positive chart is $\{y = \lambda x\}$, with convex potential $\max(y - \lambda x, 0)$. For a test function f one forms

$$(83) \quad D_f = \int_0^\infty f(\lambda) \Psi_\lambda \frac{d\lambda}{\lambda}, \quad U_f(x, y) = \int f(\lambda) \max(y - \lambda x, 0) \frac{d\lambda}{\lambda}.$$

Writing $r = y/x$, $U_f(x, y) = x u_f(r)$, and

$$(84) \quad u_f''(r) = \frac{f(r)}{r},$$

which determines U_f uniquely modulo affine functions of (x, y) . We take (84), not the superposition integral, as the defining relation; the integral is one construction method, valid on compactly supported f .

3.2. The interior carries nothing.

Theorem 3.1. *Let $U(x, y) = x u(r)$ and $V(x, y) = x v(r)$ be homogeneous potentials. Then*

$$\nabla^2 U = \frac{u''(r)}{x} \mathbf{v}_r \mathbf{v}_r^\top, \quad \mathbf{v}_r = (r, -1)^\top,$$

so every such Hessian has rank one, and the mixed density

$$U_{xx}V_{yy} + U_{yy}V_{xx} - 2U_{xy}V_{xy} = \frac{u''v''}{x^2} [r^2 + r^2 - 2r^2] = 0$$

vanishes identically.

Consequently no intersection pairing on the smooth interior of the square can be nonzero, and every nonvanishing contribution must be supported by data lost on the punctured chart: the common corner of the Frobenius rays, the periodic adelic quotient and its isotropy, and the diagonal renormalization. This is consistent with, and explains, the divergent $\log \Lambda$ term recorded in [6] for the naive diagonal self-intersection.

3.3. The renormalized corner trace.

For a test function g put

$$(85) \quad g^\vee(x) = x^{-1} \overline{g(x^{-1})}.$$

With the Mellin convention used below this is the Tate involution: $\widehat{g^\vee}(s) = \overline{\widehat{g}(1 - \bar{s})}$.

Let S be a finite set of places containing ∞ , let $X_S = \mathbb{Z}_S^\times \backslash \mathbb{A}_S$ and $\mathcal{H}_S = L^2(X_S)$, and let θ be the scaling representation of the semilocal idele class group. With P_Λ the position cutoff, $\widehat{P}_\Lambda$ its Fourier conjugate and $R_\Lambda = \widehat{P}_\Lambda P_\Lambda$, set

$$(86) \quad \mathfrak{T}_S(h) = \lim_{\Lambda \rightarrow \infty} \left(\text{Tr}_{\mathcal{H}_S} (\theta(h) R_\Lambda) - 2h(1) \log \Lambda \right).$$

The semilocal trace theorem [11, 8] gives

$$(87) \quad \mathfrak{T}_S(h) = \sum_{v \in S} \int_{\mathbb{Q}_v^\times}' \frac{h(u^{-1})}{|1-u|_v} d^\times u,$$

the local factor arising as a transverse kernel trace, $\int_{\mathbb{Q}_v} \delta((u-1)x) dx = |1-u|_v^{-1}$. For h compactly supported in $[e^{-T}, e^T]$ only primes with $p^k \leq e^T$ contribute, so the sum is finite and $\mathfrak{T}_S$ stabilizes in S . Setting $I_\partial(D_f, D_g) := \mathfrak{T}(f \star g^\vee)$ one obtains, by Tate's normalization and the global explicit formula, the value $I_\partial(D_f, D_g) = N(f \star g^\vee)$.

We stress the logical order. (86) is defined from a representation and two geometric cutoffs, before (87) and before any mention of the explicit formula; the identification with N is a theorem, not a definition.

3.4. The numerical quotient. I_∂ is Hermitian, and for any bilinear form the induced form on the quotient by its radical is well defined and nondegenerate. Hence

$$(88) \quad \bar{I}_\partial \text{ on } V := \{\text{correspondence divisors}\} / \text{rad } I_\partial$$

exists unconditionally. This is the analogue of numerical equivalence, and is available with no further hypothesis — in particular with no principal invariance.

3.5. The signature, and why it is not enough. Write $\rho' = 1 - \bar{\rho}$.

Lemma 3.2. $\rho \mapsto \rho'$ is an involution of the zero multiset of ξ preserving multiplicities, and $\rho' = \rho$ if and only if $\text{Re } \rho = \frac{1}{2}$.

So the on-line zeros are exactly the fixed points of the mirror involution and the off-line zeros exactly its two-cycles. With $\widehat{g}^\vee(s) = \overline{\widehat{g}(1 - \bar{s})}$ the explicit formula gives

$$(89) \quad I_\partial(D_f, D_g) = \widehat{f}(0)\overline{\widehat{g}(1)} + \widehat{f}(1)\overline{\widehat{g}(0)} - \sum_{\rho} m_{\rho} \widehat{f}(\rho)\overline{\widehat{g}(\rho')},$$

which is the classical Weil scalar product [17]. The form splits orthogonally: the two polar coordinates carry $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, of signature $(1, 1)$; each on-line zero contributes a negative line; each off-line mirror pair contributes a hyperbolic plane. Hence

$$(90) \quad n_+(\bar{I}_\partial) = 1 + \#P, \quad \#P = \frac{1}{2} \# \{\text{off-line zeros}\},$$

so $\text{RH} \iff n_+ = 1$, the classical Hodge-index shape.

Remark 3.3. This is an equivalence derived from the explicit formula and therefore does no work; it specifies the target statement without supplying it. The inertia count itself is due to Connes and van Suijlekom [13]. What is new here is only the packaging: the form is an intersection pairing on a numerical quotient of a divisor group, its interior density vanishes identically (Theorem 3.1), and its polar block is exactly the intersection matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ of the two rulings of a product surface — which agrees with the bidegree coefficient $\deg^+(D)\deg^+(E)$ obtained independently from a tropical tensor construction on a periodic product.

4. MELLIN DESCENT AND ITS PAIRING

4.1. The descent forces Mellin monomials. The quotient by the Frobenius action makes strict invariance unavailable.

Proposition 4.1. *There is no nonzero $f \in C_c((0, \infty))$ with $U_{n,f} - U_f$ affine for every $n \in \mathbb{N}^\times$.*

Indeed the condition reduces, by (84), to $f(r/n) = f(r)$; in exponential coordinates this is invariance under translation by $\log n$, and $\{\log n\}$ generates a dense subgroup of $\mathbb{R}$, so a continuous compactly supported invariant is constant, hence zero. The incompatible hypothesis is compact support.

Relaxing invariance to a character twist is the correct move, and it is rigid.

Theorem 4.2. *Let $f \in C((0, \infty))$ be nonzero and suppose $f(r/n) = \chi(n)f(r)$ for all $n \in \mathbb{N}^\times$. Then χ is totally multiplicative, extends without obstruction to $\mathbb{Q}_+^\times$, and there is a unique $s \in \mathbb{R}$ with $\chi(q) = q^{-s}$; moreover*

$$f(r) = cr^s, \quad c \in \mathbb{R}^\times.$$

The proof runs through exponential coordinates: continuity of f at a point where it does not vanish forces $\chi \circ \exp$ continuous at 0, hence continuous on the dense group $\Gamma = \log \mathbb{Q}_+^\times$, hence uniquely extendable to a continuous homomorphism $(\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \times)$, which is $a \mapsto e^{-sa}$. Substituting back, $F(t-a) = e^{-sa}F(t)$ for all real a, t , and setting $a = t$ gives $F(t) = F(0)e^{st}$.

Remark 4.3. There is *no* residual log-periodic factor. A genuine period would require the invariance group to be discrete of rank one; here it is dense, which forces the periodic factor to be constant. The descent therefore expels the sections from compact support into a one-parameter graded family of Mellin monomials, and the grade is the spectral variable.

4.2. The graded family is dual to the test class. Write $f_s(x) = x^{-s}$ for the graded family, so that $f_s(u^{-1}) = |u|^s$ under the module embedding.

Proposition 4.4. *The Mellin integral $\int_0^\infty x^{-s}x^w \frac{dx}{x} = \int_{\mathbb{R}} e^{t(w-s)} dt$ converges for no $w \in \mathbb{C}$. In particular f_s lies in no space A_δ of test functions whose Mellin transform is analytic on a strip about the critical line.*

Under the cutoff $[\frac{1}{T}, T]$ used in the covariance form of the explicit formula [2, 17], however,

$$(91) \quad \int_{1/T}^T x^{-s}x^w \frac{dx}{x} = \frac{2 \sinh((w-s) \log T)}{w-s},$$

which on the vertical line $w = s + iu$ is the Dirichlet kernel $2 \sin(u \log T)/u$, of total mass 2π for every T , converging weakly to $2\pi\delta$.

Corollary 4.5. $\mathcal{G} := \{f_s\}$ is not a subspace of the test class; each f_s defines instead a measure supported on the vertical line $\operatorname{Re} w = s$. The graded family is Mellin-dual to the test class, and the cutoff of the covariance form is the pairing between the two sides.

This is the structural reason the naive formalization of the preceding descent — namely substituting f_s into (87) — is a category error, and it governs everything in §4.4–§4.6.

4.3. Local terms, in closed form, on the critical strip. Monomials diagonalise convolution with the Tate involution (85):

$$(92) \quad f_s \star g^\vee = c_g(s) f_s, \quad c_g(s) = \int_0^\infty t^{1-s} \overline{g(t)} d^\times t,$$

entire in s for compactly supported g . So the problem reduces to a single monomial.

Theorem 4.6. *Decomposing $\mathbb{Q}_p^\times$ into shells $|u|_p = p^{-n}$ of $d^\times u$ -measure 1,*

$$W_p(f_s) = \frac{p^{-s}}{1-p^{-s}} + \frac{p^{s-1}}{1-p^{s-1}} + C_p, \quad C_p = \int'_{\mathbb{Z}_p^\times} \frac{d^\times u}{|1-u|_p}$$

with C_p independent of s ; the two series converge if and only if $\operatorname{Re} s > 0$ and $\operatorname{Re} s < 1$ respectively. At the archimedean place

$$W_\infty(f_s) = \int_{\mathbb{R}^\times}' \frac{|u|^{s-1}}{|1-u|} du = \frac{\pi}{\sin \pi s} + \pi \cot \pi s = \pi \cot \frac{\pi s}{2},$$

regular on $0 < \operatorname{Re} s < 1$, with value π at $s = \frac{1}{2}$, and convergent on exactly the same strip: the integrand is $|u|^{s-1}$ near 0 and $|u|^{s-2}$ near ∞ .

Corollary 4.7. *Every local term, finite and archimedean alike, converges precisely on the critical strip $0 < \operatorname{Re} s < 1$, with both boundaries excluded.*

We record, without overstating it, that the strip here arises from purely local analysis of a monomial, with no input from the zeros or the functional equation. The symmetry $s \leftrightarrow 1-s$ interchanging the two series is built into the kernel $|u|^s/|1-u|$; this is a consistency check that the construction sits in the right place, not a discovery.

4.4. The s -dependent assembly, and the divergent constant. Write $A(s) = \sum_p p^{-s}/(1-p^{-s})$ and $B(s) = A(1-s)$. Each is a Dirichlet series over prime powers, convergent respectively on $\operatorname{Re} s > 1$ and $\operatorname{Re} s < 0$ — half-planes disjoint from the strip and exchanged by $s \mapsto 1-s$. Their analytic continuations are governed by $A(s) = \sum_{N \geq 1} \frac{\varphi(N)}{N} \log \zeta(Ns)$, whose singularities come from the pole of ζ at 1.

Theorem 4.8. *$A+B$ is locally integrable on the open strip, hence defines a distribution of order zero there.*

The singular set $\{1/N\} \cup \{1-1/M\}$ is finite on every compact subinterval and accumulates only at the endpoints; each singularity is logarithmic, and $\log$ is locally integrable; the tail converges uniformly, since $\log \zeta(\sigma) \cdot 2^\sigma \rightarrow 1$ geometrically. Pointwise the two halves reinforce at $s = \frac{1}{2}$, which is the unique fixed point of $s \mapsto 1-s$ and where $B = A$ identically, so no cancellation is possible there; but pointwise divergence and local integrability are compatible, and it is the latter that Corollary 4.5 requires.

The constant is a different matter.

Theorem 4.9. *Each shell $|1-u|_p = p^{-k}$, $k \geq 1$, has measure p^{-k} and integrand p^k , hence contributes exactly 1; so $C_p = +\infty$. Moreover the $k=0$ shell contributes the finite quantity $(p-2)/(p-1) \rightarrow 1$, whose sum over primes grows like $\pi(x)$. The constant therefore diverges for two independent reasons, only the first of which is a principal-value artefact.*

4.5. Why the constant cannot be removed. We record the repairs that fail, since each localises the obstruction further.

Proposition 4.10. (1) *The counterterm $2h(1) \log \Lambda$ of (86) is linear in $\log \Lambda$, whereas the prime sum over the stabilizing set $\{p \leq e^T\}$ grows like $\pi(e^T)$. The scales are incommensurate.*

- (2) *The zeta-regularized value $-\zeta'/\zeta(0) = -\log 2\pi$ is finite, but the identification fails: the $(p-2)/(p-1)$ part is untouched by the regulator, and the continuation has $s=0$ as an accumulation point of poles.*
- (3) *Restricting g by a primitivity condition cannot help: c_g is entire, and an entire function not identically zero cannot vanish on an interval.*
- (4) *Membership in $\operatorname{rad} I_\partial$ is excluded by an explicit nonvanishing witness.*

The unifying reason is an identity constraint.

Theorem 4.11. *The finite-place principal-value local term is finite, with no residual constant, if and only if the test argument vanishes at the identity, $h(1) = 0$. Every element of the graded family violates this: $f_s(1) = 1$ for every $s \in \mathbb{C}$, because $\chi(1) = 1$ is the definition of a quasi-character.*

So the divergence is not an artefact of the shell splitting, and not a normalization one may choose away. It is a group identity. In parallel, there is no continuous extension of $\Lambda_g(f) := \mathfrak{T}(f \star g^\vee)$ from the test class to individual f_s : the functional is represented by a discrete measure supported on $\{0, 1\} \cup \{\rho\}$, and $\hat{f}_s$ is a measure on a vertical line, so the pairing is a product of two singular objects. Smearing in s does produce a convergent sum $\sum_\rho \varphi(\operatorname{Re} \rho) \overline{\hat{g}(\rho')}$, but that sum is a zero-sourced definition and is inadmissible under the constraint of §1.4.

Remark 4.12. Theorem 4.11 and the two statements following it settle, in the negative, the formulation one would naturally attempt first: extend the corner trace to individual graded elements and identify the result with the zero-side sum. That programme is not merely unachieved; it is impossible, and impossible for a reason internal to the objects — a quasi-character takes the value 1 at the identity, and finiteness of the local term requires the opposite. The construction of §4.6 is therefore not a weakened substitute but the statement the obstruction itself selects: since the divergence enters with an s -independent coefficient, it is confined to the total-mass direction and disappears on differences.

4.6. The obstruction is confined to the total-mass direction. The resolution is that the divergence enters with coefficient $f_s(1) = 1$, which by Theorem 4.11 is *independent of s* . Write $\Lambda_g(f_{s,T}) = D(T) + L_g(s, T)$ with $D(T)$ the s -independent divergent piece and L_g the assembly of Theorem 4.8.

Lemma 4.13. *For coefficients λ_i with $\sum_i \lambda_i = 0$,*

$$\sum_i \lambda_i [D(T) + L_g(s_i, T)] = D(T) \sum_i \lambda_i + \sum_i \lambda_i L_g(s_i, T) = \sum_i \lambda_i L_g(s_i, T).$$

The cancellation is an identity in the coefficients, valid at every finite T , and uses no smoothness of the combination.

It remains to know that mass-zero combinations of principal elements exist. Here the definition of “principal” must be the standard equivariant one.

Proposition 4.14. *Div is additive over sums of potentials and annihilates affine functions; it is therefore already logarithmic in the tropical dictionary, in which the multiplicative operation is addition of piecewise-linear functions, and $\operatorname{Div}(cU) = c \operatorname{Div}(U)$ is the correct power law. Requiring $\operatorname{Div}(U)$ to be strictly invariant forces the trivial character and hence weight $s = 0$, which lies on the excluded boundary of the strip and at an accumulation point of the singular set. Under the standard convention — principal divisors are the divisors of semi-invariant sections — there are nonzero principal witnesses $\operatorname{div}(U_s)$ at every real weight, in particular at every $s \in (0, 1)$.*

Theorem 4.15 (Mellin descent theorem). *Let $\operatorname{Prin}'(\mathcal{G})$ be the subgroup generated by $\{\operatorname{div}(U_s) : s \in \mathbb{R}\}$. Let $s_0, s_1 \in (0, 1)$ be distinct and off the singular set $\{1/N\} \cup \{1 - 1/M\}$, and let g be admissible. Then $\operatorname{div}(U_{s_0}) - \operatorname{div}(U_{s_1}) \in \operatorname{Prin}'(\mathcal{G})$ has total mass zero, and*

$$\Lambda_g^0(\operatorname{div}(U_{s_0}) - \operatorname{div}(U_{s_1})) = L_g(s_0) - L_g(s_1)$$

is finite, defined without reference to any zero of ξ , with the divergent constant contributing exactly zero at every finite regularization depth.

Proof. Membership holds because Prin' is by construction a subgroup, as Prin always is so that $\text{Pic} = \text{Div} / \text{Prin}$ is defined. Each witness has grade-mass 1, so the difference has mass 0. The evaluation and the exact vanishing of the constant are Lemma 4.13. The singular set is countable, so generic s_0, s_1 avoid it and L_g has honest pointwise values there by Theorem 4.8. $\square$

Corollary 4.16. *Principal invariance is a well-formed statement. The objection that the local principal subspace is trivial, so that the question could not even be posed, is answered: it is posed on $\text{Prin}'(\mathcal{G})$, and Theorem 4.15 evaluates the pairing on it.*

5. SCOPE OF THEOREM 4.15

Two of the following are limitations of the theorem; the rest fix its reading.

- **Whether principal invariance holds is not addressed here.** Theorem 4.15 makes the question askable. Answering it is the principal-difference analysis below. This is the most likely misreading of the paper and we state it first.
- **The semi-invariant convention is a revision.** Proposition 4.14 adopts the standard equivariant definition in place of a stricter one. The revision is justified — the stricter reading would trivialize the group of invariant principal divisors already on $\mathbb{P}^1$ under a torus action — but it is a change of definition and should be read as such. It is the load-bearing choice of this paper.
- **The constant is not evaluated.** $\sum_p C_p$ remains unknown under any regularization. Theorem 4.15 does not need it, but only because it works with differences; any statement requiring an absolute normalization would still be blocked.
- **Nothing here bears on RH.**

6. PRINCIPAL DIFFERENCES AND THE RADICAL

The descent question is whether the pairing of Theorem 4.15 kills principal elements. Answering it needs one ingredient not yet available: the finite part of the preceding local sum in closed form. That is Theorem 7.9 of the next section, which we use here as a forward reference; nothing in the correspondence construction depends on the present section, so there is no circularity. Granting it, the object of Theorem 4.15 acquires a closed form:

$$(93) \quad L_g(s) = c_g(s) \Phi(s), \quad \Phi(s) = \pi \cot \frac{\pi s}{2} - \frac{\zeta'}{\zeta}(s) - \frac{\zeta'}{\zeta}(1-s),$$

with Φ independent of g . Theorem 4.15 therefore reduces the descent question to the *vanishing* of Φ , and not to its being constant. Indeed, principal invariance at the pair (s_0, s_1) asks that

$$(94) \quad c_g(s_0)\Phi(s_0) - c_g(s_1)\Phi(s_1) = 0$$

for *every* admissible g . For $s_0 \neq s_1$ the Mellin evaluations $g \mapsto c_g(s_0)$ and $g \mapsto c_g(s_1)$ are linearly independent functionals, so the two coefficients in (94) must vanish separately: $\Phi(s_0) = \Phi(s_1) = 0$. Quantifying over admissible pairs, the pairing descends if and only if $\Phi \equiv 0$. A nonzero constant Φ would *not* suffice, since it leaves the nonzero residual $\Phi \cdot (c_g(s_0) - c_g(s_1))$.

Lemma 6.1. $\Phi(s) = \pi \cot(\pi s/2) + \frac{1}{2}[\psi(s/2) + \psi((1-s)/2)] - \log \pi$.

This follows from logarithmic differentiation of $\xi(s) = \xi(1-s)$, the polar terms cancelling in the symmetric combination. It is what makes Φ computable without reference to the zeros.

Theorem 6.2. (1) $\Phi \not\equiv 0$; in particular $\Phi(\frac{1}{2}) = -2.2305907\dots \neq 0$. Hence principal invariance fails and the pairing does not descend to Pic ; and the pairing is not vacuous.

- (2) Φ is not constant on $(0, 1)$. This is strictly stronger than item (1) and is not what the descent conclusion rests on; it is recorded because it also excludes the degenerate case in which the radical would be a single level set of Φ .
- (3) $\text{rad } \Lambda^0$ is spanned by the point masses at the zeros of Φ : writing $\Lambda_g^0(\sum_i \lambda_i \delta_{s_i}) = \sum_i \lambda_i c_g(s_i) \Phi(s_i)$ and using linear independence of the Mellin evaluations at distinct s_i , one gets $\lambda_i = 0$ unless $\Phi(s_i) = 0$.
- (4) On the real segment $(0, 1)$, Φ has exactly one sign change, at $s^* = 0.3016923882\dots$

Item (1) is a control that had to be run before building further, and it is now doing double duty. The functional equation gives $(\xi'/\xi)(s) + (\xi'/\xi)(1-s) \equiv 0$, so Φ vanishing identically was a genuine possibility; by the criterion above that case is exactly the one in which the pairing *would* have descended, and it is simultaneously the case in which the radical would be everything and the construction empty. The single evaluation $\Phi(\frac{1}{2}) \neq 0$ therefore settles both questions at once, in opposite directions.

That principal invariance fails is not fatal to the programme. Weil's argument does not use the pairing on Pic but on classes modulo algebraic or numerical equivalence, and the available Hodge-index statements are likewise formulated on $NS \otimes \mathbb{Q}$. A pairing descending to the quotient by a discrete radical is an object of that type. What is not established here is whether that quotient coincides with, or maps to, the numerical quotient of §3.4; the two are constructed differently and their comparison — which is what would join the two halves of the programme — is not attempted. There is no formal contradiction with Theorem 2.23: the pairing in this section is the analytic Mellin-corner candidate on completed mixed correspondences, whereas the form (66) of row (a) is the intrinsic periodic cotangent determinant on the two prime-generated rulings. The nuclear realization of Section 8 joins their contact data at the level of categorical trace. Extending the intrinsic periodic determinant and effectivity from the ruling lines to the completed mixed classes is still required for row (d); it is not a hypothesis of the nuclear Lefschetz identity.

7. WITT CORRESPONDENCES AND THEIR LOCAL LEFSCHETZ COMPLEXES

The analogue of a Frobenius graph used here is a cohomological correspondence supported by the diagonal, not an algebraic cycle with a moving support. This distinction is structural. The endomorphisms which exist over $\mathbb{Z}$ act on the Witt coefficient sheaf, and their local contact must therefore be extracted from that action. We construct it below as a derived intersection of the cyclic Witt orbit with the unit section. No finite module is appended in order to prescribe an intersection number.

7.1. Witt dynamics. Let

$$W_{\mathbb{Z}} := \bigoplus_{r \geq 1} \mathbb{Z} \phi_r$$

be Haran's global cyclotomic Witt module. We use the sheaf $\mathcal{W}_X := \underline{W}_{\mathbb{Z}}$ obtained by sheafifying the constant presheaf on the topological space of the absolute arithmetic curve X . This is an explicitly defined coefficient system on the spherical carrier; we do not identify Haran's pro-scheme category with the category of Γ -ringed spaces used in row (a). Haran's Frobenius maps on $W_{\mathbb{Z}}$ induce stalkwise sheaf endomorphisms

$$(95) \quad F_m F_n = F_{mn}, \quad F_1 = \text{id}_{\mathcal{W}_X}.$$

The constant sections give a distinguished invariant cyclic submodule

$$(96) \quad \bigoplus_{r \geq 1} \mathbb{Z} \phi_r \subseteq \mathcal{W}_X(X).$$

For the Hilbert structure defined by Haran's state, the adjoints $V_n = F_n^*$ satisfy

$$(97) \quad V_n \phi_r = \phi_{nr}, \quad V_m V_n = V_{mn}$$

[16, (12.35),(12.37)]. Haran's lambda characteristic is

$$(98) \quad \chi_T(x) := \text{tr}(\lambda_T(x)), \quad \chi_T(\phi_r) = \Phi_r(T),$$

where Φ_r is the cyclotomic polynomial [16, (13.37)]. Consequently the polynomial extracted from the lambda characteristic of the canonical cyclic vector $V_n \phi_1$ is

$$(99) \quad P_{\Gamma_n}(T) := \chi_T(V_n \phi_1) = \Phi_n(T).$$

7.2. The dynamic local contact. Put $R = \mathbb{Z}[T, T^{-1}]$. For $n \geq 1$ define the derived intersection

$$(100) \quad K_n^W := R/(P_{\Gamma_n}(T)) \otimes_R^{\mathbf{L}} R/(T-1).$$

This is the intersection of the cyclotomic lambda-characteristic locus $\text{Spec } R/(P_{\Gamma_n})$ associated with F_n and the unit section $T = 1$. We do not claim that this affine locus represents a full geometric orbit of F_n . Resolving the first quotient by $[R \xrightarrow{P_{\Gamma_n}(T)} R]$ gives the canonical model

$$(101) \quad K_n^W \simeq [\mathbb{Z} \xrightarrow{P_{\Gamma_n}(1)} \mathbb{Z}]$$

in degrees $-1, 0$. Thus the complex, its differential and its determinant are all functorially determined by F_n , the canonical vector ϕ_1 , the adjoint, and the lambda characteristic.

Proposition 7.1 (Cyclotomic local contact). *For $n > 1$, K_n^W is perfect and concentrated in cohomological degree zero, where*

$$(102) \quad P_n := H^0(K_n^W) \simeq \mathbb{Z}/\Phi_n(1)\mathbb{Z}.$$

For the identity put $P_1 = \mathbb{Z}$. Moreover

$$(103) \quad P_n \simeq \begin{cases} \mathbb{F}_p, & n = p^k, \\ 0, & n > 1 \text{ has at least two distinct prime divisors.} \end{cases}$$

Proof. Equation (101) proves perfectness. Since $\Phi_n(1) \neq 0$ for $n > 1$, its left differential is injective and its cokernel is (102). To evaluate the scalar, divide $T^n - 1 = \prod_{d|n} \Phi_d(T)$ by $T - 1$ and let $T \rightarrow 1$:

$$n = \prod_{\substack{d|n \\ d>1}} \Phi_d(1).$$

Induction gives $\Phi_{p^k}(1) = p$ along a prime-power tower. If n has at least two prime divisors, the factors belonging to its proper divisors already multiply to n , so $\Phi_n(1) = 1$. This proves both cases. For $n = 1$, $P_{\Gamma_1} = T - 1$ and the reduced degree-zero self-contact is $H^0([\mathbb{Z} \xrightarrow{0} \mathbb{Z}]) = \mathbb{Z}$, the required unit. $\square$

The reduced contacts do possess a composition law, but it is not the derived tensor product of their full perfect resolutions. Let $P(a) = \mathbb{Z}/a\mathbb{Z}$ for $a \geq 0$, with $P(0) = \mathbb{Z}$, and define

$$(104) \quad P(a) \star P(b) := H^0(P(a) \otimes_{\mathbb{Z}}^{\mathbf{L}} P(b)) = P(\gcd(a, b)).$$

The equality follows from the two-term free resolution of either cyclic module. The positive generator of the ideal (a, b) makes the displayed identification canonical. Associativity, symmetry and the unit $P(0)$ follow either from the associativity of ordinary tensor product on the degree-zero quotients or directly from the corresponding properties of gcd.

Set $a_1 = 0$ and $a_n = \Phi_n(1)$ for $n > 1$. Formula (103) gives

$$(105) \quad a_{mn} = \gcd(a_m, a_n).$$

Indeed, if both integers are powers of the same prime, all three values are that prime; in every other nonunit case the product has mixed prime support and both sides are 1. The cases involving 1 use $\gcd(0, a) = a$. Therefore

$$(106) \quad P_m \star P_n \simeq P_{mn}.$$

This is the correct reduced-contact law.

It would be false to replace $\star$ by the full derived tensor product. For example, a resolution of $\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbf{L}} \mathbb{F}_p$ has both $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{F}_p, \mathbb{F}_p)$ and $\mathbb{F}_p \otimes \mathbb{F}_p$, whereas $K_{p^2}^W$ has only one cohomology group. Thus no Künneth formula for an ambient cotangent H_1 is being assumed.

7.3. The correspondence category and its arithmetic family. We now give a category requiring no unspecified bicategorical infrastructure. Let $\mathrm{End}(\mathcal{W}_X)$ be the discrete monoidal category whose objects are Witt-sheaf endomorphisms and whose product is composition. Define $\mathrm{End}_{\mathcal{O}_C}(\mathcal{N}_{DN})$ in the same way for (78). Let $\mathrm{Pic}_{\mathrm{ext}}^{\mathrm{met}}(X)$ be the framed metrized factorized Picard category of row (a), and let $\mathrm{Cont}_{\mathrm{cyc}}$ be the symmetric monoidal category generated by the pointed cyclic modules $P(a)$ with product $\star$. Take their product monoidal category

$$(107) \quad \mathbf{E}_{W,DN}(X) := \mathrm{End}(\mathcal{W}_X) \times \mathrm{End}_{\mathcal{O}_C}(\mathcal{N}_{DN}) \times \mathrm{Pic}_{\mathrm{ext}}^{\mathrm{met}}(X) \times \mathrm{Cont}_{\mathrm{cyc}}.$$

Associators and units are componentwise, so coherence in the four factors proves coherence of (107). Its support functor sends every object below to the diagonal $X \rightarrow \mathcal{Y}_{\mathbb{S}}$, which is proper over both factors.

For $n = \prod_p p^{v_p(n)}$, let

$$D(n) = \sum_p v_p(n)[p].$$

The line $\mathcal{O}_X(D(n))$, with its canonical section, is one of the actual divisor modules of row (a). Equivalently, pull $\mathcal{L}(D(n), 0)$ back along the diagonal. Give its positive frame the Arakelov norm

$$\|\mathbf{1}_{D(n)}\| := \exp(-\deg D(n)) = n^{-1};$$

this explicitly defines the framed metrized object $\mathcal{T}_n$. Additivity of Arakelov degree and the tensor law of Proposition 2.3 give

$$(108) \quad \nu_{\infty}(\mathcal{T}_n) = \deg D(n) = \log n, \quad \mathcal{T}_m \otimes \mathcal{T}_n \simeq \mathcal{T}_{mn}.$$

On the nuclear module define ρ_n by (79). Finally set

$$(109) \quad \Gamma_n = (F_n, \rho_n, \mathcal{T}_n, P_n) \in \mathbf{E}_{W,DN}(X),$$

where the last component is not additional data: it is the output $H^0(K_n^W)$ of (100). Define the *arithmetic Witt correspondence category* $\mathrm{Corr}_{W,DN}^{\mathrm{arith}}(X)$ to be the graph-image subcategory with precisely the objects Γ_n and the componentwise isomorphisms between them. Its fourth component is therefore constrained by $P_n = H^0(K_n^W)$ and cannot be varied independently. Equations (95), (79), (108) and (106) prove that this graph-image is closed under the ambient monoidal product and contains its unit.

Theorem 7.2 (Composition and faithfulness). *There are coherent canonical isomorphisms*

$$\Gamma_m \circ \Gamma_n \simeq \Gamma_{mn}, \quad \Gamma_1 \simeq \Delta,$$

and $\Gamma_m \simeq \Gamma_n$ implies $m = n$.

Proof. The Witt component composes by (95). The nuclear component composes because $\delta_m * \delta_n = \delta_{mn}$. The metric component composes by (108), and the reduced contact component by (106). Coherence is componentwise. If two correspondences are isomorphic, their Witt maps agree; taking adjoints and applying to ϕ_1 gives $\phi_m = \phi_n$ by (97), hence $m = n$ by (96). The nuclear action gives a second proof: $\rho_n(\delta_1 e_\Gamma) = \delta_n e_\Gamma$. The metric $\nu_\infty(\mathcal{T}_n) = \log n$ gives a third. $\square$

Remark 7.3 (Exactly what Theorem 7.2 does and does not give). Four distinct composition laws are in play in (109): Witt composition, Dirichlet convolution, the tensor law of metrized lines, and the reduced product $\star$ of (104). They are genuinely different operations — in particular $\star$ is the truncation $H^0(- \otimes^{\mathbf{L}} -)$, designed to recover the gcd law (105), and *not* the derived composition of the contact complexes, as (104) and the Tor computation after it make explicit. The product category (107) makes them move together by construction, and Theorem 7.2 records that the graph-image is closed under the resulting product.

What this establishes is a faithful multiplicative family in an explicitly defined product category. It is not a construction of cohomological correspondences on $\mathcal{B}_S$ possessing pullback, derived product, pushforward, an excess-intersection formula and composition on the triple product; the bare support of every Γ_n is the same diagonal, and the information distinguishing n resides in the decorations. The faithfulness of Theorem 7.2 is accordingly recovery of the label n , not transversality of cycles. Whether these decorated objects descend to classes in an intersection theory is precisely the question row (d) would need answered, and it is not answered here.

7.4. Comparison with the perfect and nuclear structures of row (a). For $n > 1$, the realification of (101) is contractible because $\Phi_n(1) \neq 0$. Hence

$$(110) \quad \mathbb{L}_n = (\mathcal{O}_{\mathbb{Z}} \otimes_{\mathbb{Z}} K_n^W, 0, (\mathcal{O}_{\mathbb{Z}} \otimes K_n^W) \otimes_{\mathcal{O}_{\mathbb{Z}}} \mathcal{O}_{\mathbb{R}} \simeq 0) \in \text{Perf}_{DN}(\mathcal{B}_S).$$

It is perfect because its integral component is a two-term complex of finite free modules. Since its nuclear component is zero, Proposition 2.16 gives a canonical lift to Perf_{IDN} with zero periodic component. Equations (101) and (103) give the canonical equivalence

$$(111) \quad \mathbb{L}_n \simeq \begin{cases} C_p, & n = p^k, \\ 0, & n \text{ has at least two prime divisors,} \end{cases}$$

where C_p is exactly the object (71) constructed in row (a). This is an objectwise equivalence in Perf_{DN} and, after the zero-periodic lift, in Perf_{IDN} ; it is not an equality of numerical masses.

With the standard torsion metric, (101) gives

$$(112) \quad -\log \|\det_{\text{tor}} \mathbb{L}_n\| = \log \Phi_n(1) = \Lambda(n).$$

The first equality is the determinant of the dynamic intersection, and the second is (103).

Theorem 7.4 (Local Witt–Lefschetz contact). *For every $n > 1$, the derived unit contact of the cyclotomic lambda-characteristic locus associated with F_n is the perfect intrinsic Deligne–nuclear object $\mathbb{L}_n$. It is canonically equivalent to the finite contact of row (a), its determinant degree is $\Lambda(n)$, and its reduced contact is compatible with correspondence composition by (106).*

Proof. The dynamic derivation is (99)–(101); Proposition 7.1 computes its cohomology; (111) proves the comparison in Perf_{DN} and hence, by the canonical lift, in Perf_{IDN} ; and (112) computes its determinant. The last assertion is the proved gcd law (105). $\square$

The connection with the nuclear correspondence module is equally explicit. On the algebraic Witt basis define

$$(113) \quad J : \bigoplus_{r \geq 1} \mathbb{R} \phi_r \longrightarrow \mathcal{C}_{\mathbb{R}}, \quad J(\phi_r) = \delta_r.$$

It is injective, and

$$(114) \quad J(V_n \phi_r) = \delta_{nr} = \delta_n * \delta_r = \rho_n(J(\phi_r)).$$

Thus it intertwines the actual Witt dynamics on its algebraic cyclic domain with the action on the e_{Γ} summand of $\mathcal{N}_{DN}$. Applying the continuous contact functional gives

$$(115) \quad \ell(J(V_n \phi_1)) = \ell(\delta_n) = \Lambda(n) = -\log \|\det_{\text{tor}} \mathbb{L}_n\|.$$

The Witt orbit, the finite perfect contact and the nuclear action are therefore joined by explicit morphisms and determinant comparison.

7.5. Central metric normalization. This normalization can be made internal to the based metric Picard groupoid already used in row (a), without invoking an unconstructed Poincaré duality. On positive based metric lines define

$$\mathfrak{d}_n(L) := \mathcal{T}_n \otimes L^{-1}.$$

The canonical cancellations show $\mathfrak{d}_n^2(L) \simeq L$, so this is an involution. A fixed point satisfies $L^{\otimes 2} \simeq \mathcal{T}_n$. It exists by the real metric power operation $L^{(c)}$ used in Proposition 2.12, and it is unique among positive based lines because its norm is the positive square root of $\|\mathcal{T}_n\| = n^{-1}$. Consequently

$$(116) \quad w_{1/2}(\Gamma_n) = \exp(-\tfrac{1}{2}\nu_{\infty}(\mathcal{T}_n)) = n^{-1/2}$$

is the unique self-dual positive metric character. It is multiplicative because $\mathcal{T}_m \otimes \mathcal{T}_n \simeq \mathcal{T}_{mn}$, and the normalized contact coefficient is

$$(117) \quad w_{1/2}(\Gamma_n) (-\log \|\det_{\text{tor}} \mathbb{L}_n\|) = \frac{\Lambda(n)}{\sqrt{n}}.$$

This is a self-dual metric normalization. It is not asserted to be an algebraic half-Tate line or a half-integral Cartier divisor.

Corollary 7.5 (Constructed row (b)). *The family $\{\Gamma_n\}_{n \geq 1}$ is a faithful multiplicative family of properly supported Witt cohomological correspondences on the spherical square. Its local contact is derived from F_n , has a canonical lift to $\text{Perf}_{IDN}(\mathcal{Y}_{\mathbb{S}})$, agrees canonically with the contact object of row (a), and has determinant degree $\Lambda(n)$. Its faithful nuclear realization intertwines composition, and its self-dual metric normalization is $\Lambda(n)/\sqrt{n}$. Thus row (b) is constructed completely in this cohomological Deligne–nuclear sense.*

Proof. The ambient category and its arithmetic graph-image are (107) and the paragraph following (109); Theorem 7.2 proves composition and faithfulness; Theorem 7.4 constructs and computes the contact; (114) proves the nuclear comparison; and (116) proves the metric assertion. The diagonal support is proper. None of the definitions uses a zero of ξ , the explicit formula or a positivity assertion. $\square$

The conclusion is intentionally cohomological. Forgetting the coefficient maps and frames sends every Γ_n to the same diagonal, so no distinct undecorated Chow cycles are claimed. The result proved here is instead that the dynamics composes, its lambda-characteristic locus forces the perfect local Lefschetz complex, and that complex is the contact already internal to the Deligne–nuclear spherical square. Section 8 constructs the global trace of this family.

7.6. The shell functional.

Definition 7.6. For p prime and $k \in \mathbb{Z} \setminus \{0\}$ set $\Gamma_{p,k}(h) := p^{\min(k,0)} h(p^k)$, a scaled Dirac evaluation at the point $p^k \in (0, \infty)$.

Proposition 7.7. $\Gamma_{p,k}$ is linear, and on the graded family

$$\Gamma_{p,k}(f_s) = p^{-ks} \quad (k \geq 1), \quad \Gamma_{p,-m}(f_s) = p^{m(s-1)} \quad (m \geq 1).$$

Resumming the two families recovers the two closed forms of Theorem 4.6.

7.7. The normalization, and the Weil coefficient. Theorem 4.6 was stated with $d^\times u$ normalized by $\text{vol}(\mathbb{Z}_p^\times) = 1$, in which every shell has mass 1 and no factor $\log p$ arises. We adopt instead the adelic normalization, in which the valuation map pushes $d^\times u$ to the measure assigning $\log p$ to each step of the value group $p^\mathbb{Z}$ — the normalization making the finite places commensurate with the archimedean one, where the value group is $\mathbb{R}_{>0}$ with dt/t , and the one in which Weil's local terms are stated. Then $\Gamma_{p,k}^{\text{Tate}} = (\log p) \Gamma_{p,k}$.

Theorem 7.8. At the central weight,

$$\Gamma_{p,k}^{\text{Tate}}(f_{1/2}) = (\log p) p^{-k/2} = \frac{\Lambda(p^k)}{\sqrt{p^k}},$$

and the mirror shells give the same value. Thus the Weil local term

$$W_p(h) = (\log p) \sum_{k \geq 1} p^{-k/2} [h(p^k) + h(p^{-k})]$$

is recovered term by term.

The coincidence of the two shell families at $s = \frac{1}{2}$ is the fixed-point property of the mirror involution of Lemma 3.2.

7.8. The assembly.

Theorem 7.9. For $\Re s > 1$,

$$\sum_p \sum_{k \geq 1} \Gamma_{p,k}^{\text{Tate}}(f_s) = \sum_{n \geq 1} \Lambda(n) n^{-s} = -\frac{\zeta'}{\zeta}(s),$$

absolutely convergent, and the mirror family gives $-\zeta'/\zeta(1-s)$ for $\Re s < 0$.

The two half-planes are exchanged by $s \mapsto 1-s$ and are disjoint from the strip; on the strip the assembly exists only as the meromorphic continuation. This is the same phenomenon as the divergence of the naive all-places sum, now at the level of the correspondence functionals.

Remark 7.10. Theorem 7.9 by itself is only an identity between a sum of functionals and a Dirichlet series. The next section constructs the global nuclear diagonal trace which promotes it to row (c). The archimedean term is absent here by design and is supplied there by the Poisson quotient trace. Finiteness sits on $\Gamma_{p,k}$, whose support is a single point, and equivariance on the sections it is paired against; the two demands are on opposite sides of the pairing, which is what makes the construction possible at all.

8. THE NUCLEAR LEFSCHETZ CHARACTER

Row (b) produced two related, but differently typed, objects: the correspondence Γ_n , which composes multiplicatively, and its dynamic perfect contact $\mathbb{L}_n$, whose determinant degree is $\Lambda(n)$. The latter degree is not a multiplicative character. Thus a global Lefschetz formula cannot be obtained by pretending that $\mathbb{L}_n \mapsto \Lambda(n)$ is a monoidal functor. The construction in this section keeps the two types separate. It realizes Γ_n monoidally as a scaling multiplier U_n , and then forms the operator-valued contact

$$\sum_{n \geq 2} \deg_{\det}(\mathbb{L}_n) U_n = \sum_{n \geq 2} \Lambda(n) U_n.$$

The equality with the logarithmic Euler character of the Zeta operator is a theorem below. Meyer's Poisson construction then supplies a summable super-representation whose supercharacter adds the opposite finite orientation, the two polar characters and the archimedean term. This is the precise cohomological/nuclear meaning of row (c). No Cartier or Chow intersection on the spherical carrier is asserted.

Remark 8.1 (Attribution, and what is added here). The analytic engine of this section is Meyer's and is used as a black box: the weighted nuclear spaces (119), the summability of the scaling representation [21, Proposition 5.6], the Zeta operator, and the identification of the supercharacter with the two sides of the explicit formula [21, Theorem 5.8]. Nothing in that chain is reproved here, and row (c) read as an analytic identity *is* that theorem.

What this section adds is not the identity but the *provenance of its coefficients*. In Meyer's account the weights $\Lambda(n)$ enter as the coefficients of $-\zeta'/\zeta$, which is to say they are read off from the Euler product. Here they are not postulated: by Theorem 7.4 the Witt dynamic contact complex is $\mathbb{L}_n \simeq [\mathbb{Z} \xrightarrow{\Phi_n(1)} \mathbb{Z}]$, whose torsion determinant has degree exactly $\Lambda(n)$ by (112), so the Weil coefficient is the determinant degree of a contact complex attached to the correspondence Γ_n . The assembly $\sum_n \deg_{\det}(\mathbb{L}_n) \mathcal{R}(\Gamma_n)$ is then a Lefschetz sum in the shape $\Gamma_n \cdot \Delta = N_n$, rather than a resummation of a Dirichlet series. Two further points are ours: the separation of the multiplicative type Γ_n from the non-multiplicative type $\deg_{\det}$, without which the assembly is not well formed; and Proposition 1.1, which shows that the graded quotient is necessary and not merely convenient.

We claim no more than this. In particular the compatibility established here is between Meyer's trace and the Witt contact labels; it is not a derivation of Meyer's Poisson quotient from the geometry of $\mathcal{B}_{\mathbb{S}}$, and the completed object $\mathbb{L}_{\text{comp}}$ is Meyer's, not a global cohomology constructed on the carrier of row (a).

8.1. Weighted spaces and the trace category. Identify $\mathbb{R}_+^\times$ with $\mathbb{R}$ by the logarithm and write $d^*x = dx/x$. For an interval $I \subseteq \mathbb{R}$, let

$$(118) \quad \mathcal{S}_I = \{f : x^s f(x) \in \mathcal{S}(\mathbb{R}_+^\times) \text{ for every } s \in I\}.$$

For compact I this has the intersection Fréchet topology, and in general the projective-limit topology over a countable compact exhaustion. These are nuclear Fréchet spaces. Following [21, §2], put

$$(119) \quad \mathcal{H}_- := \mathcal{S}_{(-\infty, \infty)}, \quad \mathcal{S}_> := \mathcal{S}_{(1, \infty)}, \quad \mathcal{S}_< := \mathcal{S}_{(-\infty, 0)},$$

and let $\mathcal{H}_+$ be the even Schwartz functions on $\mathbb{R}$.

Our regular-representation convention is

$$(120) \quad (\lambda_t f)(x) = f(t^{-1}x), \quad \lambda(h)f = \int_0^\infty h(t) \lambda_t(f) d^*t.$$

Let $\mathbf{NFRep}$ be the exact category of smooth scaling representations on nuclear Fréchet spaces, with closed invariant embeddings having nuclear Fréchet quotient as the admissible monomorphisms. Its $\mathbb{Z}/2$ -graded enlargement is denoted $\mathbf{NFRep}^{\text{sup}}$. A graded object $E = E^{\bar{0}} \oplus E^{\bar{1}}$ is *summable* when the two integrated operators $\lambda_E(h)$ are nuclear for every $h \in C_c^\infty(\mathbb{R}_+^\times)$; then

$$(121) \quad \chi_E(h) = \text{Tr}(\lambda_{E^{\bar{0}}}(h)) - \text{Tr}(\lambda_{E^{\bar{1}}}(h))$$

is its nuclear supercharacter. Closed subspaces and quotients of nuclear Fréchet spaces are nuclear, so the objects below belong to this exact category. Meyer's summability theorem [21, Proposition 5.6] proves the additional trace-class assertion required by (121).

Let $\mathbf{Mult}(\mathcal{S}_>)$ be the one-object strict monoidal $\mathbb{C}$ -linear category whose endomorphism algebra consists of the continuous convolution multipliers of $\mathcal{S}_>$. This is the target of the individual correspondences. The supercategory $\mathbf{NFRep}^{\text{sup}}$, rather than the multiplier category, is the target of the completed trace object.

8.2. The typed realization of rows (a) and (b). Let $\mathbf{Corr}_{\mathbb{C}}^{\text{arith}}(X)$ be the one-object $\mathbb{C}$ -linear category whose endomorphism algebra is the monoid algebra of the isomorphism classes in the graph-image category of row (b). Its distinguished basis arrows are $[\Gamma_n]$ and

$$(122) \quad [\Gamma_m][\Gamma_n] = [\Gamma_{mn}].$$

This definition uses the actual composition isomorphism of Theorem 7.2; it is not a new freely chosen correspondence algebra. Put

$$(123) \quad U_n := \lambda_n^{-1}, \quad (U_n f)(x) = f(nx).$$

Since $U_m U_n = U_{mn}$, the formula

$$(124) \quad \mathcal{R}([\Gamma_n]) = U_n$$

extends uniquely to a strict monoidal $\mathbb{C}$ -linear functor

$$\mathcal{R} : \mathbf{Corr}_{\mathbb{C}}^{\text{arith}}(X) \longrightarrow \mathbf{Mult}(\mathcal{S}_>).$$

This realization is connected to both coefficient systems already constructed. The intertwiner of row (b) sends $V_n \phi_r$ to $\delta_n * \delta_r = \delta_{nr}$ in the nuclear Dirichlet algebra $\mathcal{C}_{\mathbb{R}}$. Define

$$(125) \quad \mathbf{j} : \mathcal{C}_{\mathbb{R}} \longrightarrow \mathcal{D}'(\mathbb{R}_+^\times), \quad \mathbf{j}((a_r)) = \sum_{r \geq 1} a_r \delta_{r-1}.$$

The sum is locally finite as a distribution: every compact subset of $\mathbb{R}_+^\times$ contains only finitely many r^{-1} . On a bounded set of test functions the supports lie in one compact set, so the corresponding strong-dual seminorm is bounded by a finite coordinate seminorm, and $\mathbf{j}$ is continuous. Dirichlet convolution becomes multiplicative convolution. Consequently

$$(126) \quad \mathbf{j}(J(V_n \phi_r)) = \delta_{(nr)^{-1}} = \delta_{n^{-1}} * \mathbf{j}(J(\phi_r)),$$

where J is the Witt–nuclear intertwiner (113). Acting on $\mathcal{S}_>$, convolution by $\delta_{n^{-1}}$ is precisely U_n . Thus (124) is the distributional realization of the adjoint Witt dynamics and of the e_{Γ} summand of $\mathcal{N}_{DN}$, not merely a map of labels.

The metric component gives the same label intrinsically: $\nu_\infty(\mathcal{T}_n) = \log n$. Hence either the Witt action, the nuclear point mass or the metric line recovers n , and all three determine the same multiplier U_n under (124).

The intrinsic row-(a) cohomology is compatible with this realization, with no identification of unlike types. At every finite periodic depth its nuclear frame is

$$T_{\mathcal{C}}(\mathbf{H}_{r,s}^{\text{int}}(x)) = \mathcal{O}_{\mathcal{C}} \otimes_{\mathbb{R}} V_{r,s}^{\text{per}}(x).$$

Left convolution by δ_n on the scalar factor commutes with principal translations, finite-support sums and the Cartesian-basis Künneth maps, because $\mathcal{C}_{\mathbb{R}}$ is commutative and those maps are $\mathcal{O}_{\mathcal{C}}$ -linear. Under j this scalar action is convolution by δ_{n-1} , hence the multiplier U_n . Moreover, for the dynamic contact of row (b),

$$(127) \quad \ell(\delta_n) = -\log \|\det \mathbb{L}_n\|_{\text{tor}} = \Lambda(n).$$

Thus rows (a), (b) and (c) share the same nuclear scalar action and the same derived contact determinant. The statement is a commuting comparison on the common nuclear component; it does not falsely promote the completed nuclear representation to a perfect periodic object.

For later convergence, choose the usual defining seminorms on $\mathcal{S}_a$ involving $(\log x)^j (x\partial_x)^k f$. Since $x\partial_x$ commutes with U_n while $\log x = \log(nx) - \log n$, for every such seminorm there are C, N with

$$(128) \quad q_{a,j,k}(U_n f) \leq C n^{-a} (1 + \log n)^N \sum_{j' \leq j} q_{a,j',k}(f).$$

Thus $\sum_n \Lambda(n) U_n$ converges absolutely in every defining multiplier seminorm after choosing any $a > 1$; polynomial logarithmic factors do not affect convergence.

8.3. Dynamic contact and the logarithmic Euler character. Meyer's Zeta operator is

$$(129) \quad Z = \sum_{n \geq 1} U_n, \quad Zf(x) = \sum_{n \geq 1} f(nx).$$

The coefficient sequence $(1, 1, \dots)$ is not in the rapidly decreasing algebra $\mathcal{C}_{\mathbb{R}}$. Accordingly (129) is not presented as its image. Rather, the finite sums $j(\sum_{n \leq N} \delta_n)$ converge in $\mathcal{D}'(\mathbb{R}_+^{\times})$ to the locally finite Dirac comb $\check{\zeta} = \sum_n \delta_{n-1}$: on every bounded set of test functions the pairing is eventually constant outside one finite set of indices. The operator Z is the continuous action of this distribution on $\mathcal{S}_{>}$. Proposition 3.2 of [21] proves that Z and $Z^{-1} = \sum_n \mu(n) U_n$ are mutually inverse continuous multipliers of $\mathcal{S}_{>}$, and proves the convergent Euler product

$$(130) \quad Z = \prod_p (1 - U_p)^{-1}.$$

No invertibility on $\mathcal{H}_+$ or on the quotient representations below is asserted.

For functions or distributions on $\mathbb{R}_+^{\times}$ put

$$(131) \quad (\partial f)(x) = (\log x) f(x).$$

It is a continuous derivation for multiplicative convolution. Since U_n is represented by δ_{n-1} ,

$$(132) \quad \partial U_n = -(\log n) U_n.$$

The estimates (128) justify differentiating (130) in the topology of multipliers of $\mathcal{S}_{>}$. The derivation rule and (132) give

$$(133) \quad Z\partial(Z^{-1}) = \sum_p (1 - U_p)^{-1} (\log p) U_p = \sum_p \sum_{k \geq 1} (\log p) U_{p^k} = \sum_{n \geq 2} \Lambda(n) U_n.$$

This identity is exactly compatible with the dynamic contact of row (b). Indeed, Theorem 7.4 gives, before any analytic realization,

$$(134) \quad \deg_{\det}(\mathbb{L}_n) := -\log \|\det \mathbb{L}_n\|_{\text{tor}} = \Lambda(n).$$

Combining (124), (133) and (134) yields the equality of convergent operator-valued distributions

$$(135) \quad Z\partial(Z^{-1}) = \sum_{n \geq 2} \deg_{\det}(\mathbb{L}_n) \mathcal{R}([\Gamma_n]).$$

This is a labelwise determinant realization, not the false assertion that determinant degree is a monoidal character.

For a convolution kernel put $\tau(f) = f(1)$. If $h \in C_c^\infty(\mathbb{R}_+^\times)$, apply the continuous functional $A \mapsto \tau(h * A)$ to (135). Since $\tau(h * U_n) = h(n)$, this produces

$$(136) \quad W_{\text{fin}}^+(h) = \sum_p \sum_{k \geq 1} (\log p) h(p^k).$$

Let

$$(137) \quad (\mathcal{J}h)(x) = x^{-1}h(x^{-1}).$$

Conjugating by this involution gives the opposite orientation

$$(138) \quad W_{\text{fin}}^-(h) = \sum_p \sum_{k \geq 1} (\log p) p^{-k} h(p^{-k}).$$

Both sums are finite for compactly supported h , because such a support is bounded away from 0 and ∞ .

8.4. The source-defined Poisson quotient. Use the additive Fourier transform

$$(139) \quad (\mathcal{F}f)(y) = \int_{\mathbb{R}} f(x) e^{2\pi i xy} dx,$$

fixed by the standard self-dual additive character, and set

$$(140) \quad \mathcal{H}_\cap = \{f \in \mathcal{H}_+ : f(0) = \mathcal{F}f(0) = 0\}.$$

Poisson summation gives $Zf = \mathcal{J}Z\mathcal{F}f$ on $\mathcal{H}_\cap$. Meyer constructs a scaling representation $\mathcal{H}_\cup$ fitting into the exact sequence

$$(141) \quad 0 \longrightarrow \mathcal{H}_- \longrightarrow \mathcal{H}_\cup \longrightarrow \mathbb{C}(x^0) \oplus \mathbb{C}(x^1) \longrightarrow 0.$$

His Theorem 3.3 proves that $Z : \mathcal{H}_+ \rightarrow \mathcal{H}_\cup$ is continuous, has closed range and is a topological isomorphism onto that range, and that $Zf \in \mathcal{H}_-$ exactly for $f \in \mathcal{H}_\cap$. Therefore the closed quotients

$$(142) \quad \mathcal{H}_+^0 := \mathcal{H}_+ / \mathcal{H}_\cap \simeq \mathbb{C}(x^0) \oplus \mathbb{C}(x^1), \quad \mathcal{H}_-^0 := \mathcal{H}_- / Z\mathcal{H}_\cap$$

are nuclear Fréchet scaling representations. Define the actual superobject

$$(143) \quad \mathbb{L}_{\text{comp}}^{\bar{0}} = \mathcal{H}_+^0, \quad \mathbb{L}_{\text{comp}}^{\bar{1}} = \mathcal{H}_-^0 \quad \text{in } \mathbf{NFRep}^{\text{sup}}.$$

This replaces the potentially ambiguous phrase “virtual object” by a typed $\mathbb{Z}/2$ -graded representation. This supercategorical packaging is our definition; the quotient representations and their summability are Meyer’s. Proposition 5.6 of [21] proves that it is summable. Its even part is exactly the two-dimensional polar object; its odd part is the closed quotient $\mathcal{H}_-^0$. All ingredients in (143) are defined from the Dirac comb, the additive Fourier transform and closed subquotients, before any spectral description is taken.

The archimedean distribution must retain Meyer’s normalization. For a compactly supported function h on $\mathbb{R}_+^\times$, extend it by zero at the origin and let $h_{\text{ev}}(x) = h(|x|)$. Define

$$(144) \quad W_\infty(h) := -\langle \mathcal{F}(\log |x|), y \mapsto h_{\text{ev}}(1 - y) \rangle.$$

Equivalently,

$$(145) \quad W_\infty(h) = \text{fp}_\mathcal{F} \int_0^\infty \left(\frac{h(x)}{|1-x|} + \frac{h(x)}{1+x} \right) dx.$$

Here $\text{fp}_\mathcal{F}$ is the Fourier finite-part extension, not a Cauchy principal value of an even singularity. Explicitly it is the limit after subtracting $2h(1) \log(1/\varepsilon)$ from a symmetric cutoff, plus $c_\mathcal{F}h(1)$, where the unique constant $c_\mathcal{F}$ is fixed by $\mathcal{F}^2(\log|x|)(1) = 0$. This is precisely the normalization in the proof of [21, Theorem 5.8]. The measure in (145) is dx , not d^*x .

8.5. The completed diagonal character.

Definition 8.2. For $h \in C_c^\infty(\mathbb{R}_+^\times)$, the *nuclear cohomological diagonal trace* of the completed correspondence family is

$$(146) \quad I_\Delta^{\text{nuc}}(h) := \chi_{\mathbb{L}_{\text{comp}}}(h).$$

This is the supertrace of the source-defined object (143); it is not defined by the explicit formula.

Theorem 8.3 (Arithmetic nuclear Lefschetz identity). *For every $h \in C_c^\infty(\mathbb{R}_+^\times)$,*

$$(147) \quad \boxed{I_\Delta^{\text{nuc}}(h) = \sum_p \sum_{k \geq 1} \log p (h(p^k) + p^{-k}h(p^{-k})) + W_\infty(h).}$$

The first finite orientation is $\sum \deg_{\det}(\mathbb{L}_n) \mathcal{R}([\Gamma_n])$; the second is its $\mathcal{J}$ -conjugate; the polar object is $\mathbb{C}(x^0) \oplus \mathbb{C}(x^1)$; and the last term is the Fourier–archimedean trace (144). The same character has the spectral expression

$$(148) \quad I_\Delta^{\text{nuc}}(h) = \widehat{h}(0) + \widehat{h}(1) - \sum_\rho m_\rho \widehat{h}(\rho), \quad \widehat{h}(s) = \int_0^\infty h(x) x^s d^*x,$$

where the sum is understood as the nuclear character distribution. No zero in (148) enters any definition.

Proof. The contact computation before completion is (135); it identifies the value of the positive finite trace expression in Meyer’s calculation with (136). Conjugation by $\mathcal{J}$ proves (138). Proposition 5.6 of [21] proves that the induced operators on the two closed quotients (142) are nuclear and that their trace is the difference of the corresponding ambient commutator traces. The third commutator is the Fourier commutator; the calculation in [21, Theorem 5.8 and its proof] identifies its trace with (144). Adding these three proved traces gives (147), with the displayed signs. Compact support handles the finite sums and the Fourier finite part handles the unique archimedean singularity.

For the second computation, Theorem 4.1 and Corollary 4.2 of [21] identify the transpose spectrum of $\mathcal{H}_-^0$, with algebraic multiplicity, with the nontrivial zeros of the completed zeta function. The isomorphism in (142) gives the even eigencharacters 0 and 1. The nuclear trace theorem, applied in Meyer’s Theorem 5.8, therefore gives (148): the even polar characters occur with plus sign and the odd quotient representation with minus sign. The construction order is

$$\{j(\delta_n)\}_{n \geq 1} \rightsquigarrow \zeta = \sum_{n \geq 1} \delta_{n-1}, \quad Z, \quad \text{Poisson summation}, \quad \mathcal{H}_\pm^0, \quad \chi_{\mathbb{L}_{\text{comp}}};$$

only the subsequent spectral description introduces the zeros. □

Corollary 8.4 (Self-dual metric centralization). *For $g \in C_c^\infty(\mathbb{R}_+^\times)$ put $h(x) = x^{-1/2}g(x)$. The finite part of (147) becomes*

$$\sum_p \sum_{k \geq 1} \frac{\Lambda(p^k)}{\sqrt{p^k}} (g(p^k) + g(p^{-k})).$$

For each orientation the coefficient is exactly the unique positive self-dual metric normalization (117) of row (b). No algebraic half-Tate line is used.

Corollary 8.5 (Constructed row (c)). *Row (c) is constructed completely under the following explicit contract: the source is the monoid linearization $\text{Corr}_{\mathbb{C}}^{\text{arith}}(X)$ of the isomorphism classes in $\text{Corr}_{W, DN}^{\text{arith}}(X)$; individual correspondences are realized monoidally in $\text{Mult}(\mathcal{S}_>)$; their contact weights are the determinant degrees of the dynamic complexes $\mathbb{L}_n$; and the completed diagonal trace is the nuclear supercharacter of $\mathbb{L}_{\text{comp}}$ in NRep^{sup} . Within this cohomological/nuclear category, composition, contact, convergence, both finite orientations, the polar object, the archimedean finite part and the two trace expressions are all constructed and proved compatible. This statement neither requires nor asserts a reverse functor to perfect Cartier complexes on $\mathcal{Y}_{\mathbb{S}}$ or an ordinary Chow intersection product there.*

Proof. The monoidal realization and its compatibility with rows (a) and (b) are (124)–(127). The dynamic contact comparison is (135), including its convergence proof. The source-defined summable superobject is (143). Theorem 8.3 computes its character geometrically and spectrally, and Corollary 8.4 proves compatibility with the metric normalization of row (b). These are exactly the clauses of the stated contract. $\square$

9. ROW (D): THE EXACT GEOMETRIC GATE

We now determine whether the numerical ruling signature already proved on $N_{\text{rul}}^1 \simeq \mathbb{R}^2$ can control the nuclear correspondence pairing. It cannot: the two forms live on numerical objects of different rank. This obstruction follows directly from the prime contacts, without using the spectral expression (148) or any zero of ξ .

9.1. The form and the primitive target. Let $\mathcal{T} = C_c^\infty(\mathbb{R}_+^\times)$ and define

$$(149) \quad B_{\text{nuc}}(f, g) = I_{\Delta}^{\text{nuc}}(f \star g^{\vee}).$$

The Tate involution (85) and Theorem 8.3 make this a Hermitian form. Its two-ruling primitive subspace is

$$(150) \quad \mathcal{T}^0 = \{f \in \mathcal{T} : \widehat{f}(0) = \widehat{f}(1) = 0\}.$$

In a genuine intersection category containing ruling classes F_1, F_2 with $F_i^2 = 0$ and $F_1 F_2 = 1$, the corresponding primitive class would be

$$(151) \quad M_f = D_f - \widehat{f}(1)F_1 - \widehat{f}(0)F_2, \quad M_f F_1 = M_f F_2 = 0.$$

This is the specification of a missing comparison, not a claim that D_f is already a divisor-line class on $\mathcal{Y}_{\mathbb{S}}$.

9.2. An infinite-rank obstruction independent of the zeros. Use the logarithmic coordinate $t = \log x$, and let W represent the functional I_{Δ}^{nuc} . By (147), its finite part has nonzero atoms at

$$(152) \quad t = k \log p \quad \text{and} \quad t = -k \log p \quad (p \text{ prime, } k \geq 1).$$

The Fourier–archimedean finite part is smooth away from $t = 0$, so it cannot cancel these atoms.

Theorem 9.1 (Infinite-rank contact obstruction). *The convolution map*

$$T_W : C_c^\infty(\mathbb{R}) \longrightarrow \mathcal{D}'(\mathbb{R}), \quad a \longmapsto W * a,$$

has infinite algebraic rank. Consequently (149) cannot factor through a finite-dimensional vector space, and in particular cannot factor through N_{rul}^1 .

Proof. Suppose the image lies in a finite-dimensional space E . If (η_j) is a compactly supported approximate identity, then $W * \eta_j \rightarrow W$ in $\mathcal{D}'(\mathbb{R})$. Finite-dimensional subspaces of a Hausdorff topological vector space are closed, so $W \in E$. Applying the same argument to translates of η_j shows that all translates of W lie in E . Their span E_W is therefore finite-dimensional and translation invariant.

Difference quotients converge distributionally, hence $D(E_W) \subset E_W$ for $D = d/dt$. Cayley–Hamilton gives a nonzero polynomial P with $P(D)W = 0$. Factoring P over $\mathbb{C}$ and solving the successive equations $(D - \lambda)^m u = 0$ shows that every global distributional solution is an exponential polynomial $e^{\lambda t} Q(t)$, and therefore smooth. This contradicts the nonzero point masses (152). Thus T_W has infinite rank.

If $B_{\text{nuc}}(f, g) = q(Jf, Jg)$ through a finite-dimensional E_0 , then $f \mapsto B_{\text{nuc}}(f, \cdot)$ has rank at most $\dim E_0$. The Tate involution is bijective, so this is the rank of T_W , a contradiction. Equation (77) now gives the last assertion. $\square$

The hyperbolic signature calculation on the ruling plane is exact but cannot be reused as a Hodge theorem for row (d). Forgetting the Witt map, nuclear action, metric frame and derived orbit contact makes all Γ_n the same diagonal, whereas retaining their completed contact form has infinite rank.

Corollary 9.2 (Primitive infinite rank). *The restriction of T_W to $\mathcal{T}^0$ in (150) still has infinite algebraic rank.*

Proof. The two Mellin evaluations defining $\mathcal{T}^0$ give it codimension at most two in $\mathcal{T}$. If the restricted map had finite rank, adjoining an algebraic complement of dimension at most two would make T_W finite rank on all of $\mathcal{T}$, contrary to Theorem 9.1. $\square$

9.3. The finite-support construction and its obstruction. There is a natural attempt to enlarge the ruling plane without using the explicit formula. Let S be a finite set of prime powers and put

$$(153) \quad E_S = \mathbb{R}F_1 \oplus \mathbb{R}F_2 \oplus \bigoplus_{n \in S} \mathbb{R}e_n, \quad H_S = F_1 + F_2,$$

where e_n records the decorated correspondence Γ_n . For $S \subset T$ the labelled correspondences give the evident inclusion $E_S \hookrightarrow E_T$.

The intrinsic periodic ruling form (66) factors through the two degrees and has rank two. Assigning the new symbols zero ruling degree makes all e_n radical; forgetting their decorations makes all Γ_n the same diagonal. Either operation loses the mixed sector.

Composition and reduced contact give another form which does retain that sector:

$$(154) \quad K_S(e_m, e_n) = \deg_{\text{cont}}(\Gamma_m \circ \Gamma_n, \Delta) = \Lambda(mn).$$

This is defined before the nuclear completion, using only Theorems 7.2 and 7.4. Its sign is fixed by $-\log \|\det_{\text{tor}} \mathbb{F}_p\| = \log p > 0$.

Theorem 9.3 (Finite-contact truncation obstruction). *Suppose that S contains powers of r distinct primes. On the correspondence summand of (153), K_S is the orthogonal sum*

$$\bigoplus_{p|S} (\log p) \mathbf{1}_p \mathbf{1}_p^t$$

and has inertia $(r, 0, |S| - r)$. With the ruling plane adjoined, the inertia is $(1 + r, 1, |S| - r)$. Restriction by any two ruling-degree equations leaves at least $r - 2$ positive correspondence directions. The transition inclusions preserve the forms, so the positive index is unbounded on a cofinal family.

Proof. If m, n are powers of the same prime p , then $\Lambda(mn) = \log p$; if they are supported at distinct primes, it is zero. This proves the displayed block decomposition. Each all-one block has one positive eigenvalue and all remaining eigenvalues zero. The ruling matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ adds one positive and one negative eigenvalue. Finally, a positive r -plane meets a subspace of codimension at most two in dimension at least $r - 2$, and remains positive there. Formula (154) is unchanged when S is enlarged. $\square$

Thus the two available finite constructions fail in opposite ways:

$$(155) \quad \text{ruling RR} \implies \text{mixed radical}, \quad \text{contact} \implies \text{one positive line per prime}.$$

Replacing K_S by $-K_S$ would reverse the already fixed torsion determinant and fail comparison with row (c). What is missing is a global Green term which couples distinct primes while preserving their local contact, together with sections, effectivity and a quadratic RR theorem for the e_n . None of those structures is present on (153).

9.4. The Green correction is forced. The finite part of Theorem 8.3 defines, on the test algebra,

$$(156) \quad K(f, g) = W_{\text{fin}}(f \star g^\vee), \quad W_{\text{fin}}(h) = \sum_{p, k \geq 1} \log p (h(p^k) + p^{-k} h(p^{-k})).$$

This is the completed version of the fixed contact determinants. There is no freedom in the complementary term.

Theorem 9.4 (Forced Green correction). *Let Q be a Hermitian form on the test algebra whose finite contact factor is (156) and whose final comparison with row (c) is exact, $Q = B_{\text{nuc}}$. Then*

$$(157) \quad G := Q - K = G_\infty, \quad G_\infty(f, g) = W_\infty(f \star g^\vee).$$

No different Green correction preserves both the dynamic perfect contacts and the nuclear Lefschetz identity.

Proof. The cohomological character identity (147) says $B_{\text{nuc}} = K + G_\infty$. Subtracting the prescribed K from $Q = B_{\text{nuc}}$ gives (157); two such corrections have difference zero. $\square$

The forced term has a genuine, but insufficient, energy property. Put $F(t) = e^{t/2}f(e^t)$. Mellin transformation of the Fourier finite part (144) gives

$$(158) \quad G_\infty(f, f) = \frac{1}{2\pi} \int_{\mathbb{R}} m_\infty(\tau) |\widehat{f}(\tfrac{1}{2} + i\tau)|^2 d\tau, \quad m_\infty(\tau) = \log \pi - \Re \psi(\tfrac{1}{4} + i\tau/2).$$

Here $m_\infty = -2\Re(\Gamma'_\mathbb{R}/\Gamma_\mathbb{R})$ on the critical line. The digamma series gives

$$(159) \quad m_\infty(\tau) - m_\infty(0) = - \sum_{n=0}^{\infty} \frac{(\tau/2)^2}{(n+1/4)((n+1/4)^2 + (\tau/2)^2)} \leq 0,$$

strictly away from zero. On the other hand $m_\infty(0) = \log \pi - \psi(1/4) > 0$, while the asymptotic $\psi(z) = \log z + O(z^{-1})$ gives $m_\infty(\tau) \rightarrow -\infty$. Thus G_∞ is indefinite, although subtracting its maximal constant multiplier gives a negative semidefinite translation energy.

The desired assertion is not the conditional inequality (159), but the sharp domination

$$(160) \quad (K + G_\infty)(f, f) \leq 0 \quad (f \in \mathcal{T}^0).$$

By (162), (160) is equivalent to RH. Theorem 9.4 prevents making the archimedean term more negative without changing row (c). Hence the local energy does not close row (d); a non-circular proof must derive the exact domination from a new mixed Riemann–Roch/effectivity theory.

9.5. The mixed-section forcing theorem. We finally test whether a section functor can be extracted from the objects already present. The classical index argument uses only the following numerical consequences of such a functor. On a mixed divisor space with form q , polarization H and $\deg D = q(D, H)$, suppose there are finite nonnegative, linear-equivalence-invariant quantities h^i such that

- (1) $h^0(0) = 0$ and $h^0(D) > 0$ is equivalent to strict effectivity;
- (2) every nonzero strictly effective class has positive degree;
- (3) $h^2(D) = h^0(-D)$;
- (4) for every finite-support class,

$$(161) \quad h^0(nD) - h^1(nD) + h^2(nD) = \frac{n^2}{2} q(D, D) + O_D(n).$$

These statements must be compatible with finite-support inclusions and the chosen completion. They are weaker than asking for actual section objects.

Theorem 9.5 (Mixed-section forcing). *Every package satisfying the preceding four conditions has*

$$q(D, D) \leq 0 \quad \text{whenever } q(D, H) = 0.$$

Consequently, if its independently constructed form is subsequently identified with B_{nuc} , its existence implies RH.

Proof. If D is primitive and $q(D, D) > 0$, then (161) is positive for all large n . Nonnegativity of h^1 and duality give

$$0 < h^0(nD) - h^1(nD) + h^0(-nD) \leq h^0(nD) + h^0(-nD).$$

Hence nD or $-nD$ is nonzero and strictly effective, and so has positive degree. Both have degree zero, a contradiction. The last assertion is (162). $\square$

The theorem is logically sharp at the scalar level. If RH is assumed, the numerical form is Lorentzian; orient its positive cone by $\deg > 0$, put $\chi(D) = q(D, D)/2$, assign $h_{\text{num}}^0(D) = \chi(D)$ in the future timelike cone, a positive constant on its nonzero lightlike boundary, and zero elsewhere, then set

$$h_{\text{num}}^2(D) = h_{\text{num}}^0(-D), \quad h_{\text{num}}^1 = h_{\text{num}}^0 + h_{\text{num}}^2 - \chi.$$

These scalar functions satisfy the displayed axioms. They are discontinuous at the light cone and are not dimensions of objects; this converse only shows that the numerical requirement is equivalent in strength to RH.

None of the natural categorical candidates supplies the missing functor. The Zeta operator on $\mathcal{S}_>$ is invertible, so its kernel and cokernel vanish. The positive Poisson quotient contains only the two polar characters, while the negative quotient has the zero divisor as transpose spectrum; using its eigenspaces makes effectivity spectral. Finite contact homology gives $\Lambda(n)$ and the multi-positive blocks of Theorem 9.3, not quadratic mixed sections. Finally, a nuclear trace or determinant is a linear or multiplicative spectral invariant, not a nonnegative section dimension with an effective cone; selecting its positive spectral part is precisely the circular step in (162). Thus the Poisson quotient does not currently define h^0 for mixed classes.

9.6. Audit of the two possible Hodge routes. The arithmetic Hodge-index theorem of Faltings–Hriljac–Moriwaki and its adelic extension by Yuan and Zhang [28, Theorems 1.1 and 1.3] apply to Hermitian or integrable adelic line bundles on regular proper arithmetic varieties or normal projective varieties. Their hypotheses include a generic degree-zero condition, a fixed ample or nef-and-big polarization, admissible or integrable metrics, finite arithmetic intersections and boundedness conditions for the equality statement.

Those hypotheses have not been constructed for the mixed classes D_f . The spherical ringed square is not a regular proper arithmetic variety in the sense assumed by those theorems, and a virtual nuclear Fréchet representation is not an adelic line bundle. Moreover the cotangent cohomology of Section 2 is defined on the effective external ruling sector; N_{rul}^1 cannot contain the completed mixed classes by Theorem 9.1; the nuclear character gives a trace but no effective cone or Riemann–Roch theorem for the D_f ; and no comparison sends these classes to arithmetic line bundles while preserving degree, composition, intersection and the radical. Therefore the published arithmetic index theorems cannot yet be applied.

The intrinsic Rosati candidate in the nuclear category is precisely (149). Duality proves Hermitian symmetry but not a sign. Using the separate spectral computation (148), the block calculation of §3.5 gives

$$(162) \quad B_{\text{nuc}}(f, f) \leq 0 \quad (f \in \mathcal{T}^0) \quad \Longleftrightarrow \quad \text{RH.}$$

An on-line zero contributes a negative block and an off-line mirror pair an additional hyperbolic plane. Thus (162) is an acceptance test, not a proof: selecting a negative spectral part assumes the assertion row (d) is meant to prove.

There is a genuine finite analytic result on this side, and it has a long history which we state before our own contribution to it. The localization of the Weil form was introduced by Yoshida [31], who proved that RH is equivalent to positive definiteness of Q_W on $C_c^\infty(-a, a)$ for every $a > 0$, and that this positivity *does* hold for all sufficiently small a [31, Lemma 2]. Both the localized criterion and the existence of an initial positive window are therefore due to him. Suzuki [25, Theorems 1.1–1.4] realizes each localized Weil form by the Friedrichs extension of a screw-kernel operator, proves continuity of its lowest eigenvalue, and reproves positivity, simplicity and evenness for sufficiently small windows. Positivity or nondegeneracy for every window is equivalent to RH. This does not supply the missing geometric theorem: the localized

operator is constructed from the Weil quadratic form itself, so it is a truncation of row (c), not an RR/effectivity construction independent of it. The exact analytic gate is extension from the proven small-window range to all windows without importing the spectral sign.

9.7. The primitive pullback, a certified initial interval, and the next endpoint audit.

We record the part of the analytic gate which can be closed without a global sign assumption. In logarithmic coordinates put $I_T = (-T, T)$, extend functions by zero outside I_T , and write

$$M_- F = \int_{-T}^T F(t) e^{-t/2} dt, \quad M_+ F = \int_{-T}^T F(t) e^{t/2} dt.$$

These are exactly the pullbacks of the two polar Mellin evaluations in (150): indeed $\hat{f}(s) = \int_{-T}^T F(t) e^{(s-1/2)t} dt$, so evaluation at $s = 0, 1$ gives M_-, M_+ . Let $\mathcal{P}_T = \ker M_- \cap \ker M_+$. Under $F(t) = e^{t/2} f(e^t)$, the Friedrichs operator representing $-B_{\text{nuc}}$ on compact support has the form

$$(163) \quad A_T = G_{\Gamma, T} - m_0 I - \sum_{2 \leq n < e^{2T}} \frac{\Lambda(n)}{\sqrt{n}} (S_{\log n} + S_{-\log n}), \quad m_0 = \log \pi + \gamma + \frac{\pi}{2} + 3 \log 2,$$

where $S_a F(t) = \tilde{F}(t + a)$ and $G_{\Gamma, T}$ is the zero-extension compression of the complete digamma/Gamma multiplier

$$g_{\Gamma}(\tau) = \operatorname{Re} \psi \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \psi \left(\frac{1}{4} \right).$$

The strict inequality in the sum is harmless at a threshold because the new translated supports meet in a null set. Formula (163) contains every prime power acting on the support, with the determinant and metric coefficient already fixed in (117); no prime-only truncation is made.

We shall use the following elementary operator form of Feshbach reduction.

Lemma 9.6 (Directed Feshbach criterion). *Let P be finite rank, $Q = 1 - P$, and let $A = A^*$. If*

$$(164) \quad Q A Q \geq \delta Q, \quad (Q A P)^*(Q A P) \leq H, \quad P A P - \delta^{-1} H > 0,$$

then $A > 0$.

Proof. For $x \in P\mathcal{H}$ and $y \in Q\mathcal{H}$, the complement bound and Cauchy's inequality give

$$2 \operatorname{Re} \langle Q A P x, y \rangle + \delta \|y\|^2 \geq -\delta^{-1} \|Q A P x\|^2.$$

Consequently

$$\langle A(x + y), x + y \rangle \geq \langle (P A P - \delta^{-1} H)x, x \rangle.$$

This is strictly positive when $x \neq 0$; when $x = 0$ it is at least $\delta \|y\|^2$, and is therefore strictly positive unless $y = 0$. $\square$

Theorem 9.7 (Certified primitive initial interval). *The complete operator (163) is strictly positive on $\mathcal{P}_T$ for every $0 < T \leq \frac{1}{2} \log 3$, and also at*

$$T = \log 2.$$

These are full Hilbert-space assertions, not signs of finite Galerkin matrices.

Proof. We give the enclosures and reduction which make the computer-assisted part reproducible. All elementary constants, matrix entries, quadrature nodes and remainders are evaluated as outward-rounded Arb intervals; floating point is used only to choose bases and fixed congruence matrices.

At $T = \frac{1}{2} \log 3$, the only non-null finite contact is the shift $\log 2$. A shift-compatible three-piece mesh with local Legendre degree nine has dimension 760. The first 80 Gamma-resolvent energies are assembled exactly from their exponential ODE, and the omitted positive tail is bounded below by its first Robin term. The even and odd Robin contributions are, respectively, greater than 0.0011586663336 and 0.0006199539287. Directed congruence gives a low-block lower bound -0.00050 , while exact local Grams and Taylor–Young off-cell remainders bound the Schur loss by $3.961 \cdot 10^{-6}$. Hence the complete even and odd margins are

$$0.0006547054607 \quad \text{and} \quad 0.0001159930558.$$

Lemma 9.6 proves the full-space endpoint. Extension by zero, proved after the endpoint calculations below, then proves every $0 < T \leq \frac{1}{2} \log 3$.

Remark 9.8 (What is, and what is not, new in Theorem 9.7). That *some* initial window is positive is not new: it is [31, Lemma 2], and it is reproved unconditionally, with an asymptotic expansion of the lowest eigenvalue as $a \rightarrow 0^+$, in [25, Theorem 1.4]. Theorem 9.7 is a *quantitative* refinement of that qualitative statement: it fixes the explicit endpoint $T = \log 2$, it is a full Hilbert-space assertion rather than a sign of finite Galerkin matrices, and every constant is an outward-rounded interval enclosure. The distinction matters because the open part of row (d) is the passage to *all* windows, where no quantitative initial range, however large, is by itself evidence.

At $T = \log 2$, split the interval at every translate by $\log 2$ and $\log 3$, and use local Legendre polynomials of degrees at most nine. The first 160 positive Gamma resolvents are assembled from their exact semiseparable exponential kernels. On one cell the resolvent is obtained from

$$(b^2 - \partial_t^2)u = 2bf, \quad u'(l) = bu(l), \quad u'(r) = -bu(r),$$

whose polynomial particular solution is the finite sum $2b^{-1} \sum_{j \geq 0} b^{-2j} f^{(2j)}$. The omitted positive Gamma tail is bounded below by its first Robin resolvent. The directed enclosures are

quantity	certified bound
$QA_T Q$	$> 1.5396358725 Q$
$\ QA_T P\ ^2$	$< 3.65731342 \cdot 10^{-4}$
even high Schur gap	> 0.4659
odd complement gap	> 0.0497
$\langle A_T v_2, v_2 \rangle$	> 0.0020336972824697
$\langle A_T v_o, v_o \rangle$	$> 8.13856175479 \cdot 10^{-6}$.

The remaining even direction has Rayleigh value greater than $-1.746475 \cdot 10^{-6}$, squared residual below $4.441 \cdot 10^{-24}$, and complement gap 0.0015. Its shorted positive Gamma-tail capacity is enclosed below by 1.27084620358308, whereas the adverse threshold is below 1.2111. A final application of Lemma 9.6 proves both parity restrictions positive. This completes the proof of the theorem. $\square$

The audited next endpoint. At $T = \frac{1}{2} \log 5$, use the normalized Legendre space V_{200} , with the two equations $M_- = M_+ = 0$ imposed by an interval two-by-two graph solve. A concentration estimate for the Fourier band $[-150, 150]$, its positive Gamma multiplier gap, and the exact Bessel tail of the two Tate vectors give

$$(165) \quad QA_T Q > 0.218Q, \quad Q = 1 - P_{V_{200}}.$$

Inside V_{200} perform the exact nested decomposition

$$V_{200} = D_5 \oplus S_{163} \oplus Y_{30}.$$

Interval Schur complements give

block	least centre eigenvalue	directed lower enclosure
Y_{30}	3.27356009	$0.9999999999999995 - 7.08 \cdot 10^{-17}$
S_{163} Schur	0.05844291	$1 - 8.08 \cdot 10^{-14}$
D_5 Schur	$3.11556559 \cdot 10^{-12}$	$1 - 1.72 \cdot 10^{-4}$.

It remains to control the coupling of the last five-dimensional graph to the *whole* complement. For every polynomial graph column the exact endpoint identity is

$$(166) \quad A_T F(t) = -\frac{1}{2} F(t) \log(T^2 - t^2) + U_F(t),$$

where U_F is analytic on each of the seven contact cells. It follows from

$$q(z) = \frac{ze^{z/2}}{2 \sinh z}, \quad R_1(z) = \operatorname{atanh}(e^{-z/2}) + \arctan(e^{-z/2}) + \frac{1}{2} \log z,$$

and $R'_1 = -q(z)/z + 1/(2z)$. The Taylor radius is $\pi > \log 5$. On $|z| = R = 2.5$, the Bernoulli generating function gives

$$M_R \leq \frac{e^{3R/2}}{2} \left(1 + R + 2\zeta(2) \frac{(R/\pi)^2}{1 - (R/\pi)^2} \right), \quad \|q - q_{<M}\|_{[0, \log 5]} \leq \frac{M_R (\log 5/R)^M}{1 - \log 5/R}.$$

We take $M = 140$. Every remaining integral is then a finite polynomial combination of

$$\begin{aligned} \int_0^1 x^m \log x \, dx &= -a^{-2}, & \int_0^1 x^m \log(1-x) \, dx &= -H_a/a, \\ \int_0^1 x^m \log^2 x \, dx &= 2a^{-3}, & \int_0^1 x^m \log^2(1-x) \, dx &= (H_a^2 + H_a^{(2)})/a, \\ \int_0^1 x^m \log x \log(1-x) \, dx &= H_a/a^2 - (\zeta(2) - H_a^{(2)})/a, \end{aligned}$$

where $a = m + 1$. Thus no numerical quadrature occurs in the contracted continuum Gram H_X . The five analytic-tail errors lie between $2.14 \cdot 10^{-22}$ and $5.54 \cdot 10^{-21}$; propagation of the serialized graph-coefficient balls in the stable Legendre basis gives the action errors

$$(1.6086834, 1.9477040, 2.0560371, 3.3822491, 5.0816822) \times 10^{-8}.$$

Finally the centre eigenvalues of $K_{\text{final}} - 0.218^{-1} H_X$ are

$$2.78052720 \cdot 10^{-12}, \quad 5.47152465 \cdot 10^{-10}, \quad 2.87827286 \cdot 10^{-7}, \quad 2.78898083 \cdot 10^{-5}, \quad 1.99028729 \cdot 10^{-3}.$$

They are diagnostics only. A frozen Cholesky congruence followed by Arb Gershgorin gives the rigorous lower endpoints

$$(167) \quad 0.9497247499, \quad 0.9891485575, \quad 0.9741473730, \quad 0.9360164888, \quad 0.7940881021.$$

The last inference in the former version of this calculation was not valid. The projection in (165) is the complement of the *whole* space V_{200} , whereas the five-column Gram above is formed only after eliminating two finite safe blocks. Relative to the actual orthogonal decomposition

$$\mathcal{P}_T = D \oplus S \oplus Q,$$

the missing terms are $A_S Q$ and the corrected cross

$$C_D = A_{DQ} - A_{DS} A_{SS}^{-1} A_{SQ}.$$

Writing $\kappa = \|A_{QS} A_{SS}^{-1/2}\|^2$, the valid sufficient test is

$$\kappa < 0.218, \quad K_D - (0.218 - \kappa)^{-1} C_D C_D^* > 0.$$

Neither inequality follows from the five-column calculation. Thus (165), the finite Schur enclosures and (167) remain rigorous separate statements, but they do not yet prove positivity of the complete primitive space at this endpoint.

The proof also supplies an exact nesting principle. If $T < T'$, extension by zero sends $\mathcal{P}_T$ isometrically into $\mathcal{P}_{T'}$: both Tate integrals are unchanged. The global distributions in (163) give the same quadratic value, because translations longer than the smaller support have zero correlation. Consequently positivity at a right endpoint proves every smaller window to which it applies. This observation does not turn the two endpoints of Theorem 9.7 into an unbounded cofinal sequence.

Corollary 9.9 (Certified initial interval). *The complete primitive operator is strictly positive on $\mathcal{P}_T$ for every*

$$0 < T \leq \log 2.$$

Proof. Apply the nesting principle with $T' = \log 2$ and use Theorem 9.7 at that endpoint. $\square$

9.8. The balanced channel factorization. Everything below — the threshold capacity, the reference inverse, and the one-step criterion — is expressed through a factorization of A_T into a positive reference minus a positive load. That factorization is not a modelling choice: it is forced by (163), and it is unique once the codomain convention of Definition 9.10 is fixed. We record it here so that the columns used later are exhibited rather than posited.

Definition 9.10 (Zero-extension channels). For $a > 0$ let

$$\hat{J}_{a,\pm} : L^2(I_T) \longrightarrow L^2(\mathbb{R}), \quad \hat{J}_{a,\pm} F = \frac{\tilde{F}(\cdot + a) \pm \tilde{F}}{\sqrt{2}}.$$

The codomain is $L^2(\mathbb{R})$ and not $L^2(I_T)$. This is load-bearing, not cosmetic; Remark 9.12 shows what breaks under the truncated reading.

Lemma 9.11 (Channel resolution of a compressed shift). *On $L^2(I_T)$,*

$$\hat{J}_{a,\pm}^* \hat{J}_{a,\pm} = I \pm \operatorname{Re} S_a,$$

and consequently

$$(168) \quad S_a + S_{-a} = \hat{J}_{a,+}^* \hat{J}_{a,+} - \hat{J}_{a,-}^* \hat{J}_{a,-}.$$

Proof. Translation is an isometry of $L^2(\mathbb{R})$, so $\|\tilde{F}(\cdot + a)\|_{L^2(\mathbb{R})} = \|F\|$. Since $\tilde{F}$ vanishes off I_T ,

$$\langle \tilde{F}(\cdot + a), \tilde{F} \rangle_{L^2(\mathbb{R})} = \int_{I_T} \tilde{F}(t + a) \overline{\tilde{F}(t)} dt = \langle S_a F, F \rangle_{L^2(I_T)}.$$

Hence $\|\hat{J}_{a,\pm} F\|_{L^2(\mathbb{R})}^2 = \|F\|^2 \pm \operatorname{Re} \langle S_a F, F \rangle$. Subtracting the two identities and using $S_a^* = S_{-a}$ gives (168). $\square$

Remark 9.12 (Why the codomain matters). The chain vertex at $t = r - a$ falls outside I_T , but $\hat{J}_{a,\pm} F$ does *not* vanish there: it equals $\pm \tilde{F}(r)/\sqrt{2}$. Discarding that cell replaces $\hat{J}^* \hat{J}$ by $\frac{1}{2}(I - U^*)(I - U) = I - \operatorname{Re} U - \frac{1}{2}P_{e_0}$ in the discrete chain model, which is not of the form (diagonal) minus (compressed shift) and therefore does not reproduce the shift term of A_T . With Definition 9.10 a chain of m vertices has $m + 1$ edges, Dirichlet walls at both ends, and least eigenvalue $1 - \cos \frac{\pi}{m+1}$ — which is the constant appearing in Proposition 9.18. Under the truncated reading that constant would instead be $1 - \cos \frac{\pi}{2m+1}$.

Theorem 9.13 (Balanced channel factorization). *On $L^2(I_T)$, with $w_n = \Lambda(n)/\sqrt{n}$,*

$$(169) \quad \begin{aligned} R_T &= G_{\Gamma,T} + \sum_{2 \leq n < e^{2T}} w_n \hat{J}_{n,-}^* \hat{J}_{n,-}, \\ A_T &= R_T - L_T, \\ L_T &= m_0 I + \sum_{2 \leq n < e^{2T}} w_n \hat{J}_{n,+}^* \hat{J}_{n,+}, \end{aligned}$$

where $\hat{J}_{n,\pm} = \hat{J}_{\log n, \pm}$. Both summands are positive, and they factor as $R_T = X_T^* X_T$, $L_T = Y_T^* Y_T$ with

$$(170) \quad X_T F = \left(G_{\Gamma,T}^{1/2} F, (\sqrt{w_n} \hat{J}_{n,-} F)_n \right), \quad Y_T F = \left(\sqrt{m_0} F, (\sqrt{w_n} \hat{J}_{n,+} F)_n \right),$$

as operators $L^2(I_T) \rightarrow L^2(I_T) \oplus \bigoplus_n L^2(\mathbb{R})$.

Proof. Lemma 9.11 gives $-w_n(S_{\log n} + S_{-\log n}) = w_n \hat{J}_{n,-}^* \hat{J}_{n,-} - w_n \hat{J}_{n,+}^* \hat{J}_{n,+}$. Summing over n and adjoining $G_{\Gamma,T} - m_0 I$ yields (169). For positivity of $G_{\Gamma,T}$, note that $g_{\Gamma}(\tau) = -(m_{\infty}(\tau) - m_{\infty}(0))$ with m_{∞} as in (158); so (159) says exactly $g_{\Gamma} \geq 0$, strictly away from $\tau = 0$. Since $\langle G_{\Gamma,T} F, F \rangle = \frac{1}{2\pi} \int g_{\Gamma} |\hat{F}|^2$, the compression is positive. The remaining summands are positive by construction. $\square$

Corollary 9.14. *The positive reference of Proposition 9.18 is $\widehat{\mathcal{R}}_T = R_T$, and its leading term is*

$$\mathcal{H}_{5/4} = G_{\Gamma,T}.$$

Moreover the coefficient there is $\Lambda(n)/\sqrt{n}$, and not $2\Lambda(n)/\sqrt{n}$, precisely because (168) splits the compressed shift evenly between the two signs.

Remark 9.15. The factorization is unique given the codomain convention: any other assignment of the term $\text{Re } S_a$ between the two signs breaks the equality of boundary energy received by the two channels under zero extension, which is what makes both R_T and L_T positive simultaneously.

Two consequences are used repeatedly below. First, the old-core hypothesis $0 \leq T_0 \leq I$ of §9.9, with $T_0 = R_0^{\dagger/2} L_0 R_0^{\dagger/2}$, is *exactly* the statement $A_{\text{old}} \geq 0$, since $A_{\text{old}} = R_0^{1/2} (I - T_0) R_0^{1/2}$. Second, (170) identifies the two feature maps concretely: the entries of X_T carry the complete Gamma channel and every antisymmetric contact $\hat{J}_{p^k,-}$, while the entries of Y_T carry the constant m_0 and every symmetric contact $\hat{J}_{p^k,+}$.

9.9. Threshold capacity and exact return reduction. Put $T_N = \frac{1}{2} \log N$ and $w_N = \Lambda(N)/\sqrt{N}$. At the instant a prime-power contact is born, its translated supports have null intersection. Hence

$$(171) \quad H_{N,T_N} = H_{N-1,T_N}.$$

This is worth stating in its strong form, because it is what prevents the induction from accumulating error between steps, and because it identifies the newborn directions exactly. Let $2 = q_1 < q_2 < q_3 < \dots$ be the ordered sequence of prime powers and $\tau_j = \frac{1}{2} \log q_j$; on (τ_j, τ_{j+1}) no contact is born, so by the nesting principle it suffices to prove positivity at the right endpoint of each interval, and the step is $\mathcal{P}_{\tau_j} \rightarrow \mathcal{P}_{\tau_{j+1}}$.

Theorem 9.16 (The step is exact). *Let $\mathcal{E} : \mathcal{P}_{\tau_j} \rightarrow \mathcal{P}_{\tau_{j+1}}$ be zero extension and $\mathcal{C} = \mathcal{E}(\mathcal{P}_{\tau_j})$. Then:*

- (1) *the contact list is constant, equal to $\{q_1, \dots, q_j\}$, on all of $(\tau_j, \tau_{j+1}]$;*

- (2) the birth is null on the core: *the incoming contact q_j has shift length $\log q_j = 2\tau_j$, exactly the width of the old window, so for F supported in I_{τ_j} the supports of $\tilde{F}(\cdot + \log q_j)$ and of F are disjoint and $\langle (S_{\log q_j} + S_{-\log q_j})F, F \rangle = 0$;*
- (3) *the Gamma channel is window-independent on zero-extended functions, since $\langle G_{\Gamma,T}F, F \rangle = \frac{1}{2\pi} \int g_{\Gamma} |\hat{F}|^2$ does not mention T ;*
- (4) *consequently $\mathcal{E}^* A_{\tau_{j+1}} \mathcal{E} = A_{\tau_j}$ exactly: the old-core block receives no correction from the newborn contact;*
- (5) $\mathcal{A} = \mathcal{C}^\perp$ *inside $\mathcal{P}_{\tau_{j+1}}$ decomposes as*

$$\mathcal{A} = (\text{primitives supported in the annulus}) \oplus \text{span}\{e^{-t/2}, e^{t/2}\}|_{I_{\tau_j}},$$

that is, the annulus plus exactly two directions, and those two are the Tate modes restricted to the core.

Proof. (1) The only $n \in [q_j, q_{j+1})$ with $\Lambda(n) \neq 0$ is q_j . (2) The shift $a = 2\tau_j$ maps $I_{\tau_j} = (-\tau_j, \tau_j)$ onto $(-3\tau_j, -\tau_j)$, disjoint from I_{τ_j} but for one point. (3) is the definition of the multiplier compression. (4) Subtracting the two forms, the difference is $-w_{q_j} \langle (S_{\log q_j} + S_{-\log q_j})F, F \rangle$, which vanishes by (2). (5) $F \in \mathcal{A}$ iff $F|_{I_{\tau_j}} \perp \mathcal{P}_{\tau_j}$ in $L^2(I_{\tau_j})$; since $\mathcal{P}_{\tau_j}$ is the kernel of two functionals, its orthogonal complement in $L^2(I_{\tau_j})$ is the span of the two Tate vectors restricted there. $\square$

Part (5) is used below: the two neutral Tate channels carried through the threshold are not a design choice, they are literally the two dimensions by which the enlarged primitive space exceeds the annulus.

At the end of the cell split the enlarged primitive space into the transported old core and the born two-sided annulus. If

$$H_{N,T_{N+1}} = \begin{pmatrix} A_N & B_N \\ B_N^* & D_N \end{pmatrix},$$

then exact block elimination gives

$$(172) \quad H_{N,T_{N+1}} \geq 0 \iff \text{Ran } B_N \subseteq \text{Ran } A_N^{1/2}, \quad D_N - B_N^* A_N^\dagger B_N \geq 0.$$

Thus global row (d) is equivalent to a cofinal family of source-defined shorted-capacity inequalities, together with the initial positive window.

There is a useful exact reduction of the inverse appearing in (172). Factor the old positive reference and its opposite channel as $R_0 = X_0^* X_0$, $L_0 = Y_0^* Y_0$, and put

$$T_0 = R_0^{\dagger/2} L_0 R_0^{\dagger/2}, \quad D_0 = I - T_0.$$

For the harmonic lift $H = R_0^\dagger X_0^* X_E$ and born reference S_E , introduce the two operators from the new cell to the normalized old space

$$H_c = R_0^{1/2} H S_E^{\dagger/2}, \quad Q_c = R_0^{\dagger/2} (X_0^* X_E - Y_0^* Y_E) S_E^{\dagger/2}.$$

For a fixed vector e in the supported part of the new cell, set $h_e = H_c e$, $q_e = Q_c e$. Direct block algebra gives the normalized return load

$$(173) \quad u_e = D_0 h_e - q_e.$$

Under the old-core hypothesis $0 \leq T_0 \leq I$, its scalar return moments $c_k(e) = \langle u_e, T_0^{k-1} u_e \rangle$ satisfy

$$(174) \quad c_k(e) - c_{k+1}(e) = \langle u_e, T_0^{k-1} D_0 u_e \rangle \geq 0.$$

If, in addition, $u_e \in \text{Ran } D_0^{1/2}$, the spectral theorem gives the exact sum

$$(175) \quad \sum_{k \geq 1} c_k(e) = \|D_0^{\dagger/2}(D_0 h_e - q_e)\|^2.$$

Thus the harmonic contribution telescopes; the only nontrivial inverse datum is the quadratic form $\langle q_e, D_0^{\dagger} q_e \rangle$, with its explicit cross term. Because all operators are first compressed by the two-moment projection, every returned vector remains in $\mathcal{P}_T$. Put $F_{k,e} = R_0^{\dagger/2} T_0^{k/2} q_e$. The defining balanced factorization then identifies each scalar dissipation as

$$(176) \quad \langle q_e, T_0^k q_e \rangle - \langle q_e, T_0^{k+1} q_e \rangle = -B_{\text{nuc}}(F_{k,e}, F_{k,e}).$$

It retains the complete Gamma place and every p^j , after the continuous Chebyshev term has been cancelled by the two exact Tate moments. Equation (173) and the telescoping identity (174) are unconditional algebraic identities under the old-core hypothesis. The range condition needed for (175) is precisely part of the still-open capacity estimate, so this reduction does not assume the missing enlarged-cell sign.

The return sum is not an auxiliary quantity: it *is* the Schur penalty, written in normalized coordinates.

Proposition 9.17 (The return load is the opposite residual). *Let $\ell = Y_0^* Y_E$ and $b_E = L_0 H - \ell$ be the opposite residual of (265) below, and $A_0 = R_0 - L_0$ the old primitive block. The normalized return load (173) satisfies*

$$(177) \quad u_e = -R_0^{\dagger/2} b_E S_E^{\dagger/2} e,$$

and consequently, whenever the range condition of (175) holds,

$$(178) \quad \sum_{k \geq 1} c_k(e) = \langle (b_E^* A_0^{\dagger} b_E) S_E^{\dagger/2} e, S_E^{\dagger/2} e \rangle.$$

Proof. On the supported range $R_0^{\dagger/2} R_0^{1/2}$ is the identity and $R_0^{1/2} H = R_0^{\dagger/2} R_0^{\dagger} r = R_0^{\dagger/2} r$. Hence

$$D_0 h_e = (I - T_0) R_0^{1/2} H S_E^{\dagger/2} e = R_0^{\dagger/2} (r - L_0 H) S_E^{\dagger/2} e,$$

while $q_e = R_0^{\dagger/2} (r - \ell) S_E^{\dagger/2} e$. Subtracting gives (177). For the second claim, $A_0 = R_0^{1/2} D_0 R_0^{1/2}$ gives $A_0^{\dagger} = R_0^{\dagger/2} D_0^{\dagger} R_0^{\dagger/2}$ on the supported range, so $\|D_0^{\dagger/2} u_e\|^2 = \langle D_0^{\dagger} u_e, u_e \rangle$ is (178) after substituting (177). $\square$

Thus (175) and (267) are two writings of one object, and the telescoping (174) is a convergent expansion of the Schur penalty rather than a separate estimate. The factorization $A_0^{\dagger} = R_0^{\dagger/2} D_0^{\dagger} R_0^{\dagger/2}$ used in the proof requires the supported-range hypothesis and is not automatic when $\text{Ran } R_0^{1/2}$ fails to contain the relevant vectors; that is the same hypothesis already flagged for (175).

Finally, the positive reference itself has a uniform exact coercivity which does not solve the signed problem. With the zero-extension channels of Definition 9.10, the reference is the operator R_T of Theorem 9.13,

$$\widehat{R}_T = R_T = G_{\Gamma, T} + \sum_{n \leq N} \frac{\Lambda(n)}{\sqrt{n}} \widehat{J}_{n,-}^* \widehat{J}_{n,-}, \quad N = \lfloor e^{2T} \rfloor,$$

by Corollary 9.14.

Proposition 9.18 (Shift-chain coercivity). *One has*

$$(179) \quad \widehat{\mathcal{R}}_T \geq \alpha_N I, \quad \alpha_N = \sum_{n \leq N} \frac{\Lambda(n)}{\sqrt{n}} \left(1 - \cos \frac{\pi}{\lceil 2T/\log n \rceil + 1} \right),$$

and

$$(180) \quad \alpha_N \geq \frac{2}{9(2T)^2} \sum_{n \leq N} \frac{\Lambda(n)(\log n)^2}{\sqrt{n}} \geq \left(\frac{4}{9} + o(1) \right) \sqrt{N}, \quad \alpha_N \sim \sqrt{N}.$$

The same bound holds after restriction to $\mathcal{P}_T$.

Proof. Disintegrate $(0, 2T)$ into chains $r + j \log n$. Each chain has at most $m_n = \lceil 2T/\log n \rceil$ vertices. The Hermitian part of its nilpotent unilateral shift has norm $\cos(\pi/(m_n + 1))$, proving (179) after summing and discarding the nonnegative Gamma form. Since $m_n + 1 \leq 3(2T)/\log n$ and $1 - \cos x \geq 2x^2/\pi^2$ on $[0, \pi/2]$, the first inequality in (180) follows. Keeping only primes, applying the prime number theorem to $\int_2^N (\log x)^2 x^{-1/2} d\vartheta(x)$, and using $2T = \log N + O(N^{-1})$ gives the last asymptotic. Restriction cannot weaken a quadratic lower bound. For the sharp equivalent, fix $K > 0$. On $Ne^{-K} \leq n < N$, and for sufficiently large N , the shift chain has two vertices and its bracket in (179) is exactly $1 - \cos(\pi/3) = 1/2$. The prime number theorem and partial summation give $\sum_{n \leq x} \Lambda(n)n^{-1/2} = 2\sqrt{x} + o(\sqrt{x})$, hence

$$\liminf_{N \rightarrow \infty} \frac{\alpha_N}{\sqrt{N}} \geq 1 - e^{-K/2}.$$

Letting $K \rightarrow \infty$ gives the lower limit one. Apart from a possible atom at $n = N$, of size $o(\sqrt{N})$, every bracket is at most $1/2$. The same prime-number-theorem estimate gives the matching upper limit. $\square$

The coercivity (180) invites an obvious strategy: compare the two positive halves of (169) as scalars, hoping that $R_T \geq \alpha_N I$ dominates a norm bound on L_T . That strategy is closed, and closed for a structural reason rather than for want of a sharper constant.

Proposition 9.19 (The scalar comparison fails by exactly four). *Put $N = e^{2T}$ and $\sigma_N = \sum_{2 \leq n < N} \Lambda(n)/\sqrt{n}$. Then*

$$L_T \leq (m_0 + 2\sigma_N)I, \quad R_T \geq \alpha_N I, \quad \sigma_N \sim 2\sqrt{N}, \quad \alpha_N \sim \sqrt{N},$$

so the scalar sandwich $\alpha_N \geq m_0 + 2\sigma_N$ fails, with

$$(181) \quad \lim_{N \rightarrow \infty} \frac{\alpha_N}{m_0 + 2\sigma_N} = \frac{1}{4}.$$

Proof. Each $\widehat{J}_{n,+}^* \widehat{J}_{n,+} = I + \operatorname{Re} S_n$ has norm at most 2, giving the bound on L_T . The bound on R_T is (179). The asymptotic $\sigma_N \sim 2\sqrt{N}$ is the prime number theorem with partial summation, and $\alpha_N \sim \sqrt{N}$ is (180). Dividing gives (181). $\square$

Corollary 9.20 (Unconditional no-go for the scalar route). *No improvement of the shift-chain constant can close the gap. Indeed Proposition 9.18 proves that $\alpha_N \sim \sqrt{N}$ is sharp, the upper and lower limits coinciding, so the deficit $m_0 + 2\sigma_N - \alpha_N \sim 3\sqrt{N}$ is of leading order and not a margin. Consequently the deficit must be paid by $G_{\Gamma,T}$ together with the non-scalar structure of the Schur complement, and by nothing else.*

Numerically, with $m_0 = 5.3721834192\dots$ and α_N, σ_N evaluated exactly:

N	σ_N	α_N	$m_0 + 2\sigma_N$	ratio
10	3.5375	1.4859	12.4472	0.11938
200	26.1663	11.7300	57.7048	0.20328
5000	138.7319	65.5772	282.8359	0.23186
100000	629.9414	306.4512	1265.2549	0.24221

The ratio approaches $\frac{1}{4}$ from below, as (181) requires. Each tabulated N has $\Lambda(N) = 0$, so the value of α_N does not depend on whether the birth term $n = N$ is included; at a threshold that term carries zero quadratic form in any case, by Theorem 9.16(2) below.

Corollary 9.20 is the reason the argument below is organized around a normalized transfer operator rather than around a resource inequality between norms: any route which compares R_T and L_T by size alone has already lost a factor of four before the arithmetic begins.

The arithmetic part of the return series also admits a uniform all-depth bound. Write $\Lambda_k = \Lambda^{*k}$ for Dirichlet convolution and set

$$V_{N,k} = \sum_{m \leq N} \frac{\Lambda_k(m)^2}{m}, \quad H_{N,k} = \frac{\Lambda_{2k}(N)}{\sqrt{N}}.$$

Proposition 9.21 (Uniform Witt-moment summability). *For every $\varepsilon > 0$ there are $\delta > 0$ and $C_\varepsilon < \infty$ such that, uniformly for $1 \leq k \leq \delta \log N$,*

$$(182) \quad V_{N,k} \leq C_\varepsilon \sqrt{k} e^{\varepsilon k} \frac{k!}{(2k)!} (\log N)^{2k}.$$

For every fixed $c > 0$,

$$(183) \quad \sum_{k \geq \delta \log N} \frac{V_{N,k}}{(c \log N)^{2k}} = o(1), \quad \sup_N \sum_{k \geq 1} \frac{V_{N,k} + H_{N,k}}{(c \log N)^{2k}} < \infty.$$

Thus every arithmetic Witt depth is uniformly summable. These estimates do not assert that the total capacity is at most the unit Schur budget.

Proof. Let $(z_p)_p$ be independent Haar variables on the unit circle and, for $\sigma > 0$, put

$$g_\sigma(z) = \sum_p (\log p) \sum_{j \geq 1} p^{-j(1/2+\sigma)} z_p^j.$$

The coefficient of the monomial indexed by m in g_σ^k is $\Lambda_k(m) m^{-1/2-\sigma}$. Bohr orthogonality and Rankin's inequality therefore give the exact identity and bound

$$(184) \quad \|g_\sigma\|_{2k}^{2k} = \sum_{m \geq 1} \frac{\Lambda_k(m)^2}{m^{1+2\sigma}}, \quad V_{N,k} \leq N^{2\sigma} \|g_\sigma\|_{2k}^{2k}.$$

Split $g_\sigma = P_\sigma + Q_\sigma$ into the terms with $j = 1$ and $j \geq 2$. If $\mathcal{V}(\sigma) = \sum_p (\log p)^2 p^{-1-2\sigma}$, expansion of the Steinhaus moment gives

$$\|P_\sigma\|_{2k}^{2k} \leq k! \mathcal{V}(\sigma)^k, \quad \mathcal{V}(\sigma) = \frac{1 + o(1)}{4\sigma^2}.$$

For the proper-power part,

$$|Q_{\sigma,p}| \leq \frac{(\log p)p^{-1}}{1 - p^{-1/2}}, \quad \sum_p \left(\frac{(\log p)p^{-1}}{1 - p^{-1/2}} \right)^2 < \infty.$$

Hoeffding's lemma applied separately to real and imaginary parts hence implies $\|Q_\sigma\|_{2k} \leq C\sqrt{k}$, uniformly in σ . Minkowski's inequality consequently yields, for small σ ,

$$(185) \quad \|g_\sigma\|_{2k}^{2k} \leq k! \mathcal{V}(\sigma)^k (1 + C_1 \sigma)^{2k}.$$

Take $\sigma = k/\log N$. The prime number theorem estimate for $\mathcal{V}$, (184), (185), and Stirling's inequalities prove (182), after choosing δ in terms of ε . For $k \geq \delta \log N$, instead fix a small $\sigma_0 > 0$. The same moment argument gives

$$\frac{V_{N,k}}{(c \log N)^{2k}} \leq \exp(2\sigma_0 \log N - k \log \log N + O_{c,\delta,\sigma_0}(k)),$$

uniformly up to $k \leq \log N / \log 2$; beyond that range $\Lambda_k(m) = 0$ for $m \leq N$. Summing proves the first part of (183). Finally $0 \leq H_{N,k} \leq V_{N,k}$, since $H_{N,k}$ is the off-diagonal entry of the positive two-sided word Gram whose two diagonal entries equal $V_{N,k}$. The small-depth majorant is summable because $k!/(2k)!$ beats every exponential; this proves the last assertion. $\square$

Both signs in the balanced factorization receive the same zero-extension boundary energy. Proposition 9.18 therefore controls the reference inverse but does not prove (160). The remaining global theorem is the uniform nonnegativity of (172), equivalently a Douglas factorization of the actual centered atomic-continuous cross Q_c through $D_0^{1/2}$ with the required *unit* shorted-capacity bound. Proposition 9.21 proves summability of every purely arithmetic return depth, but neither identifies this centered Douglas factor nor pays its total unit budget. This is the precise point at which the local-analysis argument stops; the Gamma-Fourier-Tate source construction below begins from this capacity defect.

Theorem 9.22 (Exact geometric row-(d) trilemma). *A non-circular geometric proof of row (d) must either:*

- (1) *factor the nuclear form through N_{rul}^1 ;*
- (2) *extend the intrinsic row-(a) periodic section-cohomology theory from the two rulings to the completed mixed correspondence module, with effectivity, Riemann-Roch, continuity and an equality theorem; or*
- (3) *prove a new Hodge-Rosati theorem intrinsically in the nuclear category before comparing its form with (149).*

The first alternative is impossible by Theorem 9.1; the data required by the second and the theorem required by the third have not been constructed.

The minimum positive result would construct an infinite-dimensional space N_D , a class H with $H^2 > 0$, a degree map and a continuous intersection form containing (151); prove from its own Riemann-Roch and effectivity theory that $M_f^2 \leq 0$, including the equality case; and only afterwards prove $M_f M_g = B_{\text{nuc}}(f, g)$. Such a theorem would close row (d). The source construction below gives a second non-circular route to the same primitive inequality: an independently positive Gamma-Euler-Tate factorization. Neither route is completed here.

9.10. The Gamma-Fourier-Tate boundary construction. The local analysis above proves the primitive inequality on a nontrivial initial interval and identifies the obstruction to propagation as a shorted old/new coupling. We now construct the arithmetic and archimedean boundary data entering that coupling. The construction is independent of the zeros of ξ . It fixes the amplitude, the moving Tate constraints and the positive Gamma storage before the remaining estimate is formulated.

9.10.1. *The finite logarithmic boundary port.* Let $\mathcal{H}_N$ be the Hilbert space with orthonormal basis $\{\delta_d : 1 \leq d \leq N\}$. For $n \leq N$, let $S_n \delta_d = \delta_{nd}$ when $nd \leq N$ and let it be zero otherwise. For $t > 0$ put

$$(186) \quad \mathcal{Z}_t = \sum_{n \leq N} n^{-t/2} S_n, \quad \mathcal{C}_t = \sum_{n \leq N} \Lambda(n) n^{-t/2} S_n, \quad Q \delta_d = (\log d) \delta_d.$$

All the sums in this subsection belong to the finite divisibility algebra. In particular, no convergence or domain assertion is implicit.

Lemma 9.23 (Finite logarithmic connection). *One has*

$$(187) \quad [Q, \mathcal{Z}_t] = \mathcal{Z}_t \mathcal{C}_t = \mathcal{C}_t \mathcal{Z}_t,$$

$$(188) \quad \mathcal{Z}'_t = -\frac{1}{2} \mathcal{Z}_t \mathcal{C}_t,$$

$$(189) \quad (\mathcal{Z}_t^{-1})' = \frac{1}{2} \mathcal{C}_t \mathcal{Z}_t^{-1}.$$

Proof. Since $[Q, S_n] = (\log n) S_n$, the coefficient of S_k in the commutator is $(\log k) k^{-t/2}$. The coefficient of S_k in $\mathcal{Z}_t \mathcal{C}_t$ is

$$k^{-t/2} \sum_{d|k} \Lambda(d) = k^{-t/2} \log k.$$

The algebra is commutative, which proves (187). Termwise differentiation proves (188); differentiating $\mathcal{Z}_t^{-1} \mathcal{Z}_t = I$ then proves (189). $\square$

We record the operator identity for which this connection is needed. Let $G(t)$ be the positive Gram family at a threshold, let $v(t) = (-h(t), I)^t$ be its harmonic residual, and let P_0 be the old coordinate projection. Write M_0 for the old block of $G(t_0)$. The normal equation is $P_0 G(t) v(t) = 0$. Hence differentiation gives

$$(190) \quad P_0 G'(t_0) v(t_0) = M_0 h'(t_0).$$

Suppose, as in the localized explicit-formula decomposition, that

$$G' = R^* \mathcal{L} R - \frac{1}{2} (QG + GQ), \quad Gv = S \delta_1,$$

where $S > 0$ is the scalar normalization. With

$$W_H = \mathcal{L}^{1/2} R v, \quad H_0 = \mathcal{L}^{1/2} R P_0^*,$$

equation (190) becomes

$$(191) \quad H_0^* W_H = M_0 h' + B_Q, \quad B_Q = \frac{S}{2} P_0 [-\mathcal{C} \delta_1 + \mathcal{Z} \mathcal{C}^* m], \quad m = \mathcal{Z}^{-1} \delta_1,$$

where the parameter is now fixed and omitted from the notation.

Define the doubled logarithmic boundary port by

$$(192) \quad W_\infty = \sqrt{\frac{S}{2}} \begin{pmatrix} \mathcal{C} \delta_1 \\ \mathcal{C}^* m \end{pmatrix},$$

$$(193) \quad H_{\infty,0} x = \sqrt{\frac{S}{2}} \begin{pmatrix} P_0^* x \\ -\mathcal{Z}^* P_0^* x \end{pmatrix}.$$

Theorem 9.24 (Exact finite Gamma–Tate amplitude). *The port (192)–(193) is defined entirely by the finite Euler transport and satisfies*

$$(194) \quad H_{\infty,0}^* W_\infty = -B_Q.$$

Consequently

$$(195) \quad (H_0^* \oplus H_{\infty,0}^*)(W_H \oplus W_\infty) = M_0 h'.$$

Proof. Taking the adjoint of (193) and applying it to (192) gives

$$H_{\infty,0}^* W_\infty = \frac{S}{2} P_0 (\mathcal{C} \delta_1 - \mathcal{L} \mathcal{C}^* m) = -B_Q$$

by (191). Adding this equality to (191) proves (195). $\square$

The theorem fixes both orientations and the normalization of the boundary amplitude. It is an amplitude identity, not an energy inequality: its finite-dimensional port cannot be identified with the complete infinite-rank Hodge factor. The continuum realization below uses it only for the arithmetic cross-incidence that it computes.

9.10.2. *The moving Tate terminal.* Let H_0, E be Hilbert spaces, let $B_0 : H_0 \rightarrow \mathbb{C}^r$ be bounded and onto, and let $B_E : E \rightarrow \mathbb{C}^r$ be bounded. Put

$$(196) \quad K_0 = \ker B_0, \quad G_0 = B_0 B_0^*, \quad C = B_0^* G_0^{-1} B_E, \quad M = I_E + C^* C = I_E + B_E^* G_0^{-1} B_E.$$

The enlarged moment map is $B(h, e) = B_0 h + B_E e$.

Theorem 9.25 (Conservative update of a moving constraint). *There is an orthogonal decomposition*

$$(197) \quad \ker B = (K_0 \oplus 0) \hat{\oplus} \{(-Ce, e) : e \in E\}.$$

Moreover

$$(198) \quad W : E \longrightarrow H_0 \oplus E, \quad Wx = \begin{pmatrix} -CM^{-1/2}x \\ M^{-1/2}x \end{pmatrix}$$

is an isometry onto the second summand. In particular,

$$(199) \quad \|M^{-1/2}x\|^2 + \|CM^{-1/2}x\|^2 = \|x\|^2.$$

Proof. The surjectivity of B_0 gives $H_0 = K_0 \hat{\oplus} \text{Ran } B_0^*$. If $h = k + r$ is the corresponding decomposition, then $B(h, e) = 0$ is equivalent to $B_0 r = -B_E e$ and hence to $r = -B_0^* G_0^{-1} B_E e = -Ce$. This proves (197). Finally,

$$\|Wx\|^2 = \langle M^{-1/2}x, (I + C^* C) M^{-1/2}x \rangle = \|x\|^2,$$

which proves both remaining assertions. $\square$

For $I_T = (-T, T)$ take the two Tate moments

$$(200) \quad B_T f = \begin{pmatrix} \langle f, e^{t/2} \rangle \\ \langle f, e^{-t/2} \rangle \end{pmatrix}.$$

Their Gram is

$$(201) \quad B_T B_T^* = 2 \begin{pmatrix} \sinh T & T \\ T & \sinh T \end{pmatrix},$$

with eigenvalues $2(\sinh T \pm T) > 0$. Applying Theorem 9.25 to $L^2(I_{T'}) = L^2(I_T) \oplus L^2(I_{T'} \setminus I_T)$ gives an explicit orthogonal update of the primitive space $\ker B_T$. At the arithmetic thresholds

$$T_N = \frac{1}{2} \log N, \quad \delta_N = T_{N+1} - T_N = \frac{1}{2} \log \frac{N+1}{N},$$

the two nonzero eigenvalues of C^*C are

$$(202) \quad \kappa_{N,\pm} = \frac{\sinh T_{N+1} - \sinh T_N \pm \delta_N}{\sinh T_N \pm T_N}.$$

Therefore

$$(203) \quad \|C\|^2 = \frac{1}{2N} + o(N^{-1}), \quad \|CM^{-1/2}\| = O(N^{-1/2}), \quad \|M^{-1/2} - I\| = O(N^{-1}).$$

Thus the motion of the two Tate constraints is an explicit conservative rank-two terminal; it creates no unidentified projection term.

We next identify the finite incidence column with its physical old/new boundary action. Put $I_N = (-T_N, T_N)$ and

$$\mathcal{E}_N = (-T_{N+1}, -T_N) \cup (T_N, T_{N+1}).$$

Let U_n be translation by $\log n$ on the zero-extension line and set

$$(204) \quad \mathcal{B}_N^{\text{ar}} = -P_{I_N} \sum_{2 \leq n \leq N} \frac{\Lambda(n)}{\sqrt{n}} (U_n + U_n^*)|_{L^2(\mathcal{E}_N)}.$$

After separating the two newborn wings, the translated intervals give an isometry

$$\mathfrak{S}_N : \ell^2\{2, \dots, N\} \hat{\otimes} L^2(\mathcal{E}_N) \longrightarrow L^2(I_N),$$

where the n th coordinate is the unique orientation entering I_N .

Proposition 9.26 (Physical incidence fibre). *If*

$$\mathbf{c}_N = \sum_{n=2}^N \frac{\Lambda(n)}{\sqrt{n}} e_n,$$

then, for every newborn profile e ,

$$(205) \quad \mathcal{B}_N^{\text{ar}} e = -\mathfrak{S}_N(\mathbf{c}_N \otimes e),$$

and

$$(206) \quad (\mathcal{B}_N^{\text{ar}})^* \mathcal{B}_N^{\text{ar}} = \left(\sum_{n=2}^N \frac{\Lambda(n)^2}{n} \right) I.$$

At word depth k , the corresponding label column is

$$(207) \quad (\log N)^{-k} \mathcal{C}_N^k \delta_1 = \left\{ \frac{\Lambda_k(n)}{\sqrt{n}(\log N)^k} \right\}_{n \leq N}.$$

Proof. For a vector on the right newborn wing only U_n enters the old interval; for a vector on the left wing only U_n^* enters it. These translated copies are pairwise disjoint, proving (205); isometry gives (206). Multiplication of an integer label by m agrees, whenever the result remains in the threshold fibre, with physical translation by $\log m$. Iterating and using Dirichlet convolution gives (207). $\square$

9.10.3. *Euler incidence and rational leakage.* On the right-oriented threshold fibre label the interval $T_N - \log m + (0, \delta_N)$ by $m \leq N$. Translation by $\log n$ sends m to mn , while translation in the opposite direction sends it to m/n . The latter remains in the integer fibre exactly when $n \mid m$.

Proposition 9.27 (Divisibility and rational-leakage decomposition). *The finite Euler boundary action is the orthogonal sum of its action on the integer divisibility fibre and an action supported on nonintegral rational labels. An integer-labelled cell never overlaps a noninteger-labelled cell in positive measure. Two rational cells labelled by m/n and m'/n' can overlap only when*

$$(208) \quad \left| \log \frac{m/n}{m'/n'} \right| < \delta_N.$$

Proof. The assertions about labels follow directly from translation of the cell positions. If $m > kn$, then

$$\frac{m}{kn} \geq \frac{N}{N-1} > \frac{N+1}{N} = e^{2\delta_N};$$

if $m < kn$, then $kn/m \geq (N+1)/N$, with equality producing at most an endpoint contact. Hence integer and noninteger cells are orthogonal. Two intervals of length δ_N overlap exactly under (208). $\square$

Write $\Lambda_k = \Lambda^{*k}$. For coprime positive integers a, b with $b > 1$ define

$$(209) \quad \ell_{N,k}(a, b) = \frac{1}{\sqrt{ab}(\log N)^k} \sum_{d \leq N/\max(a,b)} \frac{\Lambda(bd)\Lambda_k(ad)}{d}.$$

Let

$$\mathcal{R}_N^{\text{leak}} = \{a/b : (a, b) = 1, b > 1, \max(a, b) \leq N\},$$

extend $e \in E_N := L^2(0, \delta_N)$ by zero, and define

$$(210) \quad L_{N,k}e = \sum_{a/b \in \mathcal{R}_N^{\text{leak}}} \ell_{N,k}(a, b) \tau_{\log(a/b)}e, \quad \mathbf{L}_N e = \bigoplus_{k \geq 1} L_{N,k}e.$$

Only finitely many depths occur for a fixed N .

Proposition 9.28 (Exact aggregation on reduced rational labels). *The coefficient of the oppositely oriented Euler output on the reduced rational cell a/b is the sum in (209). If $\kappa_e(u) = \langle \tau_u e, e \rangle$, then*

$$(211) \quad \|\mathbf{L}_N e\|^2 = \sum_{k \geq 1} \sum_{a/b, c/d \in \mathcal{R}_N^{\text{leak}}} \ell_{N,k}(a, b) \overline{\ell_{N,k}(c, d)} \kappa_e \left(\log \frac{ad}{bc} \right).$$

Thus the Gram retains all coherent collisions between equal or nearby reduced fractions.

Proof. The equality $m/n = a/b$ with $(a, b) = 1$ is equivalent to $m = ad$, $n = bd$ for a unique positive integer d . Substitution of the Euler weight and of the k th word column gives (209). Expanding the squared norm in (210) gives (211), since the inner product of the cells labelled by a/b and c/d is $\kappa_e(\log(ad/bc))$. $\square$

9.10.4. *Positive Gamma storage and the two exterior Tate states.* For $a > 0$ and $f \in C_c^\infty(\mathbb{R})$, let $z_{a,f}$ be the causal state

$$(212) \quad z'_{a,f} = -az_{a,f} + \sqrt{2a}f, \quad z_{a,f}(-\infty) = 0,$$

and set $g_{a,f} = \sqrt{2a}z_{a,f} - f$.

Lemma 9.29 (Lossless Gamma state). *The state satisfies*

$$(213) \quad \frac{d}{dt}|z_{a,f}|^2 = |f|^2 - |g_{a,f}|^2,$$

$$(214) \quad \frac{1}{2a}\|f - g_{a,f}\|_2^2 = \frac{1}{a^2} \int_{\mathbb{R}} |z'_{a,f}(t)|^2 dt = \int_{\mathbb{R}} \frac{2\tau^2}{a(a^2 + \tau^2)} |\widehat{f}(\tau)|^2 \frac{d\tau}{2\pi}.$$

Proof. Expanding $|g_{a,f}|^2$ and using (212) proves (213). Solving (212) for f gives $f - g_{a,f} = \sqrt{2/a}z'_{a,f}$. Its Fourier multiplier has squared modulus $4\tau^2/(a^2 + \tau^2)$, proving (214). $\square$

Put $a_j = 2j + \frac{1}{2}$ and define the unreset Gamma energy

$$(215) \quad \mathcal{G}_\Gamma[f] = \sum_{j \geq 0} \frac{1}{a_j^2} \int_{\mathbb{R}} |z'_{a_j,f}(t)|^2 dt = \int_{\mathbb{R}} g_\Gamma(\tau) |\widehat{f}(\tau)|^2 \frac{d\tau}{2\pi},$$

where

$$(216) \quad g_\Gamma(\tau) = \sum_{j \geq 0} \frac{2\tau^2}{a_j(a_j^2 + \tau^2)}.$$

The causal equations are solved once on the real line; states are not reset at internal rational-cell endpoints.

For a compactly supported f define

$$(217) \quad M_+(f) = \int_{\mathbb{R}} e^{t/2} f(t) dt, \quad M_-(f) = \int_{\mathbb{R}} e^{-t/2} f(t) dt.$$

If $I = (L, R)$ contains the support of f , the Gram of the two Tate vectors is

$$(218) \quad G_I = \begin{pmatrix} e^R - e^L & R - L \\ R - L & e^{-L} - e^{-R} \end{pmatrix} > 0,$$

and the squared norm of the Tate component is

$$(219) \quad \mathcal{T}_I[f] = \begin{pmatrix} \overline{M_+(f)} & \overline{M_-(f)} \end{pmatrix} G_I^{-1} \begin{pmatrix} M_+(f) \\ M_-(f) \end{pmatrix}.$$

The moments in (217) are exactly the exterior states of (212) at $a = 1/2$: for $\text{supp } f \subset [L, R]$,

$$z_{1/2,f}(t) = e^{-t/2} M_+(f) \quad (t \geq R),$$

up to the normalization $\sqrt{2a} = 1$, and the anticausal state gives $e^{t/2} M_-(f)$ on $t \leq L$.

Take one interval containing every cell in (210), for instance

$$(220) \quad I_N = \left(-\log N, \log \frac{N}{2} + \delta_N \right).$$

The source-defined Gamma–Tate form on the physical leakage column is

$$(221) \quad \mathcal{E}_{\Gamma,N}(e) = \sum_{k \geq 1} \{ \mathcal{G}_\Gamma[L_{N,k}e] + \mathcal{T}_{I_N}[L_{N,k}e] \}.$$

Every term in this form is nonnegative and is determined by prime-power weights, the Gamma parameters and the two Tate moments. In particular, it does not use the sign of the primitive Weil form.

The first nontrivial arithmetic cell. The threshold $N = 3$ gives

$$(222) \quad T_3 = \frac{1}{2} \log 3, \quad T_4 = \log 2, \quad L_3 = \delta_3 = \frac{1}{2} \log \frac{4}{3}.$$

At word depth one the only nonzero nonintegral coefficient is

$$(223) \quad \ell_{3,1}(2, 3) = \frac{\Lambda(3)\Lambda(2)}{\sqrt{6} \log 3} = \frac{\log 2}{\sqrt{6}}.$$

There is no deeper word, because $\Lambda_k(m) = 0$ for $m \leq 3$ and $k \geq 2$. Consequently the complete rational leakage in this threshold cell is

$$(224) \quad \mathbf{L}_3 e = f_e := \frac{\log 2}{\sqrt{6}} \tau_{\log(2/3)} e, \quad \|f_e\|_2^2 = \frac{(\log 2)^2}{6} \|e\|_2^2.$$

Theorem 9.30 (Complete Gamma–Tate payment on one arithmetic cell). *For every $e \in L^2(0, L_3)$, let f_e be given by (224). Then*

$$(225) \quad \|\mathbf{L}_3 e\|_2^2 \leq \mathcal{G}_\Gamma[f_e] + \mathcal{T}_{J_3}[f_e], \quad J_3 = \log(2/3) + (0, L_3).$$

More precisely, if $\mathcal{G}_0[f]$ denotes the first Gamma-channel energy, corresponding to $a_0 = 1/2$, then

$$(226) \quad \|f_e\|_2^2 \leq \alpha_3 \mathcal{G}_0[f_e] + \eta_3 \mathcal{T}_{J_3}[f_e],$$

where

$$(227) \quad \alpha_3 = \frac{1}{4} + \frac{L_3^2}{32} < 0.251, \quad \eta_3 = \frac{1 - e^{-L_3}}{2} = \frac{2 - \sqrt{3}}{4} < 0.067.$$

Thus the Gamma interior energy and the two exterior Tate states pay the whole leakage simultaneously, with a strict positive remainder:

$$(228) \quad \begin{aligned} & \mathcal{G}_\Gamma[f_e] + \mathcal{T}_{J_3}[f_e] - \|\mathbf{L}_3 e\|_2^2 \\ & \geq (1 - \alpha_3) \mathcal{G}_0[f_e] + (1 - \eta_3) \mathcal{T}_{J_3}[f_e] \geq 0. \end{aligned}$$

Proof. Translation does not change any of the following estimates, so write $J_3 = (0, L_3)$. Let z be the causal state of the first Gamma channel,

$$z' = -\frac{1}{2}z + f_e, \quad z(0) = 0.$$

The initial value vanishes because f_e is supported on this single cell. Since $f_e = z' + \frac{1}{2}z$, integration of the cross term gives the exact identity

$$(229) \quad \|f_e\|_2^2 = \|z'\|_2^2 + \frac{1}{4}\|z\|_2^2 + \frac{1}{2}|z(L_3)|^2.$$

The anchored Cauchy–Schwarz inequality yields

$$|z(t)|^2 \leq t \int_0^t |z'(s)|^2 ds \leq t \|z'\|_2^2,$$

and hence

$$(230) \quad \|z\|_2^2 \leq \frac{L_3^2}{2} \|z'\|_2^2.$$

By (214) at $a = 1/2$,

$$(231) \quad \mathcal{G}_0[f_e] = 4 \|z'\|_2^2.$$

It remains to pay the endpoint. The exterior-state formula gives

$$z(L_3) = e^{-L_3/2} M_+(f_e).$$

The Tate terminal is the squared norm of the orthogonal projection onto $\text{span}\{e^{t/2}, e^{-t/2}\}$. It therefore dominates the projection onto the first of these one-dimensional subspaces:

$$(232) \quad \mathcal{T}_{J_3}[f_e] \geq \frac{|M_+(f_e)|^2}{\|e^{t/2}\|_{L^2(J_3)}^2} = \frac{|M_+(f_e)|^2}{e^{L_3} - 1}.$$

Consequently

$$(233) \quad \frac{1}{2}|z(L_3)|^2 \leq \frac{1 - e^{-L_3}}{2} \mathcal{T}_{J_3}[f_e] = \eta_3 \mathcal{T}_{J_3}[f_e].$$

Substitution of (230), (231) and (233) into (229) proves (226). Since the full Gamma energy contains $\mathcal{G}_0$ as its first nonnegative summand and both constants in (227) are smaller than one, (225) and (228) follow. $\square$

This theorem proves the complete one-cell assertion without a sign assumption and without resetting a state inside the cell. Its causal state starts at zero only because, at $N = 3$, the whole rational leakage consists of the single support interval J_3 . For a connected cluster at a later threshold, the outgoing state of one rational cell is the incoming state of the next; Theorem 9.30 cannot be iterated by setting those intermediate states to zero.

The same argument in fact applies to a whole connected cluster, provided it is applied once on the exterior hull rather than separately on its constituent cells.

Lemma 9.31 (Gamma–Tate payment on a finite support hull). *Let $f \in L^2(\mathbb{R})$ be supported in an interval $J = (r, r + D)$. If $D < \sqrt{24}$, then*

$$(234) \quad \|f\|_2^2 \leq \alpha(D) \mathcal{G}_0[f] + \eta(D) \mathcal{T}_J[f] \leq \mathcal{G}_\Gamma[f] + \mathcal{T}_J[f],$$

where

$$(235) \quad \alpha(D) = \frac{1}{4} + \frac{D^2}{32} < 1, \quad \eta(D) = \frac{1 - e^{-D}}{2} < 1.$$

No disjointness or orthogonality assumption is made inside J .

Proof. Translate J to $(0, D)$ and solve the first causal Gamma equation once on the complete hull:

$$z' = -\frac{1}{2}z + f, \quad z(0) = 0.$$

There is no condition on z at an internal subdivision. The proof of Theorem 9.30, with D in place of L_3 , gives

$$\begin{aligned} \|f\|_2^2 &= \|z'\|_2^2 + \frac{1}{4}\|z\|_2^2 + \frac{1}{2}|z(D)|^2, \\ \|z\|_2^2 &\leq \frac{D^2}{2}\|z'\|_2^2, \quad \mathcal{G}_0[f] = 4\|z'\|_2^2, \\ \frac{1}{2}|z(D)|^2 &\leq \frac{1 - e^{-D}}{2} \mathcal{T}_J[f]. \end{aligned}$$

Combining these identities proves the first inequality in (234). The restriction $D < \sqrt{24}$ is exactly $\alpha(D) < 1$; the second inequality then follows because the full Gamma form contains $\mathcal{G}_0$ as a nonnegative summand. $\square$

A three-cell connected cluster. Take $N = 10$, word depth $k = 1$, and

$$(236) \quad q_1 = \frac{3}{4}, \quad q_2 = \frac{7}{9}, \quad q_3 = \frac{4}{5}, \quad \delta_{10} = \frac{1}{2} \log \frac{11}{10}.$$

The three coefficients obtained from (209) are

$$(237) \quad c_1 = \frac{\log 2 \log 3}{\sqrt{12} \log 10}, \quad c_2 = \frac{\log 3 \log 7}{\sqrt{63} \log 10}, \quad c_3 = \frac{\log 5 \log 2}{\sqrt{20} \log 10}.$$

They are all nonzero. Moreover

$$(238) \quad \log \frac{q_2}{q_1} = \log \frac{28}{27} < \delta_{10}, \quad \log \frac{q_3}{q_2} = \log \frac{36}{35} < \delta_{10},$$

so the translated cells overlap in a connected chain. For $e \in L^2(0, \delta_{10})$ define their coherent physical output

$$(239) \quad f_{10,e} = \sum_{j=1}^3 c_j \tau_{\log q_j} e.$$

Its support is contained in the single exterior hull

$$(240) \quad J_{10} = \left(\log \frac{3}{4}, \log \frac{4}{5} + \delta_{10} \right), \quad D_{10} = |J_{10}| = \log \frac{16}{15} + \frac{1}{2} \log \frac{11}{10}.$$

Theorem 9.32 (Complete payment of a connected three-cell cluster). *For every $e \in L^2(0, \delta_{10})$, including profiles for which the adjacent translated copies interfere coherently,*

$$(241) \quad \|f_{10,e}\|_2^2 \leq \mathcal{G}_\Gamma[f_{10,e}] + \mathcal{T}_{J_{10}}[f_{10,e}].$$

More precisely,

$$(242) \quad \|f_{10,e}\|_2^2 \leq \alpha_{10} \mathcal{G}_0[f_{10,e}] + \eta_{10} \mathcal{T}_{J_{10}}[f_{10,e}],$$

where

$$(243) \quad \begin{aligned} \alpha_{10} &= \frac{1}{4} + \frac{1}{32} \left(\log \frac{16}{15} + \frac{1}{2} \log \frac{11}{10} \right)^2 < 0.251, \\ \eta_{10} &= \frac{1}{2} \left[1 - \exp \left(-\log \frac{16}{15} - \frac{1}{2} \log \frac{11}{10} \right) \right] < 0.054. \end{aligned}$$

Consequently

$$(244) \quad \begin{aligned} &\mathcal{G}_\Gamma[f_{10,e}] + \mathcal{T}_{J_{10}}[f_{10,e}] - \|f_{10,e}\|_2^2 \\ &\geq (1 - \alpha_{10}) \mathcal{G}_0[f_{10,e}] + (1 - \eta_{10}) \mathcal{T}_{J_{10}}[f_{10,e}] \geq 0. \end{aligned}$$

Proof. The coefficient formulas follow because, for each pair in (236), the only contributing divisor in (209) is $d = 1$. Equations (238) prove connectedness, and the extreme endpoints give (240). Numerically,

$$D_{10} = 0.1121936110 \dots < \sqrt{24}, \quad \alpha_{10} = 0.2503933564 \dots, \quad \eta_{10} = 0.0530644113 \dots$$

Apply Lemma 9.31 directly to the coherent sum $f_{10,e}$, before expanding its squared norm. Therefore every diagonal term and every interference term

$$c_i \overline{c_j} \langle \tau_{\log q_i} e, \tau_{\log q_j} e \rangle$$

is included automatically. The Gamma state is solved once from the left endpoint of J_{10} to its right endpoint and is never reset at either internal cell boundary. This proves (242); the remaining statements follow as in Lemma 9.31. $\square$

Thus the earlier warning is not an obstruction to finite clusters: it only forbids proving a cluster estimate by summing independently reset one-cell estimates. The exterior-hull argument proves every connected cluster with hull length $D < \sqrt{24}$. The following arithmetic separation lemma shows that this restriction is automatic for every connected rational-leakage cluster. The point is that the cells respect a fixed family of reciprocal half-integer bands.

Theorem 9.33 (Reciprocal half-integer band theorem). *Fix $N \geq 2$ and a word depth k . Form the intervals*

$$J_q = \log q + (0, \delta_N), \quad q = a/b \in \mathcal{R}_N^{\text{leak}}, \quad \ell_{N,k}(a, b) > 0.$$

Every connected component of $\bigcup_q J_q$ is contained in one of the bands

$$(245) \quad \left(\log \frac{m}{2}, \log \frac{m+1}{2} \right) \quad (m \geq 2), \quad \text{or} \quad \left(\log \frac{2}{m+1}, \log \frac{2}{m} \right) \quad (m \geq 2),$$

up to the possible inclusion of one boundary cell. In particular, the exterior hull of every connected component has length

$$(246) \quad D_{N,k}^{\text{comp}} \leq \log \frac{3}{2} + \delta_N \leq \frac{3}{2} \log \frac{3}{2} < \sqrt{24}.$$

Consequently every connected component C and every profile $e \in L^2(0, \delta_N)$ satisfy, for

$$f_{C,e} = \sum_{q: J_q \subset C} \ell_{N,k}(q) \tau_{\log q} e,$$

the complete estimate

$$(247) \quad \|f_{C,e}\|_2^2 \leq \mathcal{G}_\Gamma[f_{C,e}] + \mathcal{T}_{\text{hull}(C)}[f_{C,e}].$$

This assertion is uniform in N , k , the number of cells in C , and all coherent interference among them.

Proof. We first prove that a rational cell cannot cross the indicated barriers. Let $q = a/b > 1$ be reduced, with $a, b \leq N$, and let $r = m/2 > q$ be a half-integer. Since $mb - 2a$ is a positive integer,

$$\log \frac{r}{q} = \log \frac{mb}{2a} \geq \log \left(1 + \frac{1}{2a} \right) \geq \log \left(1 + \frac{1}{2N} \right).$$

But

$$(248) \quad \log \left(1 + \frac{1}{2N} \right) > \frac{1}{2} \log \left(1 + \frac{1}{N} \right) = \delta_N,$$

because $(1 + 1/(2N))^2 > 1 + 1/N$. Hence the right endpoint of J_q lies strictly to the left of $\log r$.

For $0 < q = a/b < r = 2/m$, the positive integer $2b - ma$ gives

$$\log \frac{r}{q} = \log \frac{2b}{ma} \geq \log \left(1 + \frac{1}{ma} \right).$$

Here $ma = 2b - (2b - ma) \leq 2N - 1$, and therefore the last quantity is at least $\log(1 + 1/(2N - 1))$, which is strictly larger than the left side of (248), hence larger than δ_N . Thus no cell below a reciprocal barrier $2/m$ crosses it either. Cells starting on a barrier belong only to the band on its right, so they do not join the two adjacent open bands. Notice that this proof uses only $a, b \leq N$; it is therefore valid at every word depth and does not depend on which of the coefficients happen to vanish.

The consecutive ratios of the band endpoints are

$$\frac{(m+1)/2}{m/2} = 1 + \frac{1}{m}, \quad \frac{2/m}{2/(m+1)} = 1 + \frac{1}{m}.$$

Their logarithms are at most $\log(3/2)$ for $m \geq 2$. Adding one cell length gives the first inequality in (246); the second follows from $\delta_N \leq \delta_2 = \frac{1}{2} \log(3/2)$. Lemma 9.31 now applies once to the complete coherent output on each connected component and proves (247). $\square$

Corollary 9.34 (No low-frequency fugitive inside one arithmetic band). *For a connected component C define its positive exponential polynomial*

$$P_C(\tau) = \sum_{q: J_q \subset C} \ell_{N,k}(q) q^{-i\tau}.$$

For every $0 \leq \omega < \pi/\log(3/2)$ and $|\tau| \leq \omega$ one has

$$(249) \quad c(\omega)P_C(0) \leq |P_C(\tau)| \leq P_C(0), \quad c(\omega) = \cos\left(\frac{\omega}{2} \log \frac{3}{2}\right) > 0.$$

Thus the translates belonging to a single arithmetic band cannot, at any N or word depth, form a low-frequency fugitive profile by destructive interference.

Proof. Let $x_q = \log q$ and let x_C be the midpoint of the smallest interval containing all x_q in C . The band theorem gives $|x_q - x_C| \leq \frac{1}{2} \log(3/2)$. Positivity of the coefficients yields

$$\begin{aligned} |P_C(\tau)| &\geq \Re(e^{i\tau x_C} P_C(\tau)) \\ &= \sum_{q \in C} \ell_{N,k}(q) \cos(\tau(x_q - x_C)) \\ &\geq \cos\left(\frac{\omega}{2} \log \frac{3}{2}\right) \sum_{q \in C} \ell_{N,k}(q). \end{aligned}$$

The cosine is positive by the assumed range of ω . The upper bound is the triangle inequality. $\square$

For orientation, direct enumeration of the exact nonzero coefficients at depth one gives the following largest connected hulls. These figures are not used in the proof; they display the approach to the sharp first-band width $\log(3/2)$.

N	leftmost label	rightmost label	$D_{N,1}^{\text{largest}}$
10	6/5	4/3	0.1530156...
50	33/49	48/49	0.3845948...
200	133/199	198/199	0.4004117...
1000	665/997	996/997	0.4044600...

The theorem removes the possibility of an arbitrarily long connected cluster, and Corollary 9.34 removes destructive low-frequency interference inside each such cluster. It does not by itself permit one to add (247) over disconnected bands: the unreset Gamma form and the two global Tate moments contain cross-band terms. The remaining question is therefore a recombination theorem for the separated arithmetic bands—equivalently, exclusion of cancellation between distinct band polynomials—not a further one-cell or connected-cluster estimate.

Audit of causal recombination across the gaps. There is a tempting but incorrect argument which treats decay of the causal state in a gap as a negative Schur remainder. The exact sign is the opposite. To record this point unambiguously, write

$$A_m = z(\inf C_m), \quad B_m = z(\sup C_m), \quad A_{m+1} = e^{-\Delta_m/2} B_m, \quad A_1 = 0.$$

Summing the exact cluster identities gives

$$(250) \quad \frac{1}{2} \sum_{m=1}^M (|B_m|^2 - |A_m|^2) = \frac{1}{2} |B_M|^2 + \frac{1}{2} \sum_{m=1}^{M-1} (1 - e^{-\Delta_m}) |B_m|^2.$$

Thus the gap term occurs with a plus sign, not a minus sign. Equivalently, on a gap the equation $z' = -z/2$ gives

$$4 \int_{\text{gap}} |z'|^2 = \int_{\text{gap}} |z|^2 = (1 - e^{-\Delta_m}) |B_m|^2,$$

and the two positive bulk terms are cancelled exactly by the decrease of the endpoint storage when the identity is written on the complete hull; they do not produce an additional negative remainder. Indeed the global identity remains

$$(251) \quad \|f\|_2^2 = \frac{1}{4} \mathcal{G}_0[f] + \frac{1}{4} \|z\|_2^2 + \frac{1}{2} |B_M|^2.$$

There is a second independent obstruction to that argument. The anchored estimate

$$\|z\|_{L^2(C_m)}^2 \leq \frac{|C_m|^2}{2} \|z'\|_{L^2(C_m)}^2$$

used in Lemma 9.31 requires $A_m = 0$ and cannot be reused for the restriction of the global state to later clusters. For example, the homogeneous state $z(t) = Ae^{-(t-L)/2}$ on an interval of length D has

$$\|z'\|_2^2 = \frac{1}{4} \|z\|_2^2,$$

so the displayed estimate would require $1 \leq D^2/8$, which is false for all the band lengths in (246). An arbitrarily small nonzero source perturbation gives the same failure inside a genuine later support component. Therefore causal decay by itself does not prove the required recombination; a valid proof must control the incoming states or establish the global low-frequency observability inequality below.

The single obstruction for arbitrary clusters. The preceding results permit an exact separation of the disconnected-band problem. Since every summand in (216) is strictly increasing for $\tau > 0$, while $g_\Gamma(0) = 0$ and $g_\Gamma(\tau) \rightarrow \infty$, there is a unique number $\tau_\Gamma > 0$ such that

$$(252) \quad g_\Gamma(\tau_\Gamma) = 1.$$

Numerically, $\tau_\Gamma = 0.2860886323\dots$; the numerical value is not used below. Put $\Omega_\Gamma = (-\tau_\Gamma, \tau_\Gamma)$ and define the positive and negative Gamma weights

$$w_-(\tau) = (1 - g_\Gamma(\tau))_+, \quad w_+(\tau) = (g_\Gamma(\tau) - 1)_+.$$

In fact $\tau_\Gamma < 1$, because the $j = 0$ summand alone gives $g_\Gamma(1) > 1$, and $1 < \pi/\log(3/2)$. Hence Corollary 9.34 applies on the whole deficit window Ω_Γ : no individual reciprocal half-integer band has a low-frequency zero or even an arbitrarily small normalized value there. For the k th rational leakage output write

$$(253) \quad P_{N,k}(\tau) = \sum_{a/b \in \mathcal{R}_N^{\text{leak}}} \ell_{N,k}(a, b) e^{-i\tau \log(a/b)}, \quad \widehat{L_{N,k}e}(\tau) = \widehat{e}(\tau) P_{N,k}(\tau).$$

Proposition 9.35 (Exact low-frequency reduction). *The uniform Gamma–Tate estimate (275) is equivalent to*

$$(254) \quad \begin{aligned} & \sum_{k \geq 1} \int_{\Omega_\Gamma} w_-(\tau) |\widehat{e}(\tau) P_{N,k}(\tau)|^2 \frac{d\tau}{2\pi} \\ & \leq \sum_{k \geq 1} \mathcal{T}_{I_N}[L_{N,k}e] + \sum_{k \geq 1} \int_{\mathbb{R} \setminus \Omega_\Gamma} w_+(\tau) |\widehat{e}(\tau) P_{N,k}(\tau)|^2 \frac{d\tau}{2\pi}. \end{aligned}$$

Moreover, because all coefficients $\ell_{N,k}(a, b)$ are nonnegative,

$$(255) \quad |P_{N,k}(\tau)|^2 \leq P_{N,k}(i/2) P_{N,k}(-i/2) \quad (\tau \in \mathbb{R}).$$

Thus every frequency outside the fixed compact band Ω_Γ is already paid by Gamma. The only remaining issue is the two-moment observability of the arithmetic leakage range inside that band.

Proof. For $F_k = L_{N,k}e$, Plancherel and (215) give

$$\begin{aligned} & \mathcal{G}_\Gamma[F_k] + \mathcal{T}_{I_N}[F_k] - \|F_k\|_2^2 \\ & = \mathcal{T}_{I_N}[F_k] + \int_{\mathbb{R}} (g_\Gamma(\tau) - 1) |\widehat{F}_k(\tau)|^2 \frac{d\tau}{2\pi}. \end{aligned}$$

Splitting the integral at Ω_Γ , substituting (253), and summing over k proves the equivalence. For the envelope, write $q = a/b$ and use Cauchy–Schwarz:

$$\begin{aligned} |P_{N,k}(\tau)| & \leq \sum_q \ell_{N,k}(q) \\ & = \sum_q \sqrt{\ell_{N,k}(q) q^{1/2}} \sqrt{\ell_{N,k}(q) q^{-1/2}} \\ & \leq \left(\sum_q \ell_{N,k}(q) q^{1/2} \right)^{1/2} \left(\sum_q \ell_{N,k}(q) q^{-1/2} \right)^{1/2}. \end{aligned}$$

The two factors are $P_{N,k}(i/2)^{1/2}$ and $P_{N,k}(-i/2)^{1/2}$, proving (255). $\square$

The mean/oscillation Schur decomposition. The preceding band calculation suggests a sharper way to formulate the remaining global problem. Put

$$\mathfrak{A}_N[e] = \|\mathbf{L}_N e\|_2^2, \quad \mathfrak{Q}_N[e] = \mathcal{E}_{\Gamma T, N}(e), \quad \mathfrak{D}_N = \mathfrak{Q}_N - \mathfrak{A}_N,$$

and decompose the input cell exactly as

$$(256) \quad E_N = \mathbb{C}u_N \oplus E_N^0, \quad u_N = \delta_N^{-1/2} \mathbf{1}_{(0, \delta_N)}, \quad E_N^0 = \left\{ h : \int_0^{\delta_N} h = 0 \right\}.$$

Let $d_N = \mathfrak{D}_N[u_N]$, let $b_N(h) = \mathfrak{D}_N(u_N, h)$, and let C_N be the restriction of $\mathfrak{D}_N$ to E_N^0 .

Proposition 9.36 (Exact mean/oscillation recombination criterion). *The unit Gamma–Tate estimate at threshold N holds if and only if*

$$(257) \quad C_N \geq 0, \quad b_N \in \text{Ran } C_N^{1/2}, \quad d_N - b_N^* C_N^\dagger b_N \geq 0.$$

Thus the remaining interference problem splits into a coercivity statement for zero-mean cell profiles and one scalar Schur inequality for the constant profile after all oscillatory profiles have been eliminated.

Proof. In the orthogonal decomposition (256), the defect form has block matrix

$$\mathfrak{D}_N = \begin{pmatrix} d_N & b_N^* \\ b_N & c_N \end{pmatrix}.$$

The generalized Schur-complement criterion for a closed Hermitian form is exactly (257). Positivity of $\mathfrak{D}_N$ is the unit Gamma–Tate estimate by definition. $\square$

Conjecture 9.37 (Uniform Schur angle). There is an $\varepsilon > 0$ such that for every $N \geq 3$ the mean-zero block and constant margin satisfy

$$(258) \quad C_N \geq 0, \quad d_N > 0, \quad b_N \in \text{Ran } C_N^{1/2}, \quad \rho_N := \frac{b_N^* C_N^\dagger b_N}{d_N} \leq 1 - \varepsilon.$$

At $N = 2$ the rational-leakage output is empty.

Remark 9.38 (On the constant, and a falsified stronger form). Only $\rho_N \leq 1 - \varepsilon$ is used below, and stating the conjecture at that strength is not merely prudence: the natural stronger form

$$(259) \quad \rho_N \leq \frac{1}{20 \log N}$$

is *false*, and it is already contradicted by the reconnaissance table below. At $N = 20$ that table records $\rho_N = 0.0175$ while $1/(20 \log 20) = 0.01669$; the same comparison fails at $N = 40, 80$ and 120 , so four of its five tabulated points violate (259). The companion reading “ $\rho_N \log N < 0.047$ in every sample” fails as well, the tabulated values of $\rho_N \log N$ reaching 0.0565 .

This is not a discretization artifact, and the direction of the error matters. Since ρ_N is a *supremum* over cell profiles, a finite-dimensional Galerkin evaluation returns a lower bound for it; refining the trial space can only increase the measured value. Independent recomputation on meshes from twelve to forty-eight bins reproduces the tabulated entries at twelve bins to three digits and then increases monotonically — at $N = 20$, from 0.01782 to 0.01881 — so the true ρ_N exceeds $1/(20 \log N)$ at those thresholds by a margin that grows under refinement.

What survives is the qualitative signal: ρ_N is small, of order 10^{-2} , uniformly over the sampled range, with no indication of approaching 1. That is what Conjecture 9.37 asserts, and it is all the argument consumes. Any sharper rate must be calibrated against the transfer comparison of Problem 9.44 rather than guessed.

Corollary 9.39 (Conditional completion of row (d)). *Assume Conjecture 9.37. Then the unit Gamma–Tate estimate (275) holds at every arithmetic threshold. If moreover Problem 9.44 is answered with a uniform constant $c_N \geq 1$, then the threshold Schur complements are nonnegative, the primitive inequality propagates from the certified initial interval to all windows, and row (d) follows.*

Proof. For $N \geq 3$, equation (258) gives

$$d_N - b_N^* C_N^\dagger b_N \geq \varepsilon d_N > 0.$$

Together with the asserted block positivity and range condition, Proposition 9.36 gives (275); the $N = 2$ leakage block is zero. Under the additional hypothesis, $\mathfrak{D}_E \geq \Xi_N^* \mathfrak{D}_N \Xi_N \geq 0$, so Theorem 9.43 applies at each threshold and yields $\mathcal{C}_\varepsilon \geq 0$ for every $\varepsilon > 0$, hence the range condition and (267); equivalently, by Theorem 9.42, the unit output capacity (272). Induction from Corollary 9.9, using Theorem 9.16(4) so that no error accumulates in the old-core block between steps, gives the primitive inequality for every T . $\square$

Remark 9.40. The second hypothesis is not decorative. Without it the conjecture controls a model of the threshold step rather than the step, and the conclusion is row (d) *for that model*. This is the one place where the present construction is weaker than its arrangement suggests, and Problem 9.44 is stated precisely so that the gap is measurable rather than rhetorical.

Finite-dimensional reconnaissance (audit only). To identify which part of Proposition 9.36 carries the difficulty, the exact arithmetic coefficients were evaluated on the space of twelve-bin piecewise-constant profiles on $(0, \delta_N)$. The table reports the smallest generalized quotient $\mathfrak{Q}_N/\mathfrak{A}_N$, the same quotient after imposing zero mean, and the proportion of the constant defect consumed by its Schur correction.

N	$\min \mathfrak{Q}_N/\mathfrak{A}_N$	$\min_{E_N^0} \mathfrak{Q}_N/\mathfrak{A}_N$	$b_N^* C_N^{-1} b_N/d_N$
10	5.1436	8.2823	0.0156
20	4.9506	8.9386	0.0175
40	4.8129	9.6305	0.0150
80	4.5263	10.2950	0.0116
120	4.4563	10.6917	0.0118

Mesh refinement at $N = 80$ from six to sixteen bins changes the first quotient from 4.5316 to 4.5263 and the zero-mean quotient from 10.3691 to 10.2807. These computations are neither interval-certified nor a proof for arbitrary profiles. Their consistent structural signal is that the prospective minimizer lies in the almost-constant sector, while the zero-mean sector has a large Gamma margin and the cross-sector Schur correction is small. Accordingly, the next analytic target is (257), not an unstructured estimate over all cell profiles at once.

The restriction to the arithmetic range in (254) is essential. Indeed, choose $\varphi \in C_c^\infty(-1/4, 1/4)$ with $\|\varphi\|_2 = 1$ and put

$$(260) \quad f_D(t) = D^{-1/2} \varphi(t/D), \quad I_D = (-D/2, D/2).$$

Then $\|f_D\|_2 = 1$, whereas

$$(261) \quad \mathcal{G}_\Gamma[f_D] \longrightarrow 0, \quad \mathcal{T}_{I_D}[f_D] \longrightarrow 0 \quad (D \rightarrow \infty).$$

For the first limit, substitute $\hat{f}_D(\tau) = D^{1/2} \hat{\varphi}(D\tau)$ and use $g_\Gamma(u/D) \rightarrow 0$ together with the rapid decay of $\hat{\varphi}$. For the second, the support of f_D stays a distance $D/4$ from both endpoints of I_D : its two exponential moments grow at most like $D^{1/2} e^{D/8}$, while the corresponding diagonal Tate Grams grow like $e^{D/2}$; the off-diagonal entry is only D . Hence the Tate projection tends to zero exponentially.

Consequently no support-only argument can prove the unit estimate for an aggregate whose disconnected bands have an arbitrarily long exterior hull. The one remaining theorem is precisely that outputs of the positive arithmetic polynomials $P_{N,k}$ cannot form the escaping family (260); equivalently, they must satisfy the compact low-frequency observability inequality (254) uniformly in N .

9.10.5. *The exact threshold Schur condition.* The two physical analysis factors at one threshold are not posited: they are the reference and load feature maps X_T, Y_T of (170), split by Theorem 9.16 into old-core and born-boundary source variables,

$$X = (X_0, X_E), \quad Y = (Y_0, Y_E).$$

Concretely, the entries of X carry the complete Gamma channel and every antisymmetric contact $\hat{J}_{p^k, -}$ with $p^k \leq e^{2T}$; the entries of Y carry the constant channel and every symmetric contact

$\widehat{J}_{p^k,+}$; and both are already restricted by the exact two-Tate projection, so no term can appear outside the transform below. Put

$$(262) \quad R_0 = X_0^* X_0, \quad L_0 = Y_0^* Y_0, \quad r = X_0^* X_E, \quad \ell = Y_0^* Y_E, \quad A_0 = R_0 - L_0.$$

Assume the old primitive block A_0 is nonnegative and the reference range condition $r \in \text{Ran } R_0$ holds. Define

$$(263) \quad H = R_0^\dagger r,$$

$$(264) \quad S_E = X_E^* X_E - r^* R_0^\dagger r,$$

$$(265) \quad Z_E = Y_E - Y_0 H, \quad b_E = L_0 H - \ell.$$

Theorem 9.41 (Exact threshold completion criterion). *The full old/new signed block $X^* X - Y^* Y$ is nonnegative if and only if*

$$(266) \quad b_E \in \text{Ran } A_0^{1/2}$$

and

$$(267) \quad S_E - Z_E^* Z_E - b_E^* A_0^\dagger b_E \geq 0.$$

Proof. The equation $R_0 H = r$ gives $X_0^*(X_E - X_0 H) = 0$ and

$$(X_E - X_0 H)^*(X_E - X_0 H) = S_E.$$

Hence the reference-harmonic column $v_E = (-H, I)^t$ satisfies

$$v_E^*(X^* X - Y^* Y)v_E = S_E - Z_E^* Z_E.$$

Its old cross is

$$(r - \ell) - (R_0 - L_0)H = L_0 H - \ell = b_E.$$

Congruence by the invertible triangular change from the coordinate newborn column to v_E leaves the old block equal to A_0 and changes the other entries to b_E and $S_E - Z_E^* Z_E$. The generalized Schur-complement criterion gives precisely (266) and (267). $\square$

This theorem is the fixed global target. The preceding construction determines the divergence, the moving Tate cell, the complete positive Gamma form and the exact rational leakage. It does not determine the sign in (267).

The target in normalized form. Before turning to sufficient estimates we put (267) in the normalization in which the desired bound has constant one. Apply the reference Cholesky transform: with H as in (263) and S_E as in (264), set $\tilde{X}_E = X_E - X_0 H$, so that $X_0^* \tilde{X}_E = 0$ and $S_E = \tilde{X}_E^* \tilde{X}_E$. Applying the *same* triangular source change to the load features gives the two normalized objects

$$(268) \quad \mathcal{K}_N = Y_0 R_0^{\dagger/2}, \quad y_N = (Y_E - Y_0 H) S_E^{\dagger/2},$$

and the complete comparison acquires the column form $(\mathcal{K}_N, y_N)$. Write

$$(269) \quad D_{\text{in}} = I - \mathcal{K}_N^* \mathcal{K}_N, \quad D_{\text{out}} = I - \mathcal{K}_N \mathcal{K}_N^* = I - Y_0 R_0^\dagger Y_0^*.$$

Since $\mathcal{K}_N^* \mathcal{K}_N = R_0^{\dagger/2} L_0 R_0^{\dagger/2} = T_0$ and $A_0 = R_0^{1/2} D_{\text{in}} R_0^{1/2}$, the old-core hypothesis $A_0 \geq 0$ of §9.9 is exactly $D_{\text{in}} \geq 0$, i.e. that the old comparison $\mathcal{K}_N$ is a contraction.

Theorem 9.42 (Output-defect reduction of the threshold condition). *Assume the old comparison is a contraction, $D_{\text{in}} \geq 0$. Then, on the supported ranges,*

$$(270) \quad I + \mathcal{K}_N D_{\text{in}}^\dagger \mathcal{K}_N^* = D_{\text{out}}^\dagger, \quad D_{\text{in}}^{1/2} \mathcal{K}_N^* = \mathcal{K}_N^* D_{\text{out}}^{1/2},$$

and the shorted form of the enlarged cell equals

$$(271) \quad I - y_N^* y_N - y_N^* \mathcal{K}_N D_{\text{in}}^\dagger \mathcal{K}_N^* y_N = I - y_N^* D_{\text{out}}^\dagger y_N.$$

Consequently the threshold conditions (266)–(267) hold if and only if

$$(272) \quad y_N^* D_{\text{out}}^\dagger y_N \leq I,$$

equivalently if and only if there is a contraction v_N with

$$(273) \quad y_N = D_{\text{out}}^{1/2} v_N.$$

Proof. The first identity in (270) is the push-through identity for $\mathcal{K}_N$, and the second is the Julia intertwining; both hold on the supported range with the usual kernel condition. Shorting the upper-left corner of the defect of $(\mathcal{K}_N, y_N)$ produces the left side of (271), and substituting the push-through identity gives the right side, whence (272). Douglas factorization applied to y_N and $D_{\text{out}}^{1/2}$ gives the equivalence with (273). $\square$

This locates the discrepancy precisely: it is not a residual in the cross term, but the component of the explicitly orthogonalized boundary load y_N in the small spectral layers of the *output* defect D_{out} . Note that (269) shows D_{out} is the short of the old reference energy against a prescribed load-channel output, $\langle g, D_{\text{out}} g \rangle = \|g\|^2 - \|R_0^{\dagger/2} Y_0^* g\|^2$; it is *not* the pure-Gamma Gram, and its spectrum can approach one. The Julia identity yields only $\|D_{\text{in}}^{\dagger/2} \mathcal{K}_N^* y_N\| \leq \|D_{\text{out}}^{\dagger/2} y_N\|$, so neither it nor (269) by itself proves $y_N \in \text{Ran } D_{\text{out}}^{1/2}$. That inclusion, with the unit bound, is the substantive claim.

The range condition is not an independent hypothesis. It is worth recording that the awkward half of (266) — membership in a range which is typically not closed, and which no finite computation can test — is a consequence of the quantitative estimate rather than a separate assumption.

Write $L_E = Y_E^* Y_E$ for the newborn load block and, in parallel with $D_0 = I - T_0$ of §9.9,

$$(274) \quad \mathfrak{D}_E = I - T_E, \quad T_E = S_E^{\dagger/2} L_E S_E^{\dagger/2},$$

so that the newborn block satisfies $A_{\text{new}} = S_E^{1/2} \mathfrak{D}_E S_E^{1/2}$ on its support, exactly as $A_{\text{old}} = R_0^{1/2} D_0 R_0^{1/2}$.

Theorem 9.43 (Regularized one-step criterion). *Assume $A_{\text{old}} \geq 0$, and for $\varepsilon > 0$ put*

$$\mathcal{C}_\varepsilon = \mathfrak{D}_E - Q_c^* (D_0 + \varepsilon I)^{-1} Q_c,$$

with Q_c the normalized cross of §9.9 and $\mathfrak{D}_E$ as in (274). Then $\varepsilon \mapsto \mathcal{C}_\varepsilon$ is nondecreasing in ε ; the enlarged block is nonnegative if and only if $\mathcal{C}_\varepsilon \geq 0$ for every $\varepsilon > 0$; and in that case the monotone limit $\lim_{\varepsilon \downarrow 0} Q_c^ (D_0 + \varepsilon)^{-1} Q_c$ is bounded, which forces $\text{Ran } Q_c \subseteq \text{Ran } D_0^{1/2}$ with limit $Q_c^* D_0^\dagger Q_c$.*

Proof. Monotonicity holds because $\varepsilon \mapsto (D_0 + \varepsilon)^{-1}$ is decreasing in the operator order. For $\varepsilon > 0$ the old block $A_{\text{old}} + \varepsilon$ is invertible on its support, so the ordinary Schur criterion applies; conjugating by $S_E^{\dagger/2}$ and using $A_{\text{old}} = R_0^{1/2} D_0 R_0^{1/2}$ identifies the result with $\mathcal{C}_\varepsilon$ after the corresponding rescaling of ε . Finally the block is nonnegative iff it is nonnegative after adding ε on the old summand, for every $\varepsilon > 0$, by continuity of the quadratic form. The last assertion is Douglas' criterion in shorted-operator form: the monotone supremum $\sup_{\varepsilon > 0} Q_c^* (D_0 + \varepsilon)^{-1} Q_c$ is bounded if and only if $\text{Ran } Q_c \subseteq \text{Ran } D_0^{1/2}$, and then it equals $Q_c^* D_0^\dagger Q_c$. $\square$

Accordingly, any conservative identity proposed for the transfer step should first be established in the regularized form $\mathcal{C}_\varepsilon = Z_\varepsilon^* Z_\varepsilon + E_\varepsilon$ with $E_\varepsilon \geq 0$, and only afterwards passed to the limit.

A sufficient source-level estimate. The source construction isolates a concrete sufficient estimate. Let

$$\mathcal{D}_N = \{e \in E_N : \mathcal{E}_{\Gamma T, N}(e) < \infty\}.$$

It would suffice to prove, with the same unit constant for every arithmetic threshold,

$$(275) \quad \|\mathbf{L}_N e\|^2 \leq \mathcal{E}_{\Gamma T, N}(e) \quad (N \geq 2, e \in \mathcal{D}_N).$$

Indeed, Proposition 9.27 separates the integer incidence from the noninteger leakage. The former is the finite divisibility current of Theorem 9.24; the latter is exactly $\mathbf{L}_N e$ by Proposition 9.28. The right side of (275) is the unreset positive Gamma storage plus the conservative two-dimensional Tate terminal. Thus the estimate makes the joint boundary source contractive with unit normalization.

We state plainly what is and is not established about the passage from (275) to the exact target. What Theorem 9.42 supplies is the correct normalization: the target is a *unit* capacity bound (272) in the output-defect metric, and (275) is a unit bound in the source metric. What is *not* supplied here is the comparison of those two metrics — an inequality $\Xi_N^* \mathcal{D}_N \Xi_N \leq \mathfrak{D}$ relating the Gamma–Tate source defect $\mathcal{D}_N = \mathcal{E}_{\Gamma T, N} - \mathbf{L}_N^* \mathbf{L}_N$ to the newborn-cell defect of (269), with an explicit constant and an explicit identification Ξ_N of the one-wing model cell with the two-wing annulus of Theorem 9.16(5). Until that comparison is proved, (275) is a sufficient estimate for a *model* of the threshold step, not for the step itself.

Problem 9.44 (Transfer comparison). Construct the identification $\Xi_N : E \rightarrow E_N$ of the two-wing newborn annulus of Theorem 9.16(5) with the one-wing model cell $E_N = L^2(0, \delta_N)$, and determine the best constant $c_N > 0$ in

$$\mathfrak{D}_E \geq c_N \Xi_N^* \mathcal{D}_N \Xi_N, \quad \mathcal{D}_N = \mathcal{E}_{\Gamma T, N} - \mathbf{L}_N^* \mathbf{L}_N.$$

If $c_N \geq 1$ uniformly, (275) implies the exact threshold condition (272). If $\inf_N c_N = 0$, the source route does not reach the exact target in this normalization, whatever the status of (275).

Two features of Problem 9.44 deserve emphasis. It is finite-dimensional at each N , hence decidable numerically before any analytic investment; and its answer fixes the constant that the source estimate must carry, which is otherwise unpinned.

Remark 9.45 (Measured answer, and its direction). The comparison has been evaluated directly. Assembling both sides in a common piecewise-constant Galerkin space — the physical target (267) from the balanced factorization (169), and $\mathcal{D}_N$ from the leakage construction above — and taking for Ξ_N each of the three natural identifications of the newborn corona with the model cell (restriction to one wing, and the symmetric and antisymmetric wing combinations), one finds

$$c_N < 1 \quad \text{at every arithmetic threshold } 3 \leq N \leq 37,$$

with c_N decreasing monotonically under mesh refinement at every N and decaying in N roughly like $(\log N)^{-0.6}$; at the finest meshes tested c_N ranges from 0.58 at $N = 4$ down to 0.09 at $N = 29$.

The direction is what makes this conclusive rather than indicative. Since c_N is a *minimum* over the corona, a Galerkin restriction returns an *upper* bound for it; the true constant is therefore no larger than the measured one, and $c_N < 1$ is not an artifact of the discretization.

Two consequences. First, the hypothesis $c_N \geq 1$ of Corollary 9.39 is *not* satisfied, so the unit source estimate (275) — which the same computation finds to hold comfortably, with $\mathcal{E}_{\Gamma T, N} \geq 3.6 \|\mathbf{L}_N\|^2$ throughout — does not by itself deliver the exact threshold condition. The source model is a model, and the loss in passing to it is real and grows with N . Second, the exact target itself is nevertheless satisfied on every threshold tested: the normalized quantity of Theorem 9.42

stays positive, between 0.012 and 0.075. So the evidence is that the threshold condition holds and that the route through the source model is the wrong way to prove it. Establishing (267) directly, in the output-defect metric where the target has constant one, is accordingly the live problem.

For clarity, (275) is a sufficient source-level route, not a definition of the desired Schur complement and not an additional conclusion of the identities already proved. Expanded on the test core $C_c^\infty(0, \delta_N)$, it is the explicit inequality

$$\begin{aligned}
 & \sum_{k \geq 1} \sum_{a/b, c/d \in \mathcal{R}_N^{\text{leak}}} \ell_{N,k}(a, b) \overline{\ell_{N,k}(c, d)} \kappa_e \left(\log \frac{ad}{bc} \right) \\
 & \leq \sum_{k \geq 1} \int_{\mathbb{R}} g_T(\tau) \left| \widehat{e}(\tau) \sum_{a/b \in \mathcal{R}_N^{\text{leak}}} \ell_{N,k}(a, b) e^{-i\tau \log(a/b)} \right|^2 \frac{d\tau}{2\pi} \\
 (276) \quad & + \sum_{k \geq 1} \left(\overline{M_+(L_{N,k}e)} \quad \overline{M_-(L_{N,k}e)} \right) G_{I_N}^{-1} \begin{pmatrix} M_+(L_{N,k}e) \\ M_-(L_{N,k}e) \end{pmatrix}.
 \end{aligned}$$

All quantities in (276) have been defined independently of the desired sign. The unresolved issue is its uniform validity, including coherent clusters of nearby rational labels; a cellwise argument that resets the Gamma state does not prove it.

Conditional closure ledger for row (d). Constructed and proved here: the balanced channel factorization (169) with its explicit reference and load columns; the exactness of the threshold step, Theorem 9.16, including the identification of the newborn directions as the annulus plus exactly two Tate modes; the reduction of the threshold condition to a *unit output capacity* (272) in the output-defect metric, Theorem 9.42; the regularized one-step criterion 9.43, which derives the range condition rather than assuming it; the identification of the return sum with the Schur penalty, Proposition 9.17; the Gamma–Euler–Tate boundary object, rational bands, connected-cluster payment and exact recombination block; and the certified initial interval $0 < T \leq \log 2$.

Also proved, and negative: Corollary 9.20, that no comparison of R_T against L_T by size alone can succeed — the deficit is $3\sqrt{N}$, of leading order, and the shift-chain constant is sharp.

Two independent inputs are needed, and their status now differs:

- (i) Conjecture 9.37, the uniform Schur angle bound $\rho_N \leq 1 - \varepsilon$, equivalently the unit source estimate (275). Open; measured to hold with a wide margin on every threshold tested.
- (ii) Problem 9.44, the transfer comparison $\mathfrak{D}_E \geq c_N \Xi_N^* \mathfrak{D}_N \Xi_N$ with a uniform $c_N \geq 1$, which is what would carry (i) from the source model to the exact threshold condition. **Measured, and it fails** (Remark 9.45): $c_N < 1$ at every threshold tested, decaying in N , with the Galerkin evaluation bounding the true constant from *above*.

Consequently the source route does not close row (d) as normalized, and Corollary 9.39 does not apply. What the same computation does show is that the exact target (267) itself holds on every threshold tested; the live problem is therefore to prove it directly in the output-defect metric of Theorem 9.42, where it carries constant one, rather than through the source model. The unconditional global primitive inequality, unconditional row (d), and the Riemann Hypothesis are not established in the present manuscript.

EXECUTABLE COMPANION

The source distribution includes the Google-Colab-compatible notebook `certificates/arithmetic_lefschetz_certificates.ipynb` and separate row-specific runners. Together they check the exact finite arithmetic, corrected negabinary admissible lengths, Boolean matrix-code separation, cotangent ranks, determinant limits, contact determinants, Dirichlet-convolution identities and the Lorentzian ruling matrix used above. Every cell is labelled as an exact identity, a finite model or a non-computational categorical theorem. Successful execution is evidence for the finite calculations, not a substitute for the proofs in Section 2. The same directory contains a plain-Python runner which executes all code cells without Jupyter. For the certified row-(d) endpoints of Theorem 9.7, the subdirectory `certificates/row_d_log2` contains the complete dependency closure of the $T = \log 2$ verification together with the $T = \frac{1}{2} \log 3$ assembly it is built on, and a SHA-256 manifest pinning the five load-bearing files and recording the exact reproduction commands.

That directory is organized in two tiers, and the distinction is material. The standard-library verifier `114_d_85_two_level_endpoint_budget_verify.py` runs with no dependencies and recombines the certified enclosures into the four margins quoted in the proof — the big-block gap > 0.4659 , the v_2 complement gap > 0.0020336972824697 , the capacity threshold < 1.2111 against its lower capacity, and the odd complement > 0.0497 — confirming that they compose into a strictly positive final margin. The Arb-interval scripts which *produce* those enclosures form the second tier and require `python-flint`; the manifest records their invocations verbatim. Floating point is used only to select the fixed low modes and to precondition congruences, so a poor selection can weaken a bound but cannot invalidate one.

The endpoint $T = \frac{1}{2} \log 5$ is *not* certified and no artifact claims it; the retraction and its reason are recorded in §9.7 above.

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